

Title Page

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Protocol Title:

A Phase 3b, Randomized, Multicenter, Controlled Study of Mirikizumab and Placebo or Mirikizumab Concomitantly Administered with Tirzepatide in Adult Participants with Moderately to Severely Active Ulcerative Colitis and Obesity or Overweight

Protocol Number: I6T-MC-AMCD

Amendment Number: e

Compounds: Mirikizumab (LY3074828) and tirzepatide (LY3298176)

Brief Title: Mirikizumab and Tirzepatide Concomitantly Administered in Adult Participants with Moderately to Severely Active Ulcerative Colitis and Obesity or Overweight: Phase 3b Study

Study Phase: 3b

Acronym: COMMIT-UC

Sponsor Name: Eli Lilly and Company

Legal Registered Address: Indianapolis, Indiana, USA 46285

Regulatory Agency Identifier Number(s):

IND: 174802

EU trial number: 2024-520209-38-00

Approval Date: Protocol Amendment (e) Electronically Signed and Approved by Lilly on date provided below.

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Medical Monitor Name and Contact Information will be provided separately.

Protocol Amendment Summary of Changes Table

DOCUMENT HISTORY	
Document	Date
Amendment d	10-Sep-2025
Amendment c	08-Aug-2025
Amendment b	31-Jul-2025
Amendment a	17-Apr-2025
Original Protocol	04-Feb-2025

Amendment [e]

This amendment is considered to be substantial.

The amendment is considered to be substantial because it is likely to have a significant impact on the

- safety or the rights of the study participants

Overall rationale for the amendment

Section # and Name	Description of Change	Brief Rationale
Section 1.1 Synopsis, Section 3 Objectives, Endpoints and Estimands, Section 8.1 Efficacy Assessments, Section 9.1 Statistical Hypotheses, Section 9.3.3 Secondary Endpoints/Estimands Analysis	Replaced “mucosal healing” with “histologic-endoscopic mucosal improvement plus absence of neutrophils” consistently.	Updates as per regulatory feedback.
Section 1.3 Schedule of Activities (SoA)	Review and reinforcement of diet, exercise, and lifestyle goals - screening visit removed	For clarification of lifestyle goals by site staff.
Section 1.3 Schedule of Activities (SoA)	Added Hematology and Clinical Chemistry testing for Visit 7	Correction
Section 1.3 Schedule of Activities (SoA)	Deleted mMS review at Visits 8, 15, and ED.	Correction. These visits are not required for protocol-related activities. The only site review of the mMS required is at Visit 2.
Section 2.3.1 Risk Assessment	Added risk for assessment for gallbladder disease in tirzepatide	Updated risk description in alignment with Section 8.2.14.
Section 3 Objectives, Endpoints, and Estimands	Change in the exploratory endpoint from “Baseline visceral fat index at Week 0 and visceral fat index at Week 52” to “Change in percentage of visceral fat mass of total body mass from baseline to Week 52.”	To correct the measurement of baseline visceral fat mass.

Section # and Name	Description of Change	Brief Rationale
Section 5.1 Inclusion Criteria	Added the following wording to Inclusion Criterion 4: “history of UC for greater than or equal to 8 years.”	Patients with UC should undergo colorectal cancer surveillance starting at 8 years after disease onset, as they are at higher risk of developing colorectal cancer. This timing allows for optimal prevention and early detection of colorectal cancer or dysplasia.
Section 5.1 Inclusion Criteria	Removed the following wording from Inclusion Criterion 11: “The prescribed dose has been stable for at least 2 weeks before the screening colonoscopy.” Moved the following wording to the end of the paragraph: See Section 6.9.5 for instructions on tapering of oral corticosteroid therapy.”	Deletion of duplicate text.
Section 5.2 Exclusion Criteria	Removed wording “significant active or” from Exclusion Criterion 34.	Clarification
Section 5.2 Exclusion Criteria	Removed note regarding marijuana use from Exclusion Criterion 35.	Clarification
Section 5.2 Exclusion Criteria	In Exclusion Criterion 50 for IV corticosteroids replaced “2 weeks prior to the screening colonoscopy” with “2 weeks prior to the screening colonoscopy and throughout screening”	Clarification
Section 5.2 Exclusion Criteria	Removed word “prior” from Exclusion Criterion 53.	Clarification
Section 5.2 Exclusion Criteria	Added Exclusion Criterion 61 “Have abnormal 12-lead electrocardiogram (ECG) that, in the opinion of the investigator or sponsor, increases the risks associated with participating in the study”.	Added per regulatory feedback.
Section 6.7 Continued Access to Study Intervention after the End of the Study	Added additional details for continued access to study interventions and schedule of activity.	Changes made to include flexibility to continued access and provide additional information on continued access.
Section 8.1.1.1 mMS	Changed FDA guidance from 2022 to 2016.	Correction

Section # and Name	Description of Change	Brief Rationale
Section 8.1.1.2. Endoscopy: Colonoscopy or Sigmoidoscopy	Added the details of possibility for partnership with vendor to use AI-based scoring system on endoscopy.	To incorporate the additional analyses of the endoscopy.
Section 8.1.2.3 Patient-Reported Outcomes: Daily Electronic Diary	Removed “and change” from the following sentence: “.....UC symptom severity and perception of disease severity and change:”	Patient Global Impression of Change has been removed from the list.
Section 8.2.14. Gallbladder Disease and Tirzepatide	Replaced “drugs that work like tirzepatide” with “GLP-1 receptor agonists”	Clarification
Section 10.11. Appendix 11: mMS for the Assessment of Ulcerative Colitis Disease Activity	Updated mMS appendix as per FDA Guidance 2016.	To align with data collection as per FDA Guidance 2016.
Section 11. References	Added reference to FDA guidance	Clarification

Abbreviations: mMS = modified Mayo score; UC = ulcerative colitis.

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1. Protocol Summary

1.1. Synopsis

Protocol Title: A Phase 3b, Randomized, Multicenter, Controlled Study of Mirikizumab and Placebo or Mirikizumab Concomitantly Administered with Tirzepatide in Adult Participants with Moderately to Severely Active Ulcerative Colitis and Obesity or Overweight

Brief Title: Mirikizumab and Tirzepatide Concomitantly Administered in Adult Participants with Moderately to Severely Active Ulcerative Colitis and Obesity or Overweight: Phase 3b Study

Regulatory Agency Identifier Number(s):

IND: 174802

EU trial number: 2024-520209-38-00

Rationale:

Over 50% of patients with ulcerative colitis (UC) have obesity or overweight. Obesity in patients with UC is independently associated with higher burden and costs of hospitalizations. Bariatric surgery has been shown to result in lower use of medication for inflammatory bowel disease (IBD) and improved IBD outcomes.

This study, I6T-MC-AMCD (AMCD), aims to evaluate the efficacy and safety of mirikizumab and placebo compared with mirikizumab and concomitantly administered tirzepatide in adult participants with moderately to severely active UC and obesity, or overweight and in the presence of at least 1 weight-related comorbid condition, including

- hypertension
- type 2 diabetes mellitus
- dyslipidemia
- obstructive sleep apnea, or
- cardiovascular disease.

Objectives, Endpoints, and Estimands:

Objectives	Endpoints
Primary	
To evaluate if concomitant administration of mirikizumab and tirzepatide at the MTD (up to 15 mg) is superior to mirikizumab and placebo in simultaneously achieving clinical remission and weight reduction	• Proportion of participants who simultaneously achieve clinical remission and at least 10% weight reduction at Week 52
Key Secondary	
To evaluate the effect of treatment with concomitant administration of mirikizumab and tirzepatide at the MTD (up to 15 mg) versus mirikizumab and placebo as measured by gastroenterologic outcomes	• Proportion of participants who achieve the following at Week 52: ○ clinical remission ○ endoscopic response

Objectives	Endpoints
	○ histologic-endoscopic mucosal improvement plus absence of neutrophils
Additional Secondary	
To evaluate the effect of treatment with concomitant administration of mirikizumab and tirzepatide at the MTD (up to 15 mg) versus mirikizumab and placebo as measured by gastroenterologic and patient-reported outcomes	• Proportion of participants who achieve the following at Week 24: ○ clinical remission ○ endoscopic response ○ histologic-endoscopic mucosal improvement plus absence of neutrophils • Proportion of participants who achieve the following at Week 52: ○ maintenance of clinical remission ○ corticosteroid-free remission • Proportion of participants who achieve the following at Week 24 and Week 52: ○ clinical response ○ endoscopic remission ○ Bowel Movement Urgency improvement (captured with Urgency NRS) ○ histologic improvement ○ histologic remission; that is, histologic improvement with absence of neutrophils ○ Bowel Urgency remission • Proportion of participants who achieve symptomatic remission at Weeks 4, 8, 12, 24, and 52 • Proportion of participants who achieve symptomatic response at Weeks 4, 8, 12, 24, and 52 • Proportion of participants who simultaneously achieve clinical remission and at least 5% weight reduction at Week 24 and Week 52 • Proportion of participants who simultaneously achieve clinical remission and at least 10% weight reduction at Week 24 • Change from baseline in biomarkers: CRP and fecal calprotectin, at various time points (Weeks 4, 12, 24, and 52) • Change from baseline at Week 24 and Week 52 in ○ IBDQ ○ EQ-5D-5L ○ WPAI:UC ○ PGR-S • Proportion of participants at Week 52 who had ○ resolution of EIMs among those who had EIMs at baseline ○ UC-related emergency room visits ○ UC-related hospitalizations, and ○ UC-related surgeries
To evaluate the effect of treatment with concomitant administration of mirikizumab and tirzepatide at the	• Proportion of participants who achieve at least a 10% reduction in body at Week 24 and Week 52

Objectives	Endpoints
MTD (up to 15 mg) versus mirikizumab and placebo as measured by metabolic outcomes	• Percent change in body weight at Weeks 24, 40, and 52 • Change in BMI (kg/m²) at Weeks 24, 40, and 52 • Change in waist circumference at Weeks 8, 12, 24, 40, and 52 • Change at Week 24 and Week 52 in ○ HbA1c (%), and ○ glucose (non-fasting) • Change at Week 52 in ○ total cholesterol (mg/dL) ○ TG (mg/dL) ○ LDL (mg/dL), and ○ HDL (mg/dL) (non-fasting lipids) • Change in systolic and diastolic blood pressure (mmHg) at Week 52
To describe safety and tolerability of the study interventions	• TEAEs • discontinuation of study intervention due to AEs • SAEs, and • injection site reactions

Abbreviations: AE = adverse event; CRP = C-reactive protein; BMI = body mass index; EIM = extraintestinal manifestation; HbA1c = hemoglobin A1c; HDL = high-density lipoprotein; IBDQ = Inflammatory Bowel Disease Questionnaire; LDL = low-density lipoprotein; MTD = maximum tolerated dose; NRS = numeric rating scale; PGR-S = Patient's Global Rating of Severity; SAE = serious adverse event; TEAE = treatment-emergent adverse event; TG = triglycerides; UC = ulcerative colitis; WPAI:UC = Work Productivity and Activity Impairment Questionnaire ulcerative colitis.

The primary clinical question of interest is:

In participants with moderately to severely active UC who have obesity, or are overweight and in the presence of at least 1 weight-related comorbid condition, what is the treatment difference between mirikizumab and placebo versus mirikizumab concomitantly administered with tirzepatide in simultaneously achieving clinical remission and at least 10% weight reduction at Week 52 without early permanent discontinuation of study intervention or taking specified prohibited medication?

Overall Design:

Study AMCD is a Phase 3b, randomized, multicenter study to assess the efficacy and safety of mirikizumab and placebo compared with mirikizumab concomitantly administered with tirzepatide at the MTD (up to 15 mg) in adult participants with moderately to severely active UC and obesity, or overweight and in the presence of at least 1 weight-related comorbid condition.

The maximum duration of this study is approximately 61 weeks, with the following study periods:

- Study Period I, Screening: up to 5 weeks (≤ 35 days)
- Study Period II, Induction and Dose Escalation: 24 weeks
- Study Period III, Maintenance Treatment: 28 weeks, and
- Safety follow-up: 4 weeks after the last visit.

Brief Summary:

The participants in this study will have UC and obesity, or overweight in the presence of at least 1 weight-related illness at the same time.

The main purpose of this study is to show whether in these individuals, treatment with both mirikizumab and tirzepatide, compared with treatment with mirikizumab and placebo, leads to

- decrease or disappearance of UC symptoms, and
- loss of at least one-tenth of the overall body weight.

Study details include:

- The study duration will be up to 61 weeks.
- The treatment duration will be up to 52 weeks.
- The visit frequency will be every 4 weeks during the treatment period. Between Week 28 and Week 48, some of the visits will be conducted by telephone.

Study Population:

In general, an individual may participate in this study if they

- Are ≥ 18 to ≤ 70 years of age.
- Have had an established diagnosis of UC for ≥ 3 months before Visit 2, which includes endoscopic evidence of UC and a histopathology report that supports a diagnosis of UC.
- Have
 - obesity (body mass index ≥ 30 kg/m²), or
 - overweight (body mass index ≥ 27 kg/m² to < 30 kg/m²) and in the presence of at least 1 of these weight-related comorbid conditions:
 - hypertension
 - type 2 diabetes mellitus
 - dyslipidemia
 - obstructive sleep apnea, or
 - cardiovascular disease.
- Have moderately to severely active UC as defined by an mMS of 5 to 9 points and Endoscopic Subscore of 2 to 3 (confirmed by central review) within 21 days before Visit 2.

In general, an individual may not take part in the study if they

- Have a current diagnosis of
 - Crohn's disease
 - IBD unclassified (formerly known as indeterminate colitis), or
 - primary sclerosing cholangitis.
- Have a diagnosis of type 1 diabetes or have insulin-treated type 2 diabetes mellitus.

Number of Participants:

Approximately 350 participants will be randomly assigned to mirikizumab and placebo or mirikizumab and tirzepatide used concomitantly, in a 1:1 randomization ratio.

The randomly assigned study population will include up to approximately 70% of either individuals assigned female at birth or assigned male at birth, whoever enrolls faster. The remaining sex will make up the remainder of the participant population.

Intervention Groups and Duration:

This table lists the interventions used in this study:

Intervention Name	Mirikizumab		Tirzepatide	Placebo to match tirzepatide
Study Treatment	Induction	Maintenance		
Dosage Level(s)	300 mg at Weeks 0, 4, and 8	200 mg Q4W starting at Week 12, given as 2 consecutive injections of 100 mg each	All doses QW: • 2.5 mg • 5 mg • 7.5 mg • 10 mg • 12.5 mg • 15 mg	QW
Route of Administration	IV infusion	SC (prefilled syringe)		
Administration	Onsite: • only by authorized study personnel • maximum rate of 10 mg/min by IV infusion over no less than 30 min	Initial administration: • Onsite, either ○ as self-administration by participant or caregiver, after appropriate training and under supervision of study personnel, or ○ by study personnel. Subsequent administration: • offsite (for example, at participant's home) by the participant or their caretaker after completion of training, or • onsite, by authorized study personnel (per participant's preference).		
Authorized as defined by EU Clinical Trial Regulation	Authorized and not used according to EU authorization ^a		Authorized and not used according to EU authorization ^b	Not authorized in EU

Abbreviations: EU = European Union; IV = intravenous; Q4W = every 4 weeks; QW = every week; SC = subcutaneous; T2DM = type 2 diabetes mellitus; UC = ulcerative colitis.

^a Mirikizumab is authorized in EU for the treatment of UC. In this study, clinical trial material will be used for mirikizumab.

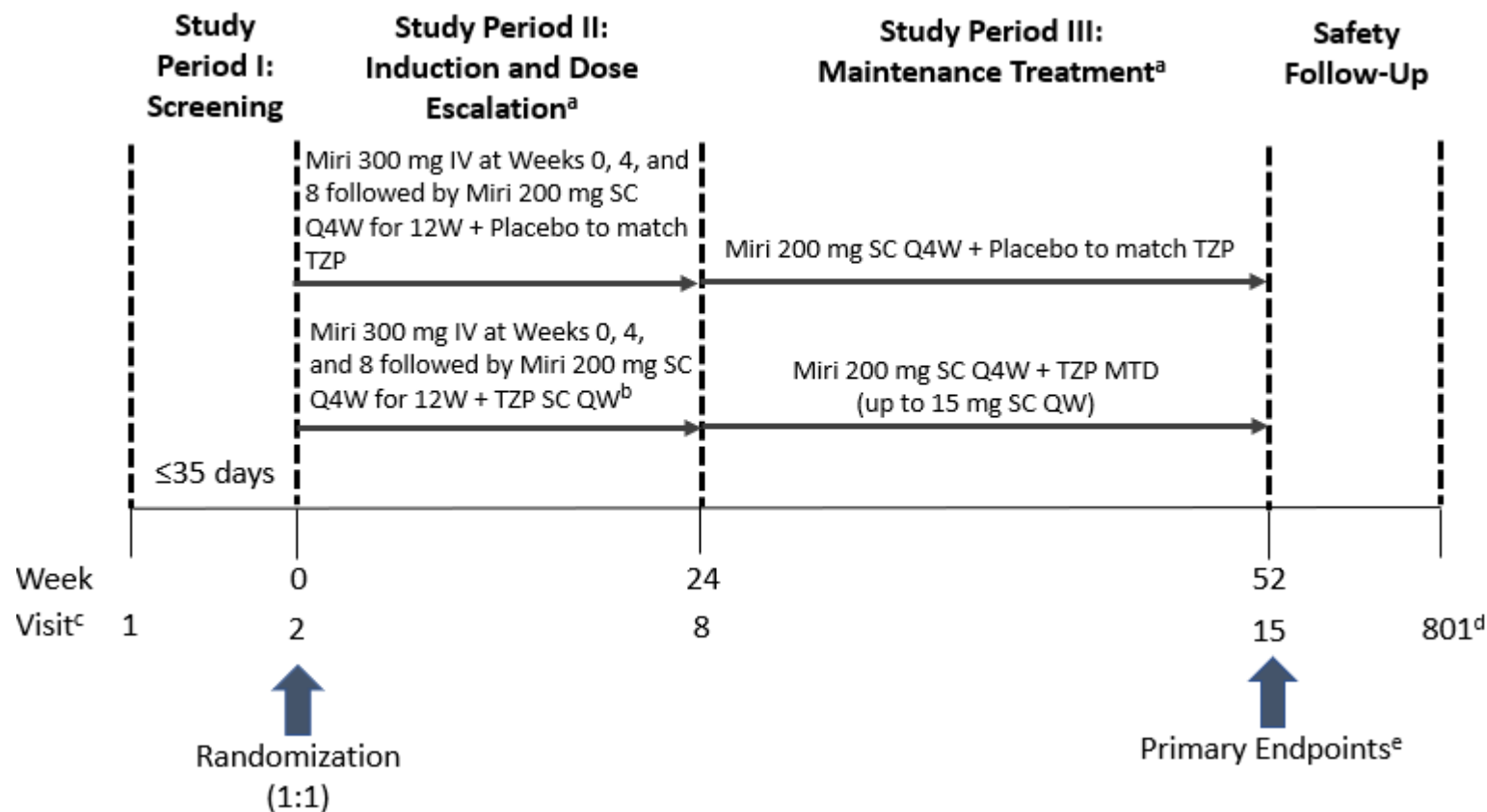
^b Tirzepatide is authorized in EU for the treatment of T2DM and for weight management, including weight loss and weight maintenance, in patients with obesity or overweight with at least one weight-related comorbid condition. In this study, clinical trial material will be used for tirzepatide.

Ethical Considerations of Benefit/Risk:

Taking into account the measures taken to minimize risk to participants participating in this study, the potential risks identified in association with mirikizumab and placebo or mirikizumab concomitantly administered with tirzepatide are justified by the anticipated benefits that may be afforded to participants with UC with obesity, or overweight and in the presence of at least 1 weight-related comorbid condition.

Data Monitoring Committee: Yes.

1.2. Schema



Abbreviations: ED = early discontinuation; IV = intravenous; Miri = mirikizumab; MTD = maximum tolerated dose; Q4W = every 4 weeks; QW = every week; SC = subcutaneous; TZP = tirzepatide; W = week.

- ^a Miri induction IV dosing at Weeks 0, 4, and 8; miri maintenance SC dosing from Week 12.
- ^b For details about TZP dose escalation, see Section [6.1.4.2](#).
- ^c Visits 2 through 15 will be conducted Q4W. During the Maintenance Period, some of the visits will be remote; for details, see the SoA (Section [1.3](#)).
- ^d Visit 801 will occur 4 weeks after the last treatment visit or ED.
- ^e For definition of the primary endpoints, see Section [3](#).

1.3. Schedule of Activities (SoA)

The SoA described below should be followed for all participants enrolled in Study AMCD. However, for those participants whose participation in this study is affected by exceptional circumstances, such as pandemics or natural disasters, please refer to Section 10.16 for additional guidance. If clinically warranted, an unscheduled visit may occur per investigator discretion.

Telehealth visits

Telehealth visits may be by telephone or other technology. Gray-shaded columns in the SoA tables represent telehealth visits.

Duration of Visit 1

Procedures for screening (Visit 1) may be conducted over more than 1 day as long as all activities are completed within the allowable visit tolerance.

1.3.1. Schedule of Activities for the Screening and Induction/Dose Escalation Period of Study I6T-MC-AMCD

Study I6T-MC-AMCD: SoA Table 1	Study Period I Screening	Study Period II Induction and Dose Escalation							Comments
Visit number	1	2	3	4	5	6	7	8	
Weeks from randomization	—	0	4	8	12	16	20	24	
Visit interval tolerance, days	≤35	—	±3	±3	±3	±3	±3	-7 to +14	
Visit detail	C	C	C	C	C	C	C	C	C = clinic
Consent and demographics									
Informed consent	X								The ICF must be signed before any protocol-specific tests or procedures are performed. See Section 10.1.3 for additional details.
Inclusion and exclusion criteria, review and confirm	X	X							Confirm inclusion and exclusion criteria before randomization and administration of first dose of study intervention.
Demographics	X								Includes ethnicity (where permissible), year of birth, sex assigned at birth, and race.
Preexisting conditions and medical history, including relevant surgical history	X								Collect all ongoing conditions and relevant past medical and surgical history.

Study I6T-MC-AMCD: SoA Table 1	Study Period I Screening	Study Period II Induction and Dose Escalation							Comments
Visit number	1	2	3	4	5	6	7	8	
Weeks from randomization	—	0	4	8	12	16	20	24	
Visit interval tolerance, days	≤35	—	±3	±3	±3	±3	±3	-7 to +14	
Visit detail	C	C	C	C	C	C	C	C	C = clinic
Prespecified medical history	X								Collect additional data for UC and comorbidities of interest, including weight-related • hypertension • T2DM • dyslipidemia • obstructive sleep apnea, or • cardiovascular disease.
Prior treatments for UC	X								
Concomitant medications	X	X	X	X	X	X	X	X	Collect all data for ongoing medications at the time of consent and concomitant medications of interest (Section 6.9.1).
Adverse events (AEs)	X	X	X	X	X	X	X	X	Any events that occur after signing the informed consent are considered AEs as defined in Section 10.3. Additional data are collected for certain AEs. See Section 8.3.3 for AESIs.
Substance abuse such as alcohol, caffeine, tobacco use	X								
Pre-colonoscopy instructions and reminders	X						X		Includes daily eDiary compliance, dates, and bowel preparation/procedure.
Physical evaluation									
Height	X								See Section 10.9.
Waist circumference	X	X	X	X	X	X	X	X	See Section 10.9.
Weight	X	X	X	X	X	X	X	X	See Section 10.9.
Complete physical examination	X								The complete physical examination excludes pelvic, rectal, and breast examinations, unless clinically indicated. All remaining physical examinations throughout the study should include a symptom-directed physical evaluation, as

Study I6T-MC-AMCD: SoA Table 1	Study Period I Screening	Study Period II Induction and Dose Escalation							Comments
Visit number	1	2	3	4	5	6	7	8	
Weeks from randomization	—	0	4	8	12	16	20	24	
Visit interval tolerance, days	≤35	—	±3	±3	±3	±3	±3	-7 to +14	
Visit detail	C	C	C	C	C	C	C	C	C = clinic
									well as an examination of the heart, lungs, abdomen, neurological system, thyroid, and visual examination of the skin. See Section 8.2.1. Assess for TB risk factors, and for signs and symptoms of active TB, including exam of peripheral lymph nodes (see Sections 8.2.1 and 8.2.10).
Symptom-directed physical assessment		X	X	X	X	X	X	X	Perform as indicated based on participant status and standard of care. May be performed by a qualified personnel per local regulations. Every symptom-directed examination should also include an examination of the heart, lungs, abdomen, neurological system, thyroid, and visual examination of the skin. At least every third month after screening, assess for TB risk factors, and for signs and symptoms of active TB, including an assessment of peripheral lymph nodes (Sections 8.2.1 and 8.2.10).
Vital signs	X	X	X	X	X	X	X	X	Include pulse rate, blood pressure, respiratory rate, and body temperature. Measured after participant has been sitting at least 5 minutes. See Sections 8.2.2 and 10.9.
Extraintestinal manifestations (EIMs)	X							X	See Section 8.2.1.
12-lead ECG (single, local)	X								Collect ECG before blood samples for laboratory testing. Local ECGs may be repeated at the investigator's discretion at any visit. See Section 8.2.3.

Study I6T-MC-AMCD: SoA Table 1	Study Period I Screening	Study Period II Induction and Dose Escalation							Comments
Visit number	1	2	3	4	5	6	7	8	
Weeks from randomization	—	0	4	8	12	16	20	24	
Visit interval tolerance, days	≤35	—	±3	±3	±3	±3	±3	-7 to +14	
Visit detail	C	C	C	C	C	C	C	C	C = clinic
Chest x-ray (posterior-anterior and if needed, lateral view; local)	X								Performed locally only if skin TB test, QuantiFERON®-TB Gold test, or T-SPOT®.TB is positive or other indications of TB are present. Interpreted and reported by a radiologist or pulmonologist. Chest x-ray is not required if one was performed within 6 months before Visit 1 and sufficient documentation exists for the TB evaluation (Sections 8.2.4 and 8.2.10). A chest CT scan can be used instead of a chest x-ray if the chest CT scan was performed within the same time window (Section 8.2.4).
DXA scan, whole body		X							See Section 8.1.3.1. If not completed, will not be considered a protocol deviation. It is recommended to perform successful baseline scan between Visits 1 and 2 and no later than Visit 3. Participants should fast for at least 8 hours without eating or drinking (except water) before each DXA scan. No water should be consumed within 2 hours before each DXA scan.
Patient-reported outcomes (electronic)									
Work Productivity and Activity Impairment Questionnaire: Ulcerative Colitis (WPAI:UC)		X						X	
Inflammatory Bowel Disease Questionnaire (IBDQ)		X						X	
EQ-5D-5L		X						X	

Study I6T-MC-AMCD: SoA Table 1	Study Period I Screening	Study Period II Induction and Dose Escalation							Comments
Visit number	1	2	3	4	5	6	7	8	
Weeks from randomization	—	0	4	8	12	16	20	24	
Visit interval tolerance, days	≤35	—	±3	±3	±3	±3	±3	-7 to +14	
Visit detail	C	C	C	C	C	C	C	C	C = clinic
Quick Inventory for Depressive Symptomatology – Self-Report 16 items (QIDS-SR16)	X				X			X	
Clinician-administered assessments (electronic)									
Review mMS		X							See Section 8.1.1.1.
Clinician-administered assessments (paper)									Collect after the assessment of AEs.
C-SSRS Screening/Baseline (Category Version)	X								
C-SSRS Since Last Assessed (Category Version)		X	X	X	X	X	X	X	
Participant education									
Injection training, tirzepatide SC		X							See Section 6.1.2. Repeat as needed.
Injection training, mirikizumab SC					X				See Section 6.1.2. Repeat as needed.
Review of contraceptive measures		X	X	X	X	X	X	X	
Review and reinforcement of diet, exercise, and lifestyle goals		X	X	X	X	X	X	X	See Section 5.3.1.
Lifestyle counseling (diet, exercise, and lifestyle goals)		X*	X*	X*	X*	X*	X*	X*	See Sections 5.3.1 and 5.3.2 *The window period for these visits should be up to 14 days
Participant dosing diary (paper)									Participants will maintain a study diary to document dosing information.
Dosing diary training		X							Repeat as needed.
Dispense dosing diary		X	X	X	X	X	X	X	
Dosing diary return			X	X	X	X	X	X	
Dosing diary review			X	X	X	X	X	X	
Participant 3-day nutrition and physical activity log (paper)									See Section 5.3.1.
Dispense 3-day log		X	X	X	X	X	X	X	
3-day log review			X	X	X	X	X	X	

Study I6T-MC-AMCD: SoA Table 1	Study Period I Screening	Study Period II Induction and Dose Escalation							Comments
Visit number	1	2	3	4	5	6	7	8	
Weeks from randomization	—	0	4	8	12	16	20	24	
Visit interval tolerance, days	≤35	—	±3	±3	±3	±3	±3	-7 to +14	
Visit detail	C	C	C	C	C	C	C	C	C = clinic
Participant daily diary (electronic)									The daily eDiary includes: • Rectal Bleeding (RB) • Stool Frequency (SF) • PGR-S, and • Urgency NRS.
Daily eDiary training	X								Repeat as needed.
Dispense daily eDiary	X								
Daily eDiary review		X	X	X	X	X	X	X	
Laboratory tests and sample collections									All laboratory tests will be performed in non-fasting state.
Hematology	X	X	X	X	X	X	X	X	
Clinical chemistry	X	X		X	X		X	X	
Lipids		X						X	
Serum pregnancy	X								Perform only for individuals of childbearing potential (IOCBP).
Urine pregnancy (local)		X	X	X	X	X	X	X	For IOCBP, pregnancy test result must be available before the first dose of study interventions. Perform additional pregnancy tests if a menstrual period is missed, if there is clinical suspicion of pregnancy, or as required by local law or regulation (Section 8.2.6).
Follicle-stimulating hormone (FSH)	X	X*							* Optional per investigator's discretion; perform as needed to confirm postmenopausal status in individuals AFAB. See Section 8.2.6.
Urinalysis	X								
Thyroid-stimulating hormone (TSH)	X								
HIV testing	X								

Study I6T-MC-AMCD: SoA Table 1	Study Period I Screening	Study Period II Induction and Dose Escalation							Comments
Visit number	1	2	3	4	5	6	7	8	
Weeks from randomization	—	0	4	8	12	16	20	24	
Visit interval tolerance, days	≤35	—	±3	±3	±3	±3	±3	-7 to +14	
Visit detail	C	C	C	C	C	C	C	C	C = clinic
Hepatitis C Virus (HCV) screening tests	X								If HCV antibody test is positive, perform HCV RNA testing. See Section 8.2.12.
Hepatitis B Virus (HBV) screening tests	X								If anti-HBc is positive, perform HBV DNA testing. See Section 8.2.11.
Hepatitis B (HBV) DNA	X			X		X		X	Only for participants who are anti-HBc positive at Visit 1. See Section 8.2.11.
Tuberculosis (TB) test	X								Participants who had a tuberculin skin test (TST) must have the test read 48 to 72 hours after placement. The test does not need to be read at the site but must be read by a trained professional and results must be presented to the site prior to first dose of study interventions. Local staff must be qualified to administer and interpret the test per the local regulation. See Sections 8.2.10 and 10.2.
Calcitonin	X								
Pancreatic amylase		X			X			X	
C-reactive protein, high sensitivity (hsCRP)		X	X		X			X	
Pancreatic lipase	X	X			X			X	
Hemoglobin A1c (HbA1c)	X				X			X	
Cystatin C	X							X	
Estimated glomerular filtration rate (eGFR)	X							X	Calculated using CKD-EPI Cystatin C Equation 2012 (NKF [WWW]).
Pharmacokinetics (PK) samples (mirikizumab; pre-dose)					X			X	Collect samples before the dosing, if dosing is scheduled. Collect additional samples at specified times relative to onset of hypersensitivity events (Section 10.2.1).

Study I6T-MC-AMCD: SoA Table 1	Study Period I Screening	Study Period II Induction and Dose Escalation							Comments
Visit number	1	2	3	4	5	6	7	8	
Weeks from randomization	—	0	4	8	12	16	20	24	
Visit interval tolerance, days	≤35	—	±3	±3	±3	±3	±3	-7 to +14	
Visit detail	C	C	C	C	C	C	C	C	C = clinic
Pharmacokinetics (PK) samples (tirzepatide; pre-dose)					X			X	Collect samples before the dosing, if dosing is scheduled. Collect additional samples at specified times relative to onset of hypersensitivity events (Section 10.2.1).
Immunogenicity (ADA) (anti-mirikizumab antibodies) samples (pre-dose)		X			X			X	Collect samples before the dosing, if dosing is scheduled. Collect additional samples at specified times relative to onset of hypersensitivity events (Section 10.2.1).
Immunogenicity (ADA) (anti-tirzepatide antibodies) samples (pre-dose)		X			X			X	Collect samples before the dosing, if dosing is scheduled. Collect additional samples at specified times relative to onset of hypersensitivity events (Section 10.2.1).
Micronutrient panels		X	X*						* Additional micronutrient testing should be done if - the PI suspects the participant is at risk having malnutrition, for example, caloric intake is significantly reduced to below 800 to 1200 kcal - a participant’s BMI falls below 22 kg/m², or - in participants with T2DM, weight loss is >20% or >4 kg/month. See Sections 8.2.13, 10.2, 10.2.2, and 10.7.
Stool tests and sample collections									
Dispense stool collection kit	X	X		X		X			
C. difficile testing	X	X							Additional C. difficile testing is allowed throughout the study at the investigator’s discretion.

Study I6T-MC-AMCD: SoA Table 1	Study Period I Screening	Study Period II Induction and Dose Escalation							Comments
Visit number	1	2	3	4	5	6	7	8	
Weeks from randomization	—	0	4	8	12	16	20	24	
Visit interval tolerance, days	≤35	—	±3	±3	±3	±3	±3	-7 to +14	
Visit detail	C	C	C	C	C	C	C	C	C = clinic
Fecal calprotectin		X	X		X			X	
Stool culture	X	X*							Additional Stool Culture testing is allowed throughout the study at the investigator’s discretion. * Additional local stool testing, for example, for ova and parasites, is allowed at the investigator’s discretion.
Stored samples									
Exploratory biomarkers (blood)		X	X		X			X	
Exploratory biomarkers (stool supernatant)		X	X		X			X	
Exploratory biomarkers (colon/rectum tissue)	X							X	Obtained from the endoscopic biopsy.
Genetics sample		X							Sample can be obtained at or after Visit 2.
Biopsy tests and sample collections									
Colonoscopy with biopsy	X							X	Sigmoidoscopy is allowed, but colonoscopy is preferred; see Section 8.1.1.2. Second colonoscopy is allowed if inadequate preparation or equipment issues. As stated in inclusion criterion [5], investigators may perform surveillance coloscopy as part of the screening colonoscopy. For post-screening colonoscopy: If the participant is not compliant with the eDiary entry (Section 8.1.1.1) within the colonoscopy visit interval, contact the Lilly medical monitor regarding rescheduling the colonoscopy. During colonoscopy and sigmoidoscopy biopsies will be obtained, and additional

Study I6T-MC-AMCD: SoA Table 1	Study Period I Screening	Study Period II Induction and Dose Escalation							Comments
Visit number	1	2	3	4	5	6	7	8	
Weeks from randomization	—	0	4	8	12	16	20	24	
Visit interval tolerance, days	≤35	—	±3	±3	±3	±3	±3	-7 to +14	
Visit detail	C	C	C	C	C	C	C	C	C = clinic
									colonic/rectal biopsies will be obtained for exploratory biomarker analysis.
Randomization and dosing-related activities									
Register visit with IWRS	X	X	X	X	X	X	X	X	
Randomization via IWRS		X							
Dispense study intervention via IWRS		X	X	X	X	X	X	X	
Dispense study intervention to participant (for at-home dosing)		X	X	X	X	X	X	X	
Administer study intervention – mirikizumab infusion		X	X	X					
Observe participant administer first dose of tirzepatide SC		X							Repeat as needed.
Observe participant administer first dose of mirikizumab SC					X				Repeat as needed.
Dispense ancillary supplies to participant		X	X*						* After Visit 2, dispense as needed.
Participant returns unused study intervention			X	X	X	X	X	X	
Assess study intervention compliance			X	X	X	X	X	X	

1.3.2. Schedule of Activities for the Maintenance Period, Early Discontinuation Visit, Unscheduled Visits, and Safety Follow-Up of Study I6T-MC-AMCD

Study I6T-MC-AMCD: SoA Table 2	Study Period III Maintenance Treatment									Safety follow-up	Comments
Visit number	9	10	11	12	13	14	15	ED	UV	801	
Weeks from randomization	28	32	36	40	44	48	52	—	—	—	
Visit interval tolerance, days	±3	±3	±3	±3	±3	±3	-7 to +14	—	—	4 weeks after last visit	
Visit detail	T	C	T	C	T	T	C	C	C	C	C = clinic T = telehealth visit (shaded columns)
Concomitant medications	X	X	X	X	X	X	X	X	X	X	Collect all data for ongoing medications at the time of consent and concomitant medications of interest (Section 6.9.1).
Adverse events (AEs)	X	X	X	X	X	X	X	X	X	X	Any events that occur after signing the informed consent are considered AEs as defined in Section 10.3. Additional data are collected for certain AEs. See Section 8.3.3 for AESIs.
Pre-colonoscopy instructions and reminders						X					Includes eDiary compliance, dates, and bowel preparation/procedure.
Physical evaluation											
Waist circumference		X		X			X	X		X	See Section 10.9.
Weight		X		X			X	X		X	See Section 10.9.
Complete physical examination										X	The complete physical examination excludes pelvic, rectal, and breast examinations, unless clinically indicated. All remaining physical examinations throughout the study should include a symptom-directed physical evaluation, as well as an examination of the heart, lungs, abdomen, neurological system, thyroid, and visual examination of the skin. See Section 8.2.1.
Symptom-directed physical assessment		X		X			X	X	X		Perform as indicated based on participant status and standard of

Study I6T-MC-AMCD: SoA Table 2	Study Period III Maintenance Treatment									Safety follow-up	Comments
Visit number	9	10	11	12	13	14	15	ED	UV	801	
Weeks from randomization	28	32	36	40	44	48	52	—	—	—	
Visit interval tolerance, days	±3	±3	±3	±3	±3	±3	-7 to +14	—	—	4 weeks after last visit	
Visit detail	T	C	T	C	T	T	C	C	C	C	C = clinic T = telehealth visit (shaded columns)
											care. May be performed by a qualified personnel per local regulations. Every symptom-directed examination should also include an examination of the heart, lungs, abdomen, neurological system, thyroid, and visual examination of the skin. At least every third month after screening, assess for TB risk factors, and for signs and symptoms of active TB, including an assessment of peripheral lymph nodes (Sections 8.2.1 and 8.2.10).
Vital signs		X		X			X	X	X	X	Include pulse rate, blood pressure, respiratory rate, and body temperature. Measured after participant has been sitting at least 5 minutes. See Sections 8.2.2 and 10.9.
Extraintestinal manifestations (EIMs)							X				See Section 8.2.1.
DXA scan, whole body							X	X*			See Section 8.1.3.1. If not completed, will not be considered a protocol deviation. Participants should fast for at least 8 hours without eating or drinking (except water) before each DXA scan. No water should be consumed within 2 hours before each DXA scan.

Study I6T-MC-AMCD: SoA Table 2	Study Period III Maintenance Treatment									Safety follow-up	Comments
Visit number	9	10	11	12	13	14	15	ED	UV	801	
Weeks from randomization	28	32	36	40	44	48	52	—	—	—	
Visit interval tolerance, days	±3	±3	±3	±3	±3	±3	-7 to +14	—	—	4 weeks after last visit	
Visit detail	T	C	T	C	T	T	C	C	C	C	C = clinic T = telehealth visit (shaded columns)
											* At ED, repeat if more than 3 months have passed since the last scan.
Patient-reported outcomes (electronic)											
Work Productivity and Activity Impairment Questionnaire: Ulcerative Colitis (WPAI:UC)							X	X			
Inflammatory Bowel Disease Questionnaire (IBDQ)							X	X			
EQ-5D-5L							X	X			
Quick Inventory for Depressive Symptomatology – Self-Report 16 items (QIDS-SR16)				X			X	X			
Clinician-administered assessments (paper)										Collect after the assessment of AEs.	
C-SSRS Since Last Assessed (Category Version)		X		X			X	X		X	
Participant education											
Review of contraceptive measures		X		X			X				
Review and reinforcement of diet, exercise, and lifestyle goals	X	X	X	X	X	X	X				See Section 5.3.1.
Lifestyle counseling (diet, exercise, and lifestyle goals)	X*	X*	X*	X*	X*	X*	X*				See Sections 5.3.1 and 5.3.2. *The window period for these visits should be up to 14 days.

Study I6T-MC-AMCD: SoA Table 2	Study Period III Maintenance Treatment									Safety follow-up	Comments
Visit number	9	10	11	12	13	14	15	ED	UV	801	
Weeks from randomization	28	32	36	40	44	48	52	—	—	—	
Visit interval tolerance, days	±3	±3	±3	±3	±3	±3	-7 to +14	—	—	4 weeks after last visit	
Visit detail	T	C	T	C	T	T	C	C	C	C	C = clinic T = telehealth visit (shaded columns)
Participant dosing diary (paper)											Participants will maintain as study diary to document dosing information.
Dispense dosing diary		X		X							
Dosing diary review	X	X	X	X	X	X	X	X	X		
Dosing diary return		X		X			X	X	X		
Participant 3-day nutrition and physical activity log (paper)											See Section 5.3.1.
Dispense 3-day log		X		X							
3-day log review	X	X	X	X	X	X	X	X			
3-day log return							X	X			
Participant daily diary (electronic)											The daily eDiary includes: • Rectal Bleeding (RB) • Stool Frequency (SF) • PGR-S, and • Urgency NRS.
Daily eDiary review		X		X			X	X			
Daily eDiary return							X	X			
Laboratory tests and sample collections											All laboratory tests will be performed in non-fasting state.
Hematology		X		X			X	X	X	X	
Clinical chemistry		X		X			X	X	X	X	
Lipids							X	X		X	
Urine pregnancy (local)	X*	X	X*	X	X*	X*	X	X		X	*For telehealth visits, pregnancy tests will be home pregnancy tests. For IOCBP, pregnancy test result must be available before the first dose of study intervention. Perform additional pregnancy tests if a menstrual period is missed, if there

Study I6T-MC-AMCD: SoA Table 2	Study Period III Maintenance Treatment									Safety follow-up	Comments
Visit number	9	10	11	12	13	14	15	ED	UV	801	
Weeks from randomization	28	32	36	40	44	48	52	—	—	—	
Visit interval tolerance, days	±3	±3	±3	±3	±3	±3	-7 to +14	—	—	4 weeks after last visit	
Visit detail	T	C	T	C	T	T	C	C	C	C	C = clinic T = telehealth visit (shaded columns)
											is clinical suspicion of pregnancy, or as if required by local law or regulation (Section 8.2.6).
Follicle-stimulating hormone (FSH)					X*						* Optional per investigator's discretion; perform as needed to confirm postmenopausal status in individuals AFAB. See Section 8.2.6.
Hepatitis B (HBV) DNA		X		X			X	X			Only for participants who are anti- HBc positive at Visit 1. See Section 8.2.11.
Pancreatic amylase				X			X	X		X	
C-reactive protein, high sensitivity (hsCRP)							X	X	X		
Pancreatic lipase				X			X	X		X	
Hemoglobin A1c (HbA1c)				X			X				
Cystatin C							X	X			
Estimated glomerular filtration rate (eGFR)							X	X			Calculated using CKD-EPI Cystatin C Equation 2012 (NKF [WWW]).
Pharmacokinetics (PK) samples (mirikizumab; pre-dose)				X			X	X		X	Collect samples before the dosing, if dosing is scheduled. Collect additional samples at specified times relative to onset of hypersensitivity events (Section 10.2.1).
Pharmacokinetics (PK) samples (tirzepatide; pre-dose)				X			X	X		X	Collect samples before the dosing, if dosing is scheduled. Collect additional samples at specified times relative to onset of hypersensitivity events (Section 10.2.1).

Study I6T-MC-AMCD: SoA Table 2	Study Period III Maintenance Treatment									Safety follow-up	Comments
Visit number	9	10	11	12	13	14	15	ED	UV	801	
Weeks from randomization	28	32	36	40	44	48	52	—	—	—	
Visit interval tolerance, days	±3	±3	±3	±3	±3	±3	-7 to +14	—	—	4 weeks after last visit	
Visit detail	T	C	T	C	T	T	C	C	C	C	C = clinic T = telehealth visit (shaded columns)
Immunogenicity (ADA) (anti-mirikizumab antibodies) samples (pre-dose)							X	X		X	Collect samples before the dosing, if dosing is scheduled. Collect additional samples at specified times relative to onset of hypersensitivity events (Section 10.2.1).
Immunogenicity (ADA) (anti-tirzepatide antibodies) samples (pre-dose)							X	X		X	Collect samples before the dosing, if dosing is scheduled. Collect additional samples at specified times relative to onset of hypersensitivity events (Section 10.2.1).
Micronutrient panels				X*							* Additional micronutrient testing should be done if - the PI suspects the participant is at risk having malnutrition, for example, caloric intake is significantly reduced to below 800 to 1200 kcal - a participant's BMI falls below 22 kg/m², or - in participants with T2DM, weight loss is >20% or >4 kg/month. See Sections 8.2.13, 10.2, 10.2.2, and 10.7.
Stool tests and sample collections											
Dispense stool collection kit				X*							* Dispense as needed.
<i>C. Difficile</i> testing				X							Additional <i>C. Difficile</i> testing may be done at the investigator's discretion.

Study I6T-MC-AMCD: SoA Table 2	Study Period III Maintenance Treatment									Safety follow-up	Comments
Visit number	9	10	11	12	13	14	15	ED	UV	801	
Weeks from randomization	28	32	36	40	44	48	52	—	—	—	
Visit interval tolerance, days	±3	±3	±3	±3	±3	±3	-7 to +14	—	—	4 weeks after last visit	
Visit detail	T	C	T	C	T	T	C	C	C	C	C = clinic T = telehealth visit (shaded columns)
Fecal calprotectin							X	X	X		
Stool culture				X*							Additional stool testing is allowed throughout the study at the investigator's discretion *Additional local stool testing (for example, ova and parasites) is allowed at the investigator's discretion.
Stored samples											
Exploratory biomarkers (blood)							X	X			
Exploratory biomarkers (stool supernatant)							X	X	X		
Exploratory biomarkers (colon/rectum tissue)							X	X			Obtained from the endoscopic biopsy.

Study I6T-MC-AMCD: SoA Table 2	Study Period III Maintenance Treatment									Safety follow-up	Comments
Visit number	9	10	11	12	13	14	15	ED	UV	801	
Weeks from randomization	28	32	36	40	44	48	52	—	—	—	
Visit interval tolerance, days	±3	±3	±3	±3	±3	±3	-7 to +14	—	—	4 weeks after last visit	
Visit detail	T	C	T	C	T	T	C	C	C	C	C = clinic T = telehealth visit (shaded columns)
Biopsy tests and sample collections											
Colonoscopy with biopsy							X	X**	X*		Sigmoidoscopy is allowed, but colonoscopy is preferred; see Section 8.1.1.2. If the participant is not compliant with the eDiary entry (Section 8.1.1.1) within the colonoscopy visit interval, contact the Lilly medical monitor regarding rescheduling the colonoscopy. *Unscheduled sigmoidoscopy or colonoscopy per investigator's discretion (Section 6.9.6) ** No need to repeat colonoscopy if completed up to 3 months prior. Second colonoscopy is allowed if inadequate preparation or equipment issues. During colonoscopy and sigmoidoscopy biopsies will be obtained, and additional colonic/rectal biopsies will be obtained for exploratory biomarker analysis.
Dosing-related activities											
Register visit with IWRS	X	X	X	X	X	X	X	X		X	
Dispense study intervention via IWRS		X		X							
Dispense study intervention to participant (for at-home dosing)		X		X							

Study I6T-MC-AMCD: SoA Table 2	Study Period III Maintenance Treatment									Safety follow-up	Comments
Visit number	9	10	11	12	13	14	15	ED	UV	801	
Weeks from randomization	28	32	36	40	44	48	52	—	—	—	
Visit interval tolerance, days	±3	±3	±3	±3	±3	±3	-7 to +14	—	—	4 weeks after last visit	
Visit detail	T	C	T	C	T	T	C	C	C	C	C = clinic T = telehealth visit (shaded columns)
Dispense ancillary supplies to participant		X*		X*							* Dispense as needed.
Participant returns unused study intervention		X		X			X	X			
Assess study intervention compliance	X	X	X	X	X	X	X				

Abbreviations: AFAB = identified female at birth; AESIs = adverse events of special interest; anti-HBc = anti-hepatitis B core antibody; BMI = body mass index; C-SSRS = Columbia-Suicide Severity Rating Scale; CKD-EPI = Chronic Kidney Disease Epidemiology Collaboration; CT = computed tomography; DXA = dual-energy x-ray absorptiometry; ECG = electrocardiogram; ED = early discontinuation; eDiary = electronic diary; ICF = informed consent form; IWRS = interactive web-response system; mMS = modified Mayo score; NRS = numeric rating scale; PI = principal investigator; PGR-S = Patient's Global Rating of Severity; SC = subcutaneous; T2DM = type 2 diabetes mellitus; TB = tuberculosis; UC = ulcerative colitis; UV = unscheduled visit.

2. Introduction

2.1. Study Rationale

Over 50% of patients with UC have obesity or overweight. Obesity in patients with UC is independently associated with higher burden and costs of hospitalizations. Bariatric surgery has been shown to result in lower use of medication for IBD and improved IBD outcomes.

This study, I6T-MC-AMCD (AMCD), aims to evaluate the efficacy and safety of mirikizumab and placebo compared with mirikizumab and concomitantly administered tirzepatide in adult participants with moderately to severely active UC and obesity, or overweight and in the presence of at least 1 weight-related comorbid condition, including

- hypertension
- T2DM
- dyslipidemia
- obstructive sleep apnea, or
- cardiovascular disease.

Mirikizumab was first approved in Japan on 27 March 2023. Mirikizumab has been approved in multiple geographies for the treatment of moderately to severely active UC in patients 18 years of age and above (Omvoh™ package insert, 2024; Omvoh summary of product characteristics).

Tirzepatide is approved as an adjunct to a reduced-calorie diet and increased physical activity for chronic weight management in adults with obesity, or overweight in the presence of at least 1 weight-related comorbid condition, in the United States and EU, with applications either approved or under review in other regions (Zepbound® package insert, 2024; Mounjaro® summary of product characteristics).

2.2. Background

Mirikizumab

Mirikizumab is a humanized IgG4-variant monoclonal antibody that binds to the p19 subunit of IL-23 without binding to other members of the IL-12 cytokine family. Mirikizumab is being developed for the treatment of immune-mediated diseases in which the IL-23 pathway is thought to have a pathogenic role.

Mirikizumab Phase 3 induction Study I6T-MC-AMAN (AMAN) and maintenance Study I6T-MC-AMBG (AMBG) results demonstrated the consistent and robust efficacy of mirikizumab in the treatment of moderately to severely active UC.

Tirzepatide

Tirzepatide is a GIP receptor and GLP-1 receptor agonist. Tirzepatide is indicated as an adjunct to a reduced-calorie diet and increased physical activity for chronic weight management in adults with obesity, or overweight and in the presence of at least 1 weight-related comorbid condition (Zepbound package insert, 2024; Mounjaro summary of product characteristics).

Unmet medical need

Approximately 55.4% of patients with UC have obesity or overweight (Newberry and Lichtenstein 2013; Johnson et al. 2021).

Obesity leads to more rapid clearance and lower trough concentrations of biologic agents in patients with IBD and is associated with decreased treatment responsiveness to biologics (Bassi and Singh 2022). In the mirikizumab UC Phase 3 Study AMAN, for clinical remission and endoscopic response, there was a numerical trend for participants with obesity to have reduced response to mirikizumab compared with participants who had a normal weight. This numerical trend was unrelated to mirikizumab exposure.

In addition, obesity in patients with IBD, including UC, is independently associated with higher burden and costs of hospitalizations (Nguyen et al. 2019; El-Dallal et al. 2021). Similarly, bariatric surgery has been shown to be safe in patients with IBD and results in lower use of medication for IBD and improved IBD outcomes (Wise et al. 2022).

GLP-1 and inflammation

GLP-1 is involved in modulating inflammation by inhibiting mucosal cytokines and gut epithelial architecture in animal models of colitis, leading to mucosal healing suggesting that GLP-1 analogs may have an anti-inflammatory effect in patients with IBD colitis (Anbazhagan et al. 2017).

GLP-1 receptor agonists and IBD

A Danish registry of patients with IBD and T2DM showed that there was a decreased risk of IBD-related hospitalization and IBD-related surgery among users of GLP-1-based therapies compared with non-GLP-1-based therapies (Villumsen et al. 2021).

Alchirazi et al. (2024) conducted a US cohort propensity-matched study with over 14,000 patients with IBD, which included both CD and UC cohorts. In patients with UC, comparing patients who did or did not receive GLP-1 receptor agonists, those who did were less likely to

- be hospitalized
- undergo colectomy
- require steroids, or
- have colon cancer.

In a recent study, patients with T2DM in an Israeli nationwide cohort of patients with IBD and GLP-1 analog exposure for at least 6 months had improved outcomes in IBD, specifically in patients with obesity (Gorelik et al. 2024).

In a population-based cohort study in 10,855 adult patients using administrative health data from British Columbia, patients with immune-mediated inflammatory diseases such as IBD and T2DM taking GLP-1 receptor agonists had fewer major adverse cardiovascular events and a lower chance of death compared with those taking DPP-4 inhibitors (Karacabeyli et al. 2024).

Among an IBD cohort comprised of patients with diabetes and patients without diabetes, GLP-1 receptor agonists were well tolerated and were not associated with major adverse IBD outcomes. Additionally, GLP-1 receptor agonists were effective and associated with significant decreases in BMI and hemoglobin A1c. These changes were not significantly different from those observed in

a control group without IBD who initiated a GLP-1 receptor agonist during the same time. The study's findings therefore suggest that GLP-1 receptor agonists can be comfortably prescribed for patients with IBD (Levine et al. 2024).

2.3. Benefit/Risk Assessment

Detailed information about the known and expected benefits and risks and reasonably expected AEs of mirikizumab and tirzepatide may be found in the relevant IBs and prescribing information.

2.3.1. Risk Assessment

2.3.1.1. Study Interventions

Mirikizumab

Evaluation of safety data from mirikizumab clinical studies examining psoriasis, UC, and CD shows a safety profile consistent with the anti-IL-23p19 antibody class. The risks (adverse drug reactions) found to be associated with mirikizumab include

- injection site reactions
- infusion-related hypersensitivity reactions
- upper respiratory tract infections
- ALT and AST increases
- headache, and
- rash.

In addition, the following special warnings and precautions for use are recommended when using mirikizumab:

- **Serious hypersensitivity reactions:** Serious hypersensitivity reactions, including anaphylaxis, may occur with mirikizumab administration. If a serious hypersensitivity reaction occurs, discontinue mirikizumab immediately and initiate appropriate therapy.
- **Infections:** Mirikizumab may increase the risk of infection. Treatment with mirikizumab should not be initiated in patients with a clinically important active infection until the infection resolves or is adequately treated. Consider the risks and benefits of treatment prior to initiating the use of mirikizumab in patients with a chronic infection or a history of recurrent infection. Instruct patients to seek medical advice if signs or symptoms of clinically important acute or chronic infection occur. If a serious infection develops, consider discontinuation of mirikizumab until the infection resolves. Mirikizumab should not be given to patients with active TB. Consider anti-TB therapy before starting mirikizumab in patients with latent TB.
- **Hepatic enzyme elevations:** Elevations of aminotransferases have been reported in patients receiving mirikizumab. Evaluate liver enzymes as clinically indicated. If

increases in ALT or AST are observed and drug-associated liver injury is suspected, discontinue mirikizumab until this diagnosis is excluded (Section 8.2.9).

- Immunizations: Avoid the use of live vaccines in patients treated with mirikizumab. No data are available on the response to live or non-live vaccines.

Tirzepatide

Tirzepatide has not been studied in patients with severe GI disease.

Gastrointestinal adverse reactions

Tirzepatide has been associated with GI adverse reactions, sometimes severe, which include nausea, vomiting, and diarrhea. These adverse reactions may lead to dehydration, which could lead to a deterioration in renal function, including acute renal failure. Patients treated with tirzepatide should be advised of the potential risk of dehydration, due to the GI adverse reactions and take precautions to avoid fluid depletion and electrolyte disturbances. This should particularly be considered in the elderly, who may be more susceptible to such complications.

Potential malnutrition

Events related to malnutrition have been reported, including severe, in patients receiving tirzepatide. Risks associated with malnutrition include, but are not limited to, vitamin and mineral deficiency, protein deficiency, and low body weight.

Gallbladder disease

Gallbladder problems including gallstones have been reported uncommonly in patients taking tirzepatide and GLP-1 receptor agonists. In the tirzepatide SURMOUNT trials, cholelithiasis was reported in 1.1%, cholecystitis was reported in 0.7%, and cholecystectomy was reported in 0.2% of tirzepatide-treated patients. Risk may be increased if patients have diabetes or obesity and in cases of rapid weight loss. Some gallbladder problems may need surgery.

Potential risk of aspiration during anesthesia or deep sedation

An increasing safety concern has emerged regarding the use of GLP-1 receptor agonists in the perioperative setting, particularly due to the risk of delayed gastric emptying and subsequent aspiration during endoscopic and surgical procedures. Reports indicate instances of pulmonary aspiration of gastric contents in patients on GLP-1 receptor agonists undergoing sedation or general anesthesia, prompting a multi-society clinical practice guidance published on 29 October 2024, by various medical organizations including the American Gastroenterological Association and the American Society of Anesthesiologists (Kindel et al. 2024). It is recommended that investigators follow their local guidelines about GLP-1 agonists and deep sedation. Adhering to these recommendations can enhance patient safety during endoscopic and surgical procedures (Kindel et al. 2024). See Section 10.15 for recommendations for the safe use of GLP-1 receptor agonists during anesthesia or deep sedation, per the multisociety clinical practice guidance (Kindel et al. 2024).

Concomitant use of mirikizumab and tirzepatide

There are no contraindications per approved labeling for the use of both medications concomitantly. This Phase 3b study will use the marketed dosing regimens per prescribing information for mirikizumab and tirzepatide to evaluate the relative benefit/risk profile

associated with concomitant use of mirikizumab and tirzepatide in participants with UC with obesity or overweight.

Drug-drug interactions

There is accumulated safety data individually for mirikizumab and tirzepatide (see IBs). The concern for drug-drug interactions (anti-IL-23-p19 antibody with a small molecule like tirzepatide) is thought to be minimal but has not been assessed clinically. Tirzepatide is not considered immunosuppressive.

2.3.1.2. Study Procedures

Risks due to Common Study Procedures in Study AMCD		
Potential Risks of Clinical Significance	Summary of Data/ Rationale for Risk	Mitigation Strategy
<i>Risk associated with the endoscopic procedures</i>		
Potential risk of aspiration during anesthesia or deep sedation	These risks are considered rare but are recognized by the US and EU labelling information (Zepbound package insert; Mounjaro, summary of product characteristics).	See “Potential risk of aspiration during anesthesia or deep sedation” above and Section 10.15.
Bleeding and colonic perforation	These risks are considered rare but are recognized risks for endoscopic procedures (Rabeneck et al. 2008; Arora et al. 2009).	Trained and experienced endoscopists will be performing endoscopic assessments.

2.3.1.3. Management of Risks

The potential risks will be proactively monitored. Sections 5.1, 5.2, 7.1, 8.2, and 8.3 address known potential risks associated with mirikizumab and tirzepatide or the study procedures.

2.3.2. Benefit Assessment

Mirikizumab

Mirikizumab is approved for the treatment of moderately to severely active UC (Section 2.1). At the time of this benefit/risk assessment, mirikizumab has demonstrated efficacy in Phase 2 and Phase 3 placebo-controlled studies in adults with

- psoriasis (Reich et al. 2019; Papp et al. 2020)
- UC (Sandborn et al. 2018; D’Haens et al. 2019, 2022), and
- CD (Sands et al. 2022; Ferrante et al. 2024).

In the Phase 3 Studies AMAN and AMBG, over periods of 12 weeks and 40 weeks respectively, mirikizumab therapy had efficacy in both induction and maintenance phases across clinical,

symptomatic, endoscopic, and histologic measures of disease, even after treatment failure with conventional immunosuppressive agents, biologic therapies, or tofacitinib.

Participants in Study AMCD have obesity or are overweight. The prevalence of NASH in patients with morbid obesity has been reported to be 37% (Machado et al. 2006). Those with NASH, especially in the presence of fibrosis, are at a higher risk for adverse outcomes, including overall mortality, liver transplantation, or liver-related events (Angulo et al. 2015). The beneficial effects of weight loss in patients with NASH/MASH are well characterized (Promrat et al. 2010; Vilar-Gomez et al. 2015; Loomba et al. 2024). Treatment with tirzepatide results in glucose lowering, weight loss, and improved metabolic health in patients with T2DM (Frias et al. 2018).

Tirzepatide

SURMOUNT-1 and SURMOUNT-2 trials demonstrated the superiority of tirzepatide to placebo as an adjunct to life intervention in achieving body weight reduction and improving cardiometabolic measures (Zepbound package insert, 2024). Weight reduction in “tirzepatide-treated” participants started to differ from those in placebo as early as Week 4. The rate of weight reduction was higher in the first 24 weeks in tirzepatide-treated participants, with most of the weight reduction (approximately 12 to 17% of baseline weight) achieved at Week 36, as compared with 3% in those on placebo (Jastreboff et al. 2022). The weight reduction continued until the study was completed at Week 72.

2.3.3. Overall Benefit Risk Conclusion

Taking into account the measures taken to minimize risk to participants in this study, the potential risks identified in association with mirikizumab and placebo or mirikizumab concomitantly administered with tirzepatide are justified by the anticipated benefits that may be afforded to participants with UC with obesity, or overweight and in the presence of at least 1 weight-related comorbid condition.

3. Objectives, Endpoints, and Estimands

For definitions of the UC-specific endpoints, see Section 8.1.

Objectives	Endpoints
Primary	
To evaluate if concomitant administration of mirikizumab and tirzepatide at the MTD (up to 15 mg) is superior to mirikizumab and placebo in simultaneously achieving clinical remission and weight reduction	Proportion of participants who simultaneously achieve clinical remission and at least 10% weight reduction at Week 52
Key Secondary	
To evaluate the effect of treatment with concomitant administration of mirikizumab and tirzepatide at the MTD (up to 15 mg) versus mirikizumab and placebo as measured by gastroenterologic outcomes	- Proportion of participants who achieve the following at Week 52: - clinical remission - endoscopic response - histologic-endoscopic mucosal improvement plus absence of neutrophils
Additional Secondary	
To evaluate the effect of treatment with concomitant administration of mirikizumab and tirzepatide at the MTD (up to 15 mg) versus mirikizumab and placebo as measured by gastroenterologic and patient-reported outcomes	- Proportion of participants who achieve the following at Week 24: - clinical remission - endoscopic response - histologic-endoscopic mucosal improvement plus absence of neutrophils - Proportion of participants who achieve the following at Week 52: - maintenance of clinical remission - corticosteroid-free remission - Proportion of participants who achieve the following at Week 24 and Week 52: - clinical response - endoscopic remission - Bowel Movement Urgency improvement (captured with Urgency NRS) - histologic improvement - histologic remission; that is, histologic improvement with absence of neutrophils - Bowel Urgency remission - Proportion of participants who achieve symptomatic remission at Weeks 4, 8, 12, 24, and 52 - Proportion of participants who achieve symptomatic response at Weeks 4, 8, 12, 24, and 52 - Proportion of participants who simultaneously achieve clinical remission and at least 5% weight reduction at Week 24 and Week 52 - Proportion of participants who simultaneously achieve clinical remission and at least 10% weight reduction at Week 24

Objectives	Endpoints
	• Change from baseline in biomarkers: CRP and fecal calprotectin, at various time points (Weeks 4, 12, 24, and 52) • Change from baseline at Week 24 and Week 52 in ○ IBDQ ○ EQ-5D-5L ○ WPAI:UC ○ PGR-S • Proportion of participants at Week 52 who had ○ resolution of EIMs among those who had EIMs at baseline ○ UC-related emergency room visits ○ UC-related hospitalizations, and ○ UC-related surgeries
To evaluate the effect of treatment with concomitant administration of mirikizumab and tirzepatide at the MTD (up to 15 mg) versus mirikizumab and placebo as measured by metabolic outcomes	• Proportion of participants who achieve at least a 10% reduction in body weight at Week 24 and Week 52 • Percent change in body weight at Weeks 24, 40, and 52 • Change in BMI (kg/m²) at Weeks 24, 40, and 52 • Change in waist circumference at Weeks 8, 12, 24, 40, and 52 • Change at Week 24 and Week 52 in ○ HbA1c (%), and ○ glucose (non-fasting) • Change at Week 52 in ○ total cholesterol (mg/dL) ○ TG (mg/dL) ○ LDL (mg/dL), and ○ HDL (mg/dL) (non-fasting lipids) • Change in systolic and diastolic blood pressure (mmHg) at Week 52
To describe safety and tolerability of the study interventions	• TEAEs • discontinuation of study intervention due to AEs • SAEs, and • injection site reactions
Exploratory	
Exploratory objectives and endpoints will also evaluate all the key secondary and additional secondary measures listed above at additional time points as noted in the SoA.	
To demonstrate the effect of treatment with concomitant administration of mirikizumab and tirzepatide at the MTD (up to 15 mg) versus mirikizumab and placebo as measured by efficacy and inflammatory biomarkers	• Proportion of participants who achieve at least 5%, and at least 15% weight reduction at Week 24 and Week 52 • Change in percentage of visceral fat mass of total body mass from baseline to Week 52^a • Shift in waist-to-height ratio categories (<0.5, 0.5-0.59, ≥0.6) at Week 24 and Week 52 • Changes in exploratory biomarkers
To characterize the PK of mirikizumab and tirzepatide	Trough serum concentrations of mirikizumab and trough plasma concentrations of tirzepatide

Abbreviations: AE = adverse event; CRP = C-reactive protein; BMI = body mass index; DXA = dual-energy absorptiometry; EIM = extraintestinal manifestation; HbA1c = hemoglobin A1c; HDL = high-density lipoprotein; IBDQ = Inflammatory Bowel Disease Questionnaire; LDL = low-density lipoprotein; MTD = maximum tolerated dose; NRS = numeric rating scale; PGR-S = Patient's Global Rating of Severity; PK = pharmacokinetics; SAE = serious adverse event; SoA = schedule of activities; TEAE = treatment-emergent adverse event; TG = triglycerides; UC = ulcerative colitis; WPAI:UC = Work Productivity and Activity Impairment Questionnaire ulcerative colitis.

^a Assessed via DXA scan.

Primary estimand

The primary clinical question of interest is:

In participants with moderately to severely active UC who have obesity, or are overweight and in the presence of at least 1 weight-related comorbid condition, what is the treatment difference between mirikizumab and placebo versus mirikizumab concomitantly administered with tirzepatide in simultaneously achieving clinical remission and at least 10% weight reduction at Week 52 without early permanent discontinuation of study intervention or taking specified prohibited medication?

The estimand is described by the following attributes:

- Population - Participants with moderately to severely active UC and
 - obesity, or
 - overweight and in the presence of at least 1 weight-related comorbid condition:
 - hypertension
 - T2DM
 - dyslipidemia
 - obstructive sleep apnea, or
 - cardiovascular disease.
- Endpoint – Simultaneously achieve clinical remission and at least 10% weight reduction at Week 52.
- Treatment condition - Mirikizumab and placebo versus mirikizumab concomitantly administered with tirzepatide.
- ICE – Early permanent treatment discontinuation and specified prohibited medication use are addressed by the composite strategy. Participants will be considered as nonresponders after the first occurrence of these ICEs. Detailed definitions of specified prohibited medication will be provided in the SAP.
- Population-level summary - Difference in proportion of participants with response.

Rationale for estimand: The primary estimand strategy assumes the following:

- If a participant used specified prohibited medication, the participant was not receiving sufficient benefits from study intervention.
- If a participant permanently discontinued study intervention early, the participant experienced a burden of study intervention that outweighed its benefits.

Additional estimand(s) may be defined in the SAP.

Secondary estimand(s)

The clinical questions of interest and estimand strategies for the key secondary objectives are the same as for the primary objective, except for the endpoints to be evaluated. These endpoints are as follows:

- clinical remission at Week 52
- endoscopic response at Week 52, and
- histologic-endoscopic mucosal improvement plus absence of neutrophils at Week 52.

Additional estimand(s) for other secondary objectives will be provided in the SAP.

4. Study Design

4.1. Overall Design

Study AMCD is a Phase 3b, randomized, multicenter study to assess the efficacy and safety of mirikizumab and placebo compared with mirikizumab concomitantly administered with tirzepatide at the MTD (up to 15 mg) in adult participants with moderately to severely active UC and obesity, or overweight and in the presence of at least 1 weight-related comorbid condition.

The maximum duration of this study is approximately 61 weeks, with the following study periods:

- Study Period I, Screening: up to 5 weeks (≤ 35 days)
- Study Period II, Induction and Dose Escalation: 24 weeks
- Study Period III, Maintenance Treatment: 28 weeks, and
- Safety follow-up: 4 weeks after the last visit.

For randomization ratios and stratification factors, see Section 6.3.

4.2. Scientific Rationale for Study Design

Study rationale

Concomitant administration of mirikizumab and tirzepatide in patients with UC with obesity or overweight is expected to improve gastroenterologic and/or metabolic outcomes compared with treatment with mirikizumab and placebo.

Sample size

Approximately 350 participants who meet entry criteria will be randomly assigned at Visit 2 in a 1:1 ratio to either the

- mirikizumab and placebo treatment arm, or
- mirikizumab and tirzepatide at the MTD (up to 15 mg) concomitant administration treatment arm.

Refer to Section 9.5 for the rationale for the sample size.

Choice of primary endpoint

The multi-component primary endpoint at Week 52 includes clinical remission and at least 10% weight reduction.

Clinical remission

Clinical remission, as defined in Section 8.1, was chosen based on regulatory guidance from FDA (2016) and EMA (2018). Both recommend a composite endpoint for both symptomatic and endoscopic remission.

At least 10% weight reduction

A body weight reduction of at least 5% is considered the threshold for clinically meaningful effect on the basis of improvement in metabolic health. Greater weight reductions yield additional clinical benefits and may be required for improvement in certain weight-related

complications (Garvey et al. 2016) and are often more desired therapeutic goals in clinical practice.

Weight reduction of at least 10% was chosen based on SURMOUNT-1 data that all 3 doses of tirzepatide (5, 10, and 15 mg) had significantly higher percentages of participants achieving at least 10% weight reduction targets compared to placebo. Over 90% of the participants receiving 15 mg achieved at least 5% weight reduction at 72 weeks (Jastreboff et al. 2022).

Blinding

Mirikizumab will be open label, tirzepatide and placebo to match will be blinded.

Data monitoring committee (DMC)

See Section 10.1.5.1.

Rationale for collection of race and ethnicity data

In this study, collection of demographic information includes race and ethnicity. The scientific rationale is based on the need to assess variable response in safety and/or efficacy based on race or ethnicity. This question can be answered only if all the relevant data are collected.

Race and ethnicity will be collected using a standardized self-reporting process. Participants will be asked to identify their own race and ethnicity. The process will use a 2-step procedure:

1. Ethnicity: “Hispanic or Latino,” “Not Hispanic or Latino,” and “Not Reported.”
2. Race: Permitted categories include American Indian or Alaskan Native, Asian, Black or African American, Native Hawaiian or Other Pacific Islander, White, or Not Reported.

Justification for use of placebo

The AMCD study design ensures that all participants receive an appropriate standard of care for UC. All participants will receive mirikizumab, which is increasingly becoming a standard of care for UC. Furthermore, all participants will receive lifestyle coaching, including dietary and exercise advice, which will lead to weight reduction in participants who are obese or overweight in both arms of the study.

Participants in the control arm receiving a placebo are not receiving placebo alone, but rather receiving a placebo in addition to standard-of-care treatment. The inclusion of a placebo in the trial is scientifically and methodologically justified to isolate the effects of tirzepatide when combined with mirikizumab. This approach ensures that any observed differences between the 2 groups can be attributed solely to tirzepatide, thereby allowing a rigorous evaluation of its efficacy and safety when used concomitantly with mirikizumab in UC. The use of a placebo does not compromise safety in participants who are overweight and obese.

4.2.1. Patient Input into Design

Efforts to reduce participant burden include

- allowing telephone visits
- focused clinician and participant measures at key time points, and
- laboratory monitoring that is consistent with standard of care for participants with moderately to severely active UC.

Site identification included evaluation of diversity by geography and investigator sites.

4.3. Justification for Dose

4.3.1. Mirikizumab

Mirikizumab will be dosed per approved product labeling. Detailed information about the efficacy, safety, and dosing rationale for mirikizumab may be found in the IB and prescribing information.

The induction and maintenance dose levels and regimens in Study AMCD were based on analyses of PK, safety, and efficacy data from the following studies of mirikizumab in participants with moderately to severely active UC (D’Haens et al. 2023):

- Study I6T-MC-AMAC: Phase 2 study with a 12-week IV induction and ≤92-week SC maintenance
- Study AMAN (LUCENT-1): Phase 3 induction study with a 12-week IV treatment duration
 - Active dosing regimen: 300 mg mirikizumab IV at Weeks 0, 4, and 8, and
- Study AMBG (LUCENT-2): Phase 3 maintenance study with a 40-week SC treatment duration.
 - Active dosing regimen: 200 mg mirikizumab SC Q4W.

4.3.2. Tirzepatide

Tirzepatide at the MTD (up to 15 mg) administered SC QW per prescribing information will be evaluated in this study. Participants may be treated with a lower maintenance dose of 5 mg or 10 mg if they do not achieve full dose-escalation to 15 mg and/or do not tolerate 15 mg. Tirzepatide has been associated with GI adverse reactions, which include nausea, vomiting, and diarrhea. In general, these events were mild or moderate in severity and transient, with few severe episodes.

Dosing algorithms starting at a low dose of 2.5 mg QW accompanied by dose-escalation of 2.5-mg increments Q4W should permit time for development of tolerance to GI events and are predicted to minimize GI tolerability concerns.

Tirzepatide doses of 5 mg, 10 mg, and 15 mg as MTD are included in the prescribing information, and were selected based principally on the following criteria:

- each dose provides robust weight reduction relative to placebo
- the percent of participants achieving ≥10% weight reduction is higher with 15 mg than 10 mg or 5 mg, and
- safety and tolerability were supported by Phase 3 results in T2DM and obesity or overweight with weight-related comorbidity or in participants without diabetes mellitus.

The proposed tirzepatide maintenance dose of up to 15 mg using a dose-escalation regimen is expected to safely maximize the potential for weight reduction while minimizing GI tolerability concerns as has been achieved for approved GLP-1 receptor agonist drugs. The subsequent benefit of weight reduction should result in clinically meaningful improvements in participants with UC.

4.4. End of Study Definition

The end of the study is defined as the date of the last visit of the last participant in the study or last scheduled procedure shown in the SoA for the last participant in the trial.

A participant is considered to have completed the study if the participant has completed all periods of the study, including the last visit or the last scheduled procedure shown in the SoA.

5. Study Population

Participant eligibility for enrollment in the study is based on the criteria listed in this section. The inclusion and exclusion criteria used to determine eligibility should only apply at screening or other specified visits, and not continuously throughout the study.

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, is not permitted.

5.1. Inclusion Criteria

Participants are eligible to be included in the study only if all of the following criteria apply:

Age

1. Must be ≥ 18 to ≤ 70 years of age at the time of signing the ICF.

Type of participant and disease characteristics

2. Have had an established diagnosis of UC for ≥ 3 months before baseline (Visit 2/randomization), which includes endoscopic evidence of UC and a histopathology report that supports a diagnosis of UC.
3. Have moderately to severely active UC as defined by an mMS of 5 to 9 points and ES of 2 to 3 (confirmed by central review) within 21 days before baseline.

Notes: Participants with proctitis at baseline (Visit 2) may participate in the study if they meet all the eligibility criteria for the study, including endoscopic and rectal bleeding criteria for moderate-to-severe disease. The number of such participants will be capped at 15% of the total number of participants enrolled in the study.

4. Are up-to date on colorectal cancer surveillance per local guidelines in case of
 - a family history of colorectal cancer
 - personal history of increased colorectal cancer risk
 - age >45 years,
 - history of UC for greater than or equal to 8 years or
 - other known risk factors.
5. Participants with a history of UC for greater than or equal to 8 years who have had a surveillance colonoscopy completed within 1 year prior to baseline must have documented negative results for colorectal dysplasia and cancer.

Note: Acceptable methods of colonoscopic surveillance include

- standard-definition colonoscopy with 4 quadrant colon biopsies every 10 cm
- high-definition white-light colonoscopy with narrow-band imaging and targeted colon biopsies, and
- chromoendoscopy with methylene blue or indigo carmine dye spray and targeted colon biopsies.

Investigators may perform surveillance colonoscopy as part of the screening colonoscopy if they choose to do so.

6. Are willing to consult with a dietitian or nutritionist, or equivalent qualified designee or delegate* according to local standards, to receive lifestyle management counseling during

trial. (*Dietary counseling will be performed by a registered dietitian or nutritionist or equivalent qualified designee or delegate trained in collecting and assessing dietary intake and providing individualized dietary counseling according to the local standard).

Weight

7. Have

- [7a] obesity ($\text{BMI} \geq 30 \text{ kg/m}^2$), or
- [7b] overweight ($\text{BMI} \geq 27 \text{ kg/m}^2$ to $< 30 \text{ kg/m}^2$) and in the presence of at least 1 of these weight-related comorbid conditions:
 - hypertension
 - T2DM
 - dyslipidemia
 - obstructive sleep apnea, or
 - cardiovascular disease.

Sex assigned at birth and contraceptive/barrier requirements

8. Individuals AMAB and AFAB may participate in this trial, including both IOCBP and INOCBP.

Contraceptive use by participants should be consistent with local regulations regarding the methods of contraception for those participating in clinical studies. For the contraception requirements of this protocol, see Section 10.4.

Informed consent

9. Are capable of giving signed informed consent as described in Section 10.1, which includes compliance with the requirements and restrictions listed in the ICF and in this protocol.

Prior medication failure

10. Have an inadequate response to, loss of response to, or intolerance to at least 1 of the medications described in inclusion criterion [10a] or [10b]. Documentation of dose, frequency, route of administration, and duration of the prior failed treatment is required or could be reported by the participant.

[10a] **Conventional-failed participants:** Participants who

- have had an inadequate response to or a loss of response to, or are intolerant to at least 1 of the following medications:
 - oral corticosteroids
 - oral AZA or 6-MP, or
 - oral aminosaliclates (for example, mesalamine, sulfasalazine, olsalazine, and balsalazide)

AND

- have neither failed nor demonstrated an intolerance to an advanced therapy indicated for the treatment of UC, for example, a biologic or biosimilar medication, such as anti-TNF, anti-integrin antibody, or a JAK inhibitor.

[10b] **Advanced therapy-failed participants:** Participants who have an inadequate response to or a loss of response to, or are intolerant to advanced therapy for UC, defined as:

- a biologic or biosimilar medication such as anti-TNF antibodies
- anti-integrin antibodies
- JAK inhibitors such as tofacitinib or upadacitinib
- sphingosine 1-phosphate receptor 1 inhibitors such as etrasimod or ozanimod, or
- anti-IL-12p40 antibodies, for example, ustekinumab.

Investigators must be able to document an adequate clinical trial of the medication if available or could be provided by patient reported history.

Notes:

- Participants previously exposed to advanced therapy who do not meet criterion [10b] must still meet criterion [10a].
- Participants previously exposed to investigational therapies for the treatment of UC must still meet criterion [10a] OR [10b].
- Participants who meet both criteria [10a] and [10b] will be considered to be “advanced therapy-failed” for the purpose of this study.
- Oral methotrexate use will be allowed during the study; however, prior or current use of oral methotrexate is insufficient for inclusion into the study.

Ulcerative colitis medication dose stabilization

11. Are on a stable dose, defined as listed here, if using these medications:

Medication	Comment
• Oral 5-ASAs for UC	The prescribed dose has been stable for at least 2 weeks prior to the screening colonoscopy.
• Oral corticosteroids for UC: ○ prednisone ≤ 20 mg per day or equivalent ○ budesonide extended-release tablets up to 9 mg per day (budesonide MMX), or ○ beclomethasone dipropionate (gastro-resistant prolonged-release tablet) up to 5 mg per day	Participants on oral corticosteroids need to have been on corticosteroids for at least 4 weeks prior to screening but the prescribed dose must be stable for at least 2 weeks before the screening colonoscopy. See Section 6.9.5 for instructions on tapering of oral corticosteroid therapy.
• AZA • 6-MP, or • methotrexate	The prescribed dose has been stable for at least 8 weeks before the screening colonoscopy.

Diagnostic assessments

12. Have clinically acceptable central laboratory test results at screening (retesting is allowed for hematology and chemistry), as assessed by the investigator, including

[12a]	Hematology
	absolute neutrophil count $\geq 1.2 \times 10^9/L$ ($\geq 1.2 \times 10^3/\mu L$ or ≥ 1.2 GI/L)
	platelet count $\geq 100 \times 10^9/L$ ($\geq 100 \times 10^3/\mu L$ or ≥ 100 GI/L)

	hemoglobin ≥ 8.5 g/dL (≥ 85 g/L) for individuals AMAB and > 8.0 g/dL (> 80 g/L) for individuals AFAB
	lymphocyte count ≥ 750 cells/ μ L ($> 0.75 \times 10^3/\mu$ L or > 0.75 GI/L)
	total WBC $\geq 2.5 \times 10^9$ /L ($\geq 2.5 \times 10^3/\mu$ L or ≥ 2.5 GI/L)
[12b]	Chemistry
	serum creatinine ≤ 2 times ULN
	TBL ≤ 1.5 times ULN
	ALT and AST ≤ 2 times ULN
	ALP ≤ 1.5 times ULN
[12c]	estimated glomerular filtration rate (eGFR) ≥ 60 mL/min/1.73 m ² calculated by Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) Cystatin C Equation 2012 (NKF [WWW])
[12d]	calcitonin ≤ 35 ng/L (≤ 35 pg/mL)

Note: Participants with an established diagnosis of Gilbert's syndrome can be included with bilirubin levels ≤ 3 times ULN. The Gilbert's syndrome diagnosis requires source documentation showing unconjugated hyperbilirubinemia with no evidence of hemolysis.

Other inclusion criteria

13. Are reliable and willing to make themselves available for the duration of the study and are willing to follow study procedures as required, including, but not limited to
 - colonoscopies
 - self-injecting intervention or injection by a caregiver, when applicable
 - maintaining a diary
 - completing patient-reported questionnaires, and
 - storing and taking provided study interventions as directed.

5.2. Exclusion Criteria

Participants are excluded from the study if any of the following criteria apply:

Gastrointestinal medical conditions

14. Have a current diagnosis of
 - Crohn's disease
 - IBD unclassified (formerly known as indeterminate colitis), or
 - primary sclerosing cholangitis.
15. Have had or will need bowel resection or intestinal or intra-abdominal surgery, including but not limited to
 - extensive colonic surgery for UC, for example, subtotal colectomy, ostomy, or ileoanal pouch
 - any small bowel or colonic surgery within 6 months prior to baseline (Visit 2), or
 - any non-intestinal intra-abdominal surgery within 3 months of baseline.
16. Are likely to require surgery for the treatment of UC during the study.

Note: Participants who have had limited surgery for UC, for example, segmental colonic resection, may be allowed in the study, if this does not affect the assessment of efficacy. Discussion with the sponsor should occur prior to screening of such participants.

17. Have evidence of toxic megacolon or stricture or stenosis within the colon that cannot be traversed by a sigmoidoscope or colonoscope.

Adenoma, dysplasia, or gastrointestinal cancer

18. Have any history or current evidence of cancer of the GI tract.
19. Have had prior or current GI dysplasia, other than a completely removed low-grade dysplastic lesion, before the screening endoscopy.
20. Have any adenomatous polyp without dysplasia occurring in areas of the colon and rectum, that has not been removed.

Note: If the polyp(s) has been completely removed, this criterion would no longer apply.

21. Have a diagnosis of any of the following at screening or during the study:
 - cancer other than squamous cell or basal cell carcinoma of the skin (for details, see exclusion criterion [32])
 - any colonic adenomatous polyp with low-grade dysplasia not completely removed
 - any adenomatous polyp with high-grade dysplasia, or
 - evidence of dysplasia in nonpolypoid mucosa (indefinite for dysplasia, low-grade or high-grade dysplasia) in the GI tract.

Other medical conditions

22. Have a diagnosis of T1DM or insulin-treated T2DM.
23. Have any current or past history, or a diagnosis of proliferative diabetic retinopathy, diabetic maculopathy, or nonproliferative diabetic retinopathy that requires acute treatment.
24. Have a history of severe hypoglycemia and/or hypoglycemia unawareness within the 6 months prior to Visit 1 (screening).
25. Have a history of ketoacidosis or hyperosmolar state or coma in the last year.
26. Have a self-reported change in body weight greater than 5% (gain or reduction) within 3 months prior to Visit 1.
27. Have a prior or planned surgical treatment for obesity, such as gastric bypass (bariatric) surgery or restrictive bariatric surgery (for example, Lap-Band®), excluding liposuction or abdominoplasty if performed more than 1 year prior to Visit 1.
28. Have a known clinically significant gastric emptying abnormality (for example, severe gastroparesis or gastric outlet obstruction).
29. Have a family or personal history of medullary thyroid cancer (MTC) or multiple endocrine neoplasia (MEN) syndrome type 2.
30. Have a current or past history of chronic or acute pancreatitis.
31. Have acute or chronic hepatitis, or signs and symptoms of any other liver disease other than nonalcoholic fatty liver disease.
32. Have a diagnosis or history of malignant disease within 5 years prior to baseline (Visit 2).

Exceptions:

- basal cell or squamous epithelial carcinomas of the skin that have been resected with no evidence of metastatic disease for 3 years, and

- cervical carcinoma in situ that has been resected with no evidence of recurrence for at least 5 years prior to baseline (Visit 2).
33. Have a significant uncontrolled illness that in the opinion of the investigator may compromise the participant's safety or interfere with interpretation of data; such illnesses or conditions include, but are not limited to
- cerebrocardiovascular; for example, myocardial infarction, unstable angina, unstable arterial hypertension, moderate-to-severe (New York Heart Association class III/IV) heart failure, or cerebrovascular accident
 - respiratory
 - hepatic
 - renal
 - GI (excluding UC)
 - endocrine
 - hematologic or lymphoproliferative
 - neurologic
 - serious infection
 - awaiting or having a history of organ transplant (except keratoplasty), or
 - evidence of active vasculitis or uveitis.
34. Have a history of unstable major depressive disorder (MDD), suicidal ideation, or other severe psychiatric disorder (for example, schizophrenia, bipolar disorder, or other serious mood or anxiety disorder) within the last 2 years.
- Note:** Participants with MDD or generalized anxiety disorder whose disease state is considered stable for the past 2 years and expected to remain stable throughout the course of the study, in the opinion of the investigator, may be considered for inclusion.
35. Have a history of chronic alcohol abuse, IV drug abuse, or other illicit drug abuse within 1 year before screening colonoscopy.
36. Are, in the judgment of the investigator, actively suicidal and therefore deemed to be at significant risk for suicide.
37. Have answered "yes" to either Question 4 or Question 5 on the "Suicidal Ideation" portion of the C-SSRS **and** the ideation occurred within the past month.
- OR**
- Have answered "yes" to any of the suicide-related behaviors on the "suicidal behavior" portion of the C-SSRS **and** the behavior occurred within the past month.
38. Have had extra-abdominal surgery and have not recovered fully following surgery, including complete wound healing, before screening.
39. Have
- a known or suspected allergy or hypersensitivity to mirikizumab or tirzepatide, any of their excipients, or any other GLP-1 receptor agonists or excipients, that would pose an unacceptable risk to the participant in this study
 - clinically significant multiple or severe drug allergies, or

- a history of severe post-treatment hypersensitivity reactions, including, but not limited to, erythema multiforme major, linear IgA dermatosis, toxic epidermal necrolysis, or exfoliative dermatitis.

Infections and infectious disease

40. Have a current or recent acute, active infection. For at least 30 days before screening (Visit 1) and up to the randomization visit (Visit 2), participants must have no symptoms or signs of confirmed or suspected infection, and must have completed any appropriate anti-infective treatment.

Note: Participants who have an upper respiratory infection, a vaginal candida infection, or an oral candida infection and who are being treated only symptomatically and not requiring systemic anti-infectives may be considered for enrollment if other study eligibility criteria are met. Enrollment of participants with other uncomplicated local infections should be discussed with the sponsor's designated medical monitor.

41. Have had any of the following types of infection, including chronic or recurring extraintestinal infections, within 3 months prior to the screening visit (Visit 1) or develops any of these infections before the randomization visit (Visit 2):

- Serious (requiring hospitalization, or IV or equivalent oral antibiotic treatment, or both)
- Opportunistic (as defined in Winthrop et al. 2015)

Note: Herpes zoster is considered active and ongoing until all vesicles are dry and crusted over.

- Chronic (duration of symptoms, signs, and/or treatment of 6 weeks or longer)
- Recurring (including, but not limited to, herpes simplex, herpes zoster, recurring cellulitis, and chronic osteomyelitis).

Note: Participants with only recurrent mild and uncomplicated orolabial herpes, or genital herpes, or both may be discussed with the sponsor's designated medical monitor and considered for enrollment if other study eligibility criteria are met.

42. Have a nonserious extraintestinal infection that is not adequately treated prior to screening and randomization.

Note: A participant who develops an extraintestinal infection during the screening period must be adequately treated before receiving study intervention.

43. Have active TB or untreated LTBI.

[43a] Have active TB (Section 8.2.10).

[43b] Have had LTBI that has not been treated with a complete course of appropriate therapy as defined by the WHO and the United States CDC, unless such treatment is underway (Section 8.2.10).

Note: Participants diagnosed with LTBI at screening may be allowed to re-screen for the study, provided they fulfil the criteria described in Section 8.2.10. Participants who have a documented history of completing an appropriate TB prophylaxis regimen with no history of risk of re-exposure since their treatments were completed and no evidence of active TB are eligible to participate in the study.

44. Have HIV infection.

45. Have a current infection with HBV; that is, positive for HBsAg and/or PCR positive for HBV DNA (Section 8.2.11).
46. Have a current infection with HCV; that is, positive for HCV RNA (Section 8.2.12).
47. Have a known active COVID-19 infection.

If a participant has signs/symptoms suggestive of COVID-19, they should undergo molecular (for example, PCR) testing to rule out SARS-CoV-2 infection. Participants who do not meet COVID-19 eligibility criteria must be screen failed and may only rescreen after they meet the following COVID-19 criteria:

- Symptomatic participants: At least 14 days have passed since recovery, defined as resolution of fever without use of antipyretics and improvement in symptoms
 - Asymptomatic participants: At least 14 days have passed since the first positive molecular (for example, PCR) test result.
48. Have *C. difficile* or other intestinal infection within 30 days of screening colonoscopy or test positive at screening for *C. difficile* or for other intestinal pathogens.
 49. Have a confirmed diagnosis of cytomegalovirus-associated colitis.

Note: Participants may be eligible if they have had adequate treatment of this condition and have resolution of symptoms at least 3 months prior to screening colonoscopy, provided all other study eligibility criteria are met.

Prior or concomitant therapy

50. Have received any of the following for the treatment of UC within the timeframes listed in this table:

Treatment of UC	Within This Time Frame	Comment
Rectally administered topical therapies, for example: - corticosteroid or 5-ASA enemas - corticosteroid or 5-ASA suppositories - corticosteroid foam applications	2 weeks prior to the screening colonoscopy	
IV corticosteroids	2 weeks prior to the screening colonoscopy and throughout screening	Participants requiring systemic corticosteroids for non-UC conditions, except corticosteroids to treat adrenal insufficiency, are excluded.
Immunomodulatory medications, including - oral cyclosporine - IV cyclosporine - mycophenolate mofetil - thalidomide - JAK inhibitors, for example, tofacitinib, baricitinib, upadacitinib, filgotinib - tacrolimus; see exception in the Comment column	2 weeks prior to the screening colonoscopy	AZA, 6-MP, and methotrexate are allowed. Topical use tacrolimus for non-UC conditions is allowed and not subject to a washout period before the screening colonoscopy. Other immunomodulatory medications should be discussed with the sponsor prior to screening.

Treatment of UC	Within This Time Frame	Comment
Budesonide standard formulation	2 weeks prior to the screening colonoscopy	Oral budesonide extended-release tablet formulation (budesonide MMX) is allowed.
Leukocyte apheresis (leukapheresis, for example, Adacolumn®)	3 weeks prior to screening colonoscopy	
Any investigational nonbiologic therapy	4 weeks prior to the screening colonoscopy or within 5 half-lives prior to the screening colonoscopy, whichever is longer	
Infliximab or infliximab biosimilars	7 weeks prior to screening colonoscopy	This exclusion does not apply if the participant has a result from a validated drug-level test demonstrating undetectable drug level in the period between last dose and the screening visit.
Adalimumab or adalimumab biosimilars	10 weeks prior to screening colonoscopy	This exclusion does not apply if the participant has a result from a validated drug-level test demonstrating undetectable drug level in the period between last dose and the screening visit.
Golimumab or golimumab biosimilars	10 weeks prior to screening colonoscopy	This exclusion does not apply if the participant has a result from a validated drug-level test demonstrating undetectable drug level in the period between last dose and the screening visit.
Any investigational biologic therapy, not otherwise mentioned	8 weeks prior to the screening colonoscopy or 5 half-lives prior to the screening colonoscopy, whichever is longer	
Interferon therapy	8 weeks prior to screening colonoscopy	
S1P receptor agonist, for example, ozanimod or etrasimod	8 weeks prior to screening colonoscopy	
Anti-integrin antibodies, for example • vedolizumab • approved biosimilars	12 weeks prior to the screening colonoscopy	This exclusion does not apply if the participant has a result from a validated drug-level test demonstrating undetectable drug level in the period between last dose and the screening visit.
Agents that deplete B or T cells, for example • rituximab • alemtuzumab • visilizumab • approved biosimilars	12 months prior to the screening colonoscopy	Participants are excluded if there is evidence of persistent targeted lymphocyte depletion at the time of screening colonoscopy.
Anti-IL-23p19 antibodies, for example • mirikizumab • risankizumab	At any time prior to screening	Only individuals who have never received these medications can be enrolled

Treatment of UC	Within This Time Frame	Comment
• brazikumab • guselkumab • tildrakizumab • approved biosimilars		
Anti-IL-12p40 antibodies, for example, ustekinumab or ustekinumab biosimilars	16 weeks prior to screening colonoscopy	This exclusion does not apply if the participant has a result from a validated drug-level test demonstrating undetectable drug level in the period between last dose and the screening visit

51. Have failed 4 or more classes of advanced therapies approved for the treatment of UC.

52. Have received any of the medications within the required washout period described in this table.

Prohibited Medication	Required Washout Period
<i>Antihyperglycemic medications</i>	4 weeks or 5 half-lives, whichever is longer, prior to Visit 2
• sulfonylurea medications • amylin mimetics • DPP-4 inhibitors • glinides, and/or • insulin	Note: As stated in Section 6.9.2, antihyperglycemic medications for the management of T2DM are permitted; for example, metformin, thiazolidinediones, and SGLT2 inhibitors.
<i>Medications that may promote weight loss</i>	3 months prior to Visit 2
• GLP-1 receptor agonists • GIP/GLP-1 receptor agonists • orlistat • sibutramine • phenylpropanolamine • mazindol • phentermine • lorcaserin • phendimetrazine • phentermine/topiramate combination • naltrexone/bupropion, or • oral superabsorbent hydrogel	
<i>Medications that may cause significant weight gain</i>	3 months prior to Visit 2
• tricyclic antidepressants • atypical antipsychotics, and • mood stabilizers. Examples: • imipramine • mirtazapine • paroxetine • phenelzine • chlorpromazine • thioridazine • clozapine • olanzapine • valproic acid (and its derivatives), or • lithium	Exceptions: • Selective serotonin reuptake inhibitors other than paroxetine are permitted. Amitriptyline is permitted, as a medication commonly prescribed in IBD.
<i>Vaccinations</i>	4 weeks before Visit 2

Prohibited Medication	Required Washout Period
Live attenuated vaccination(s), including BCG	Notes: - Participants should be excluded if they intend to receive any vaccination during the study or within 3 months after receiving the last dose of study intervention. - The following are not considered live vaccines: RNA vaccines, vaccines with inactive viral elements, and/or nonreplicating viral vector vaccines.

53. Have a previous failure or intolerance to GLP-1 receptor agonists or tirzepatide.

Prior or concurrent clinical study experience

54. Are currently enrolled in any other clinical study involving an investigational product or any other type of medical research judged not to be scientifically or medically compatible with this study.
55. Have previously completed or discontinued from this study or any other study investigating mirikizumab or tirzepatide. This criterion does not apply to patients undergoing rescreening procedures.

Other exclusions

56. Are Lilly employees or employees of third-party organizations involved with the study.
57. Are investigator site personnel directly affiliated with this study and/or their immediate families. Immediate family is defined as a spouse, parent, child, or sibling, whether biological or legally adopted.
58. Are unwilling or unable to comply with the use of a data collection device to directly record data from the participant daily for the duration of this study or are unable to complete other study procedures.
59. Are unsuitable for inclusion in the study in the opinion of the investigator or sponsor for any reason that may compromise the participant's safety or confound data interpretation.
60. Are pregnant, or intend to become pregnant or to breastfeed during the study or within 10 weeks after receiving the last dose of the study intervention.
61. Have abnormal 12-lead electrocardiogram (ECG) that, in the opinion of the investigator or sponsor, increases the risks associated with participating in the study.

5.3. Lifestyle Considerations

5.3.1. Health Promotion and Lifestyle Program

Per the SoA (Section 1.3), participants will consult a dietitian or nutritionist, or equivalent qualified designee or delegate* according to local standards, to receive lifestyle management counseling. (*Dietary counseling will be performed by a registered dietitian or nutritionist or equivalent qualified designee or delegate trained in collecting and assessing dietary intake and providing individualized dietary counseling according to the local standard).

Healthy dietary patterns and physical activity goals established during the lifestyle consultation and the importance of adherence to the lifestyle component of the study will be reinforced at each visit by site staff. Ensuring participants have optimal nutritional intake is highly recommended. In addition to Section 5.3.2, nutrition support and lifestyle resources are available in the investigator resources.

To encourage adherence, it is recommended that the participants complete their 3-day nutrition and physical activity log before each counseling visit. During each visit, the participant's lifestyle behaviors are reviewed and advice to maximize adherence is provided as needed.

5.3.2. Meals and Dietary Considerations

Individualized counseling for a healthy dietary pattern

At Visit 2 and subsequent visits, study participants will receive individualized counseling regarding a healthy dietary pattern with the goal of optimizing nutritional status while on pharmacotherapy. The counseling will be performed by a dietitian or nutritionist, or equivalent qualified designee, according to the local standard.

Counseling considerations

The focus of the counseling should facilitate healthier dietary choices and promote mindful eating.

It should include instruction on customized nutrient-dense food and beverage choices to reflect personal preferences, cultural traditions, and budgetary considerations.

Consider the following principles in counseling sessions:

- Focus on high quality, nutrient rich, minimally processed foods. Limit foods high in solid fats, added sugar, and salt.
- Include fiber rich foods.
- For each meal (on an 8-10 inch [20-25 cm] plate), consider including approximately ½ plate of fruits/vegetables, ¼ plate of whole grains, and ¼ plate of protein. Incorporate 1 to 2 servings of healthy, unsaturated fats on most days (for example, nuts, avocado, olive oil, and salmon).
- Encourage participants to optimize daily protein intake to reach calculated daily protein goal based on ideal body weight and tailored to individual health and nutrition requirements. General daily recommended protein requirements are 60 to 120 g/day for women and 75 to 120 g/day for men. This should be individualized based on each participant's unique needs such as medical conditions, physical activity levels, etc.
- Drink enough water and/or unsweetened or decaffeinated fluids to stay well hydrated. Fluid intake of at least 2 L/day may be necessary for optimal hydration. Modifications to daily fluid amount should be individualized based on each participant's unique needs such as medical conditions, hydration status, physical activity levels, etc.
- Follow hunger cues. If not feeling hungry, prioritize protein intake first, plan to eat smaller and more frequent (every 3-4 hours) meals/snacks and avoid skipping meals to meet daily nutrition requirements as calculated based on individual needs.

5.3.3. Monitoring Nutritional Needs

Caloric restriction is a standard component of weight-reduction interventions, and there is potential that a small portion of participants will have significantly reduced caloric intake.

Medical staff should monitor the participant's nutritional and hydration status and attention should be given to ensure participants are eating high-quality foods that include adequate micronutrients and macronutrients. These recommendations should be personalized based on participants' comprehensive medical conditions at the investigator's discretion.

Consider additional assessment as described in Sections [8.2.13](#) and [10.2.2](#) if

- the PI suspects the participant may have malnutrition, for example, caloric intake is significantly reduced to below 800 to 1200 kcal
- a participant's BMI falls below 22 kg/m², or
- in participants with T2DM, weight loss is >20% or >4 kg/month.

5.3.4. Physical Activity

At Visit 3 and all subsequent visits, participants will be advised regarding achieving a healthy physical activity level. At least 150 minutes per week of moderate intensity aerobic exercise, and muscle strengthening (resistance) exercise at least 3 times per week is recommended.

Counseling in lifestyle modification should be completed according to the SoA and must be documented in the participant's medical record.

Activity before blood collections

Participants will abstain from strenuous exercise for 24 hours before each blood collection for clinical laboratory tests.

5.3.5. Blood Donation

Study participants should be instructed not to donate blood or blood products during the study.

5.4. Screen Failures

A screen failure occurs when a participant who consents to participate in the clinical study is not subsequently assigned to study intervention. A minimal set of screen failure information is required to ensure transparent reporting of screen failure participants to meet the CONSORT publishing requirements and to respond to queries from regulatory authorities. Minimal information includes demography, screen failure details, and any SAE.

Individuals who do not meet the criteria for participation in this study (screen failure) may be rescreened once or twice (see Section [5.4.2](#)).

5.4.1. Allowed Retesting of Screening Investigations

These screening investigations can be retested 1 time at the discretion of the investigator:

- **Screening hematology and chemistry blood tests:** where results are outside the acceptable range for inclusion in the study but may be within the acceptable range for inclusion on retesting due to test-retest variability

- **Stool testing:** if there is a technical difficulty in performing or reporting the *C. difficile* or stool culture assays (see Section 5.4.1.1)
- **Retesting or confirmatory testing with an IGRA:** for example, QuantiFERON®-TB Gold or T-SPOT® assay (see Section 8.2.10)
- **Colonoscopy:** where the endoscopist is unable to adequately visualize the mucosa, for example, due to poor bowel preparation or technical issues with equipment, or where the central reader is unable to determine the centrally read ES, for example, due to failure of the recording equipment.

The retesting described above is not considered rescreening.

Retesting of other screening investigations should be discussed with the sponsor's designated medical monitor prior to retesting.

5.4.1.1. Screening Stool Testing

Stool culture

Participants must have a negative stool culture from which no enteric pathogens are isolated to participate in the study.

Retesting is allowed within the screening period if there is a technical difficulty in performing or reporting the stool culture assay, as stated in Section 5.4.1.

Participants who have a "positive" stool culture result can be rescreened once, provided that the following conditions have been met:

- the participant has been adequately treated, and
 - if antibiotics were prescribed, participant has been off antibiotics for 30 days, or
 - if antibiotics were not prescribed, 30 days or more has elapsed since resolution of acute symptoms and signs associated with the underlying intestinal infection.

To confirm that the reason for screen failure (positive stool culture) has resolved, it is recommended that the investigator performs a local stool culture test before rescreening. Upon rescreen (Visit 1), a negative stool culture result from the central laboratory is required to meet this eligibility criterion.

Additional local stool culture/testing is allowed at the investigator's discretion or based on country requirements.

C. difficile toxin testing

This assay tests for the presence of *C. difficile* toxin protein, followed by a confirmatory test for *C. difficile* toxin gene expression in the stool sample.

Participants should test negative for *C. difficile* to receive mirikizumab. Retesting is allowed within the same screening period if there is a technical difficulty in performing or reporting the *C. difficile* assays, or if in the judgment of the investigator, the participant's symptoms or signs are not consistent with *C. difficile* infection.

Participants who test positive at screening for *C. difficile* can be rescreened once for the study, once they have been adequately treated and off antibiotics for 30 days.

Participants who have been adequately treated for *C. difficile* with fecal microbial transplantation or IV immunoglobulin therapy can be rescreened once for the study 30 days after completing their therapy.

To confirm that the reason for screen failure (positive *C. difficile*) has resolved, it is recommended that the investigator performs a local *C. difficile* test before rescreening. Upon rescreen (Visit 1), a negative *C. difficile* result from the central laboratory is required to meet this eligibility criterion.

5.4.2. Rescreening of Individuals Who Fail Screening

Informed consent for rescreening

Each time an allowed rescreening is performed, the potential participant must sign a new ICF and will receive a new unique identifier (Sections 10.1.3 and 10.1.4).

Procedures not required to be repeated during rescreening

If a participant's protocol-required screening chest x-ray and TB tests were performed within 90 days before the date of signing the ICF for rescreening, these procedures are not required to be repeated. However, these procedures can be repeated at the discretion of the investigator.

Rescreening after failure to meet study entry criteria

This table describes conditions under which a potential study participant may be rescreened. If a participant's reason for screen failure is not listed in the table, the participant may not be rescreened.

If a participant is not eligible because of this entry criterion...	then the participant may be rescreened when...	Frequency of allowed rescreening for the specified criterion
2, 3, 5, 10-13, 15, 16, 32, 33, 35, 38, 41, 42, 48, 50, 52 (regarding vaccinations), 54, 58-60	the reason for the screen failure has resolved. For <i>C. difficile</i> -related rescreening, see Section 5.4.1.1.	up to 2 times, for a maximum total of 3 screenings
43b	participant is completely treated for LTBI.	up to 2 times, for a maximum total of 3 screenings
49	the reason for the screen failure has resolved.	1 time

The interval between the allowed rescreening should be at least 4 weeks unless a shorter interval has been agreed with the sponsor's designated medical monitor. The investigator should confirm that the participant has a negative *C. difficile* stool toxin or stool culture or stool ova parasite (as applicable) before performing additional rescreening investigations.

5.5. Criteria for Temporarily Delaying Enrollment of a Participant

This section is not applicable to this study.

6. Study Intervention(s) and Concomitant Therapy

Study intervention is defined as any medicinal product(s) or medical device(s) intended to be administered to or used by a study participant according to the study protocol.

6.1. Study Intervention(s) Administered

This table lists the interventions used in this clinical study. Both of the study interventions will be administered according to their label.

For simplicity, in the context of dosing, the term “tirzepatide” refers to both “tirzepatide” and “matching placebo” in this protocol.

Intervention Name	Mirikizumab		Tirzepatide	Placebo to match tirzepatide
Study Treatment	Induction	Maintenance		
Dosage Level(s)	300 mg at Weeks 0, 4, and 8	200 mg Q4W starting at Week 12, given as 2 consecutive injections of 100 mg each	All doses QW: • 2.5 mg • 5 mg • 7.5 mg • 10 mg • 12.5 mg • 15 mg	QW
Route of Administration	IV infusion	SC (prefilled syringe)		
Administration	Onsite: • only by authorized study personnel • maximum rate of 10 mg/min by IV infusion over no less than 30 min	Initial administration: • Onsite, either ○ as self-administration by participant or caregiver, after appropriate training and under supervision of study personnel, or ○ by study personnel. Subsequent administration: • offsite (for example, at participant’s home) by the participant or their caretaker after completion of training, or • onsite, by authorized study personnel (per participant’s preference).		
Authorized as defined by EU Clinical Trial Regulation	Authorized and not used according to EU authorization ^a		Authorized and not used according to EU authorization ^b	Not authorized in EU

Abbreviations: EU = European Union; IV = intravenous; Q4W = every 4 weeks; QW = every week; SC = subcutaneous; T2DM = type 2 diabetes mellitus; UC = ulcerative colitis.

^a Mirikizumab is authorized in EU for the treatment of UC. In this study, clinical trial material will be used for mirikizumab.

^b Tirzepatide is authorized in EU for the treatment of T2DM and for weight management, including weight loss and weight maintenance, in patients with obesity or overweight with at least on one weight-related comorbid condition. In this study, clinical trial material will be used for tirzepatide.

Packaging and labeling

Study interventions will be supplied by the sponsor or its designee in accordance with current Good Manufacturing Practice. Study interventions will be labeled as appropriate for country requirements.

6.1.1. Participant Monitoring and Observation After Study Intervention**Initial monitoring and observation**

Participants will be monitored for 1 hour after IV administration of mirikizumab. Longer monitoring is allowed when required by investigator practice or local standard of care. Sites must have resuscitation equipment, emergency medications, and appropriately trained personnel available during the infusion and post-infusion monitoring period.

In addition, participants should be observed after tirzepatide dosing at randomization (Visit 2) and the initial administration of mirikizumab SC at the investigator's discretion for any AEs.

Subsequent monitoring and observation

Subsequent monitoring and observation will be performed according to investigator practice or local standard of care.

6.1.2. Self-Administration Training and Dosing Diary**Training for tirzepatide self-administration**

Site personnel will train the participant and/or caregiver to administer tirzepatide using a prefilled syringe at Visit 2. It is strongly recommended that the trained participant or caregiver administers the first dose of tirzepatide at Visit 2 under supervision of the site personnel. Thereafter, study intervention should be administered by the participant or caregiver. If additional training is necessary, an injection may be administered by the participant and/or caregiver under the supervision of the site personnel. If the participant or their caregiver do not feel comfortable administering the injection, it can be done by the study personnel. The site personnel must document in the source documents that injection training has been provided to the participant or caregiver verbally and in written form at the first dispensing visit. Refresher training will be provided as needed throughout the study. Injection training is the responsibility of the investigator or other qualified staff.

Training for mirikizumab self-administration

In this protocol, self-administration is defined as the SC injection of 200 mg mirikizumab, where either the participant injects themselves or a caregiver administers the injection to the participant.

Site personnel will train the participant and/or caregiver to administer mirikizumab SC using prefilled syringes at Visit 5.

For considerations regarding observation of the first self-administration and documentation of mirikizumab injection training, see subsection "Training for tirzepatide self-administration."

Dosing diary

Dosing diary will be dispensed to each participant as indicated in the SoA, for recording pertinent data about each injection.

6.1.3. Timing and Location of Doses**Timing of concomitant administration of mirikizumab and tirzepatide or mirikizumab and placebo**

When both study interventions are due on the same date, it is recommended that they are administered within a reasonable time frame from each other (that is, around the same time of day, such as both in the morning) and in a different location (see “Dosing locations” subsection). During mirikizumab induction dosing, it is recommended that the administration of tirzepatide or placebo occur at least 30 minutes after the end of the mirikizumab infusion. Refer to the appropriate directions for use provided by the sponsor for the study intervention.

Timing of doses for tirzepatide

There are no restrictions on the time of day each weekly dose of tirzepatide is given, but it is advisable to administer the tirzepatide injections on the same day and time each week, with or without meals.

If a dose of tirzepatide is missed, the participant must take it as soon as possible within 4 days (96 hours) after the missed dose. If more than 4 days have passed, that dose must be skipped and the next dose must be taken on the regular scheduled day. In each case, participants can then resume their regular QW dosing schedule.

Timing of doses for mirikizumab

There are no restrictions on the time of day each Q4W dose of mirikizumab is given. If a mirikizumab SC dose is missed, administer it as soon as possible and resume dosing Q4W thereafter. If the participant has a scheduled visit >14 days from the last administered dose, continue dosing per the original dosing schedule per the SoA.

Dosing locations

For both tirzepatide and SC administration of mirikizumab, injection sites include

- the abdomen (stomach area)
- thigh (upper leg), and
- back of the upper arm.

Participants will self-inject the SC study intervention in the abdomen or thigh.

If the injection is given by somebody else, they will administer the injection either in the upper arm, abdomen, or thigh.

When mirikizumab and tirzepatide are administered concomitantly, use an anatomically distinct site on the opposite side of the body for injecting each study intervention; for example, if mirikizumab is injected in the right side of the abdomen, tirzepatide should be injected in the left thigh or the back of the left upper arm. Do not inject in areas where the skin is tender, bruised, erythematous, or indurated.

A new prefilled syringe will be used for each SC injection.

6.1.4. Dose Escalation

6.1.4.1. Mirikizumab

No dose escalations are applicable.

6.1.4.2. Tirzepatide

Tirzepatide treatment will follow a fixed dose-escalation as follows:

- the starting dose will be 2.5 mg QW. After 4 weeks, the dose is increased to 5 mg, and
- dose is increased by 2.5 mg Q4W until the 15-mg dose is reached or to the highest maintenance dose tolerated by the participant (5 mg, 10 mg, or 15 mg).

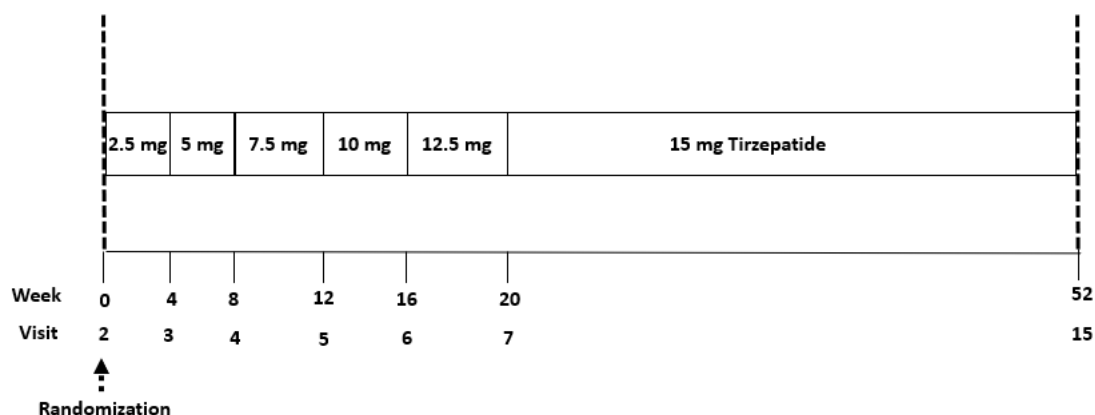
This table describes the dose-escalation, which allows 20 weeks to escalate to 15 mg MTD, if tolerated.

Visit (Week)	Weekly Tirzepatide
Visit 2 (Week 0) up to Visit 3 (Week 4)	2.5 mg QW
Visit 3 (Week 4) up to Visit 4 (Week 8)	5 mg QW
Visit 4 (Week 8) up to Visit 5 (Week 12)	7.5 mg QW
Visit 5 (Week 12) up to Visit 6 (Week 16)	10 mg QW
Visit 6 (Week 16) up to Visit 7 (Week 20)	12.5 mg QW
Visit 7 (Week 20) up to Visit 15 (Week 52)	15 mg QW

The investigator should try to escalate participants to the 15-mg dose during the initial dose-escalation. The below figure describes the dose-escalation over 20 weeks and to up to 52 weeks.

The allowed maintenance doses are 5 mg, 10 mg, and 15 mg. Intermediate doses of 2.5 mg, 7.5 mg, and 12.5 mg are used only during dose escalations.

Tirzepatide dose escalation



Intolerance to dose escalation

If a participant does not tolerate the dose-escalation, refer to Section 6.6.2 for additional instructions.

6.2. Preparation, Handling, Storage, and Accountability

Sponsor or its designee will supply study intervention to the investigator for dispensing at Visit 2.

The investigator or designee must confirm appropriate storage conditions have been maintained during transit for all study intervention received and any discrepancies are reported and resolved before use of the study intervention.

Only participants enrolled in the study may receive study intervention. Only authorized study personnel may supply, prepare, or administer study intervention.

At the study site, all study intervention must be stored in a secure, environmentally controlled, and monitored (manual or automated) area in accordance with the labeled storage conditions with access limited to the investigator and authorized study personnel. For off-site (at-home) administration, sponsor will provide storage instructions.

The investigator or authorized study personnel are responsible for study intervention accountability, reconciliation, and record maintenance, that is, receipt, reconciliation, and final disposition records.

Further guidance and information for the final disposition of unused study interventions are provided in the Pharmacy Manual.

6.3. Assignment to Study Intervention**Method of treatment assignment**

All eligible participants will be centrally assigned to randomized study intervention using the IWRS. Before the study is initiated, the login information and directions for the IWRS will be provided to each site.

Randomization ratios and stratification factors

Participants will be randomly assigned in a 1:1 ratio to either the

- mirikizumab and placebo treatment arm, or
- mirikizumab and tirzepatide concomitant administration treatment arm.

The randomization will be stratified by

- sex (AFAB vs. AMAB)
Note: The randomly assigned study population will include up to approximately 70% of either individuals AFAB or AMAB, whoever enrolls faster. The remaining sex will make up the remainder of the participant population.
- BMI ≥ 27 kg/m² to <30 kg/m² versus ≥ 30 kg/m²
- previous advanced therapy failure status (yes/no), and
- baseline disease activity (mMS: [5-6] or [7-9]).

For participants with BMI ≥ 27 kg/m² to < 30 kg/m², the presence of at least 1 weight-related comorbid condition is required to be randomized.

6.4. Blinding

Mirikizumab treatment will be open label.

Tirzepatide treatment, and matching placebo, will be blinded for both study participants and study personnel.

Emergency unblinding

Emergency unblinding for AEs may be performed through the IWRS. This option may be used ONLY if the participant's well-being requires knowledge of the participant's treatment assignment. All actions resulting in an unblinding event are recorded and reported by the IWRS.

If, because of emergency unblinding, a participant's tirzepatide assignment is unblinded to the participant or to blinded site personnel who are performing study assessments, including the investigator, the participant must be discontinued from tirzepatide (Section 7.1.1).

In case of an emergency, the investigator has the sole responsibility for determining if unblinding of a participants' intervention assignment is warranted. Participant safety must always be the first consideration in making such a determination. If the investigator decides that unblinding is warranted, it is the responsibility of the investigator to promptly document the decision and rationale and notify the sponsor as soon as possible.

6.5. Study Intervention Compliance

When the participant and/or caregiver administers study intervention(s) at home, compliance with study interventions will be assessed at visits indicated in the SoA (Section 1.3). Compliance will be assessed by

- direct questioning
- counting of unused study drug and/or empty cartons returned
- review of the dosing diary, and
- documented in the source documents.

Participants will be instructed to return any unused prefilled syringes at the times specified in the SoA.

Treatment compliance for each visit interval is defined as taking at least 75% of the required doses of study interventions. Similarly, a participant will be considered significantly noncompliant if he or she is judged by the investigator to have intentionally or repeatedly taken more than the prescribed amount of medication (more than 125%).

In addition to the assessment of a participant's compliance with the study intervention administration, other aspects of compliance with the study treatments will be assessed at each visit based on

- the participant's adherence to the visit schedule
- completion of the dosing diary, and
- any other parameters the investigator considers necessary.

Participants considered to be poorly compliant with their medication and/or the study procedures will receive additional training and instruction, as required, and will be reminded of the importance of complying with the protocol.

A record of the number of mirikizumab, placebo, and tirzepatide prefilled syringes dispensed to and taken by each participant must be maintained and reconciled with study intervention and compliance records. The date and time of each dose administered, location of administration, and reason if dose was not administered will be recorded in the dosing diary.

6.6. Dose Modification

6.6.1. Mirikizumab

Dose modification is not applicable to mirikizumab in this study.

6.6.2. Tirzepatide

For dose escalation information, refer to Section 6.1.4.2.

During the dose-escalation period, the investigator should make every effort to escalate participants to the MTD and continue participants on this dose for the duration of the treatment period.

Dose modification for tirzepatide is not permitted, except for management of

- intolerable GI symptoms (Section 6.6.2.1), and
- participants with perceived excessive weight reduction (Section 6.6.2.2).

6.6.2.1. Management of Participants with Gastrointestinal Symptoms

The dose-escalation scheme is designed to minimize the development of intolerable GI symptoms. As stated in the table below, participants who experience intolerable GI symptoms (for example, nausea, vomiting, or diarrhea) at any time during the study, should first be counselled on dietary behaviors that may help mitigate nausea and vomiting, for example, eating smaller meals, splitting 3 daily meals into 4 or more smaller ones, and stopping eating when they feel full. If symptoms persist, the participant should be prescribed, at the investigator's discretion, symptomatic medication, for example, anti-emetic or anti-diarrheal medication temporarily. Prior to initiation of anti-diarrheal and while on anti-diarrheal medication, participants should be evaluated for signs of concurrent infection or colonic dilation. For any worsening of GI symptoms, investigators should also consider evaluating for a flare/worsening of a participant's UC. This may include evaluating for *C. difficile* infection, objective evidence of a disease flare and toxic megacolon.

A temporary interruption of study drug for 1 dose is permitted, provided the participant has taken the last 3 weekly dose. If the participant responds to symptomatic medications, study treatment should be resumed at the assigned dose immediately, either alone or in combination with symptomatic medication, which can also be utilized to manage symptoms. If intolerable GI symptoms persist despite the above measures, the investigator should consider dose de-escalation with subsequent re-escalation and may contact the sponsor for questions. A maximum of 2 dose de-escalation or re-escalations per participant will be permitted during the study. If intolerable GI symptoms persist despite dietary counseling, symptomatic treatment, temporary drug

interruption for one dose, and resumption at a lower dose of study drug, the participant should be discontinued from tirzepatide and consider evaluating the participant for UC exacerbation. All participants who discontinue tirzepatide should continue to attend scheduled study visits. All dose adjustments will be managed through IWRS.

This table summarizes steps the investigator should follow to mitigate GI symptoms (for example, nausea, vomiting, or diarrhea) at any time during the study for the tirzepatide intervention group.

Management of participants with intolerable GI AEs	
Step	Guidance
Step 1	Advise participants to eat smaller meals, for example, splitting 3 daily meals into 4 or more smaller meals, and to stop eating when they feel full and avoiding foods that trigger symptoms.
Step 2	Continue STEP 1 + If symptoms persist, treat with symptomatic medication, for example, anti-emetic or anti-diarrheal medication temporarily, per local availability and individual participant needs. Use of symptomatic medication should be captured as concomitant medication in the CRF.
Step 3	Continue STEP 1 + STEP 2 + If Steps 1 and 2 do not alleviate intolerable GI symptoms during the first 40 weeks of the study, consider temporary interruption of tirzepatide . Temporary interruption of tirzepatide is defined as • omission of 1 tirzepatide dose, and • taken the last 3 weekly doses The data related to temporary interruption of study treatment should be documented in source documents and entered on the CRF. In these instances, the participant will only hold tirzepatide; mirikizumab may be administered as scheduled. Participant should be on stable maintenance dose from Week 40 through Week 52. After the interruption, the investigator should restart the dose or escalate the dose as required, with the participant taking medication to alleviate their GI symptoms. If tirzepatide is temporarily interrupted for 2 or more consecutive doses, please refer to Section 7.1.2 for restart instructions. The data pertaining to temporary interruption of study intervention should be documented in source documents. Note: When restarting tirzepatide, the study personnel should notify the sponsor at least 2 weeks before the dosing visit to ensure adequate intervention supply at the site.
Step 4	If intolerable GI symptoms persist despite the above measures, the investigator should consider dose de-escalation with subsequent re-escalation and may contact the sponsor for questions.

Dose de-escalation and subsequent re-escalation schedule

As stated earlier in this section, participants unable to tolerate 2.5 mg during dose escalation of tirzepatide, despite measures taken as described in the table above, will be discontinued from tirzepatide.

For participants unable to tolerate any dose-escalation between 5 mg and 15 mg inclusive, despite measures taken as described earlier in this section, the investigator should contact Lilly medical monitor to consider dose de-escalation to the next lower dose with subsequent re-escalation to reach the 15-mg, 10-mg, or 5-mg dose by Week 40 as described in this table:

If the participant tolerates...	Then the dose is increased by...
5 mg to 2.5 mg for 4 weeks	2.5 mg Q4W up to MTD (5 mg, 10 mg, or 15 mg)
7.5 mg to 5 mg for 4 weeks	
10 mg to 7.5 mg for 4 weeks	
12.5 mg to 10 mg for 4 weeks	
15 mg to 12.5 mg for 4 weeks	

If a participant experiences GI tolerability issues despite the measures listed above, then the investigator should follow the guidance detailed below for dose reduction to an appropriate MTD by Week 40. Only up to 2 de-escalation or re-escalation attempts are allowed. The re-escalation attempt can continue past Week 40 as long as the temporary dose de-escalation was initiated at or before Week 40.

Guidance for dose reduction to an MTD

Participants should reach an MTD of 5, 10, or 15 mg by Week 40. For GI intolerability, investigator should first follow the steps described earlier in this section prior to considering a dose reduction. If de-escalation or re-escalation does not work (up to a maximum of 2 attempts), then permanently reduce dose as listed in the following table. Participants who are not tolerating their dose at Week 40 after de-escalation or re-escalation attempt(s) will have the following MTD from Week 40 forward.

If participant tolerates ...	but after the dose de-escalation/re-escalation attempt(s) does not tolerate	then participant continues with their MTD/maintenance dose from Week 40 on as....
15 mg	-	15 mg
12.5 mg	15 mg	10 mg
10 mg	12.5 or 15 mg	10 mg
5 mg	7.5 or 10 mg	5 mg
2.5 mg	5 mg	Participant will be discontinued from tirzepatide but remains in the study for continued follow-up.

6.6.2.2. Management of Participants with Perceived Excessive Weight Reduction

If participants reach a BMI ≤ 22 kg/m² any time during the study, the investigator or designated provider should initially evaluate the participant's dietary intake to confirm that the participant is meeting dietary intake guidance.

For participants who have perceived excessive weight loss or have reached a BMI of ≤ 22 kg/m² and if there are concerns that the degree of body weight reduction may lead to participant decision to discontinue from study intervention, the investigator should contact the sponsor to consider a tirzepatide dose reduction as outlined in Section 6.6.2.3.

In addition, participants should be evaluated for malnutrition (see Sections 8.2.13 and 10.2.2) if

- the PI suspects the participant is at risk of malnutrition, for example, caloric intake is significantly reduced to below 800 to 1200 kcal
- a participant's BMI falls below 22 kg/m², or

- in participants with T2DM, weight loss is >20% or >4 kg/month.

As stated in Section 7.1.1, participants whose BMI falls below 19 kg/m² during the study will be discontinued from tirzepatide.

6.6.2.3. Permanent Dose Reduction

The investigator may reduce the tirzepatide dose at any time due to

- recurrent intolerable GI AEs (Section 6.6.2.1)
Notes: If participants have had 2 dose de-escalation or re-escalation attempts and continues to have intolerable GI AEs, discontinue tirzepatide treatment. If participants have had no or only 1 dose de-escalation during the treatment period and develop intolerable GI AEs beyond Week 40, their tirzepatide dose will be permanently reduced through Week 52 (see table below).
- perceived excessive weight loss (see table below), and
- BMI $\leq$ 22 kg/m² (see table below).

This table provides guidance for reducing a tirzepatide dose.

If a participant is currently taking this dose of tirzepatide...	Then...
2.5 mg	discontinue tirzepatide.
5 mg	discontinue tirzepatide.
7.5 mg	reduce dose to 5 mg.
10 mg	reduce dose to 5 mg.
12.5 mg	reduce dose to 10 mg.
15 mg	reduce dose to 10 mg.

6.7. Continued Access to Study Intervention after the End of the Study

At the end of study participation, participants should be treated according to local practice for moderately to severely active UC and obesity or overweight. Treatment decisions can be made by the participant's primary physician but may also be made by the study investigator.

The sponsor may provide continued access to mirikizumab for 1 year to participants in selected countries where mirikizumab will not be launched after marketing authorization. Data collection will be limited to those items as noted in the SOA. The participant should follow the below SOA:

							Safety Follow-Up
Visit number	501	502	503	504	505	506	v901
Week from randomization	52 (same visit as last maintenance treatment visit)	64	76	88	100	104	8 weeks after last dose
Visit interval tolerance days		+/-7	+/-7	+/-7	+/-7	+/-7	
Complete informed consent for Continued Access	X						
Collect SAE		X	X	X	X	X	X
Dispense mirikizumab	X	X	X	X	X		

Abbreviations: SAE = serious adverse event; v = visit.

6.8. Treatment of Overdose

For this study, any dose of study intervention greater than the specified number of injections assigned through randomization will be considered an overdose. Tirzepatide overdose is defined as any amount of tirzepatide over the intended dose in less than 72 hours.

Refer to the mirikizumab and tirzepatide IBs and product labeling for further information.

In the event of an overdose, the investigator or treating physician should:

- contact the medical monitor immediately
- evaluate the participant to determine, in consultation with the medical monitor, whether study intervention should be interrupted or whether the dose should be reduced
- closely monitor the participant for any AE or SAE and laboratory abnormalities as medically appropriate until, for example, study intervention no longer has a clinical effect and report the overdose event as an AE and document it, and
- initiate appropriate supportive treatment according to the participant's clinical signs and symptoms.

6.9. Prior and Concomitant Therapy

6.9.1. General Considerations

Participants should be instructed to consult authorized site personnel before taking any new medications or supplements during the study.

Authorized site personnel should consult the sponsor's designated medical monitor if there are any questions about concomitant therapies during the study.

Participants are permitted to use concomitant medications that they require during the study, except certain medications specifically excluded in the protocol (Section 5.2) and those that may interfere with the assessment of efficacy and safety characteristics of the study interventions (Sections 6.9.2 and 5.2).

Tirzepatide delays gastric emptying and, thereby, has the potential to impact the absorption of concomitantly administered oral medications. Caution should be exercised when oral medications are concomitantly administered with tirzepatide. Monitor participants on oral medications dependent on threshold concentrations for efficacy and those with a narrow therapeutic index (for example, warfarin) when concomitantly administered with tirzepatide.

Concomitant therapy data collection

Any medication or vaccine, including over-the-counter or prescription medicines, vitamins, and/or herbal supplements, or other specific categories of interest, that the participant is receiving at the time of enrollment or receives during the study must be recorded along with the

- reason for use
- dates of administration including start and end dates, and
- dosage information including dose and frequency for concomitant therapy of special interest.

For this study, the concomitant medications of special interest include

- antihyperglycemic medications (Section 6.9.2), and
- prior treatments for IBD (Section 5.2, inclusion criteria [10] and [11], and Sections 6.9.4 and 10.8).

The medical monitor should be contacted if there are any questions regarding concomitant or prior therapy.

6.9.2. Concomitant Antihyperglycemic Medications

After randomization, discontinuation, dose modification, or formulation change of anti-hyperglycemic medications would be at the guidance of primary physician or healthcare provider qualified to manage diabetes mellitus.

Participants with T2DM are allowed to use concomitant antihyperglycemic medications that they require during the study, except

- sulfonylurea medications
- amylin mimetics
- DPP-4 inhibitors
- GLP-1 receptor agonists
- GIP/GLP-1 receptor agonists
- glinides, and
- insulin.

Antihyperglycemic medications for the management of T2DM are permitted; for example, metformin, thiazolidinediones, and SGLT2 inhibitors.

Participants who develop persistent severe hyperglycemia as determined by the investigator or healthcare provider caring for participant's diabetes may be candidates for addition of or dose increase in permitted antihyperglycemic therapies. The choice of new antihyperglycemic therapy (with the exception of amylin mimetics, GLP-1 receptor agonists, GIP/GLP-1 receptor agonists,

DPP-4 inhibitors, glinides, insulin, or sulfonylureas) or amount of dose increase will be at the discretion of the investigator or the participant's usual diabetes care physician, according to local guidelines for treatment of T2DM.

Management and reporting of incident diabetes

Participants will be monitored throughout the study for incident diabetes. If, in the opinion of the investigator, incident diabetes is suspected or occurs after randomization, participant should be monitored by investigator or referred to healthcare provider with appropriate specialization in management of diabetes per local guidelines. In general, diabetes mellitus is considered if HbA1c $\geq 6.5\%$, random plasma glucose ≥ 200 mg/dL, or diagnosis of diabetes mellitus (ADA 2024).

Participants who develop diabetes during the study will be educated on the signs and symptoms of hypoglycemia and its treatment, and monitor hypoglycemic episodes as directed by local standard of care under direction of the participant's healthcare professional. Any event of hypoglycemia should be reported to the site.

6.9.3. Prohibited Concomitant Medications

For prohibited antihyperglycemic medications, see Section 6.9.2. For other prohibited concomitant medications, see Section 5.2.

Administration of prohibited UC medications, approved or investigational, is not allowed. The use of such medications should not be withheld if, in the opinion of the investigator, failure to prescribe them would compromise participant safety. Participants who require a prohibited medication to treat their UC must be discontinued from study intervention (Section 7.1.1).

A participant who requires a prohibited medication for a non-UC indication should be discussed with the sponsor's designated medical monitor. A participant who initiates a prohibited medication for a non-UC indication may either discontinue the study intervention (Section 7.1.1) or discontinue the prohibited medication.

If a concomitant medication is needed to treat an AE or for appropriate medical management, the investigator should base decisions on the participant and clinical factors, considering prohibited medications.

6.9.4. Permitted Concomitant Medications

For permitted antihyperglycemic medications, see Section 6.9.2.

Participants should maintain their usual medication regimens for concurrent conditions or diseases throughout the study, unless those medications are specifically excluded (Section 5.2).

Participants taking permitted prescription UC concomitant medications, other than oral corticosteroids, are to keep the doses stable unless modifications are needed due to AEs or dose modification is specified in Section 10.8.

Continued use of nonprescription herbal supplements or medicines that are taken specifically for the treatment of UC should be discussed with the sponsor's designated medical monitor.

Monitoring of participants enrolled on 6-MP and azathioprine

The sponsor recommends monitoring the thiopurine metabolites levels of 6TG and 6-MMP for any participants enrolled on 6-MP and azathioprine. The rationale for this recommendation includes (Klein et al. 2024):

- Slower gastric and duodenal transit time, mainly due to the GLP-1 receptor agonist effects, may allow for increased 6-MP absorption.
- Anti-inflammatory properties of GLP-1 receptor agonist may augment bowel healing and facilitate improved thiopurine absorption.
- A patient's weight loss on GLP-1 therapy such as tirzepatide may affect weight-based dosing.

6.9.5. Corticosteroid Use and Taper

Local administration of corticosteroids, for example, intranasal, inhaled, intra-articular, or topical, is allowed (Section 10.8).

Participants who enter the study on oral corticosteroid therapy for treatment of UC should maintain a stable dose of corticosteroid during the first 12 weeks of treatment.

Corticosteroid tapering will be initiated at Week 12. For participants who cannot tolerate the corticosteroid taper without recurrence of clinical symptoms, corticosteroid taper may be paused, or the corticosteroid dose may be increased back to the original dose at induction baseline, but not higher. If doses higher than baseline are needed, consult the sponsor's designated medical monitor.

In situations where the steroid tapering is paused, the tapering regimen described below should be reinitiated within 2 weeks of interruption of taper, if possible, with a goal to complete tapering no later than Week 24.

The recommended tapering schedule for oral corticosteroids (other than budesonide extended -release tablets [budesonide MMX] or beclomethasone dipropionate [gastro-resistant prolonged release tablet]) is as follows.

- Dose >10 mg per day prednisone or equivalent: taper daily dose by 5 mg per week until receiving 10 mg per day, and then continue tapering at 2.5 mg per week until 0 mg per day.
- Dose ≤10 mg per day prednisone or equivalent: taper daily dose by 2.5 mg per week until 0 mg per day.

The recommended tapering schedule for oral budesonide MMX (up to 9 mg per day) is to decrease frequency of tablets from daily to every other day for 2 weeks, followed by every third day for 2 weeks, and then discontinue.

The recommended tapering schedule for participants receiving oral beclomethasone dipropionate (gastro-resistant prolonged-release tablet) (up to 5 mg per day) is to reduce tablets to 5 mg every other day for 4 weeks, and then discontinue.

6.9.6. Disease Progression and Rescue Therapy

The participants with exacerbations of their UC symptoms will have an opportunity to be evaluated at a scheduled or unscheduled visit for UC disease worsening. Worsening of underlying UC will be defined at a scheduled or unscheduled visit based on

- evaluation of symptoms in electronic diary entries, hematology, chemistry, C-reactive protein, fecal calprotectin, and calculation of the combined stool frequency and rectal bleeding score in UC, and
- exclusion of other causes of worsening disease activity unrelated to underlying UC, such as infections.

At the investigator's discretion, to further evaluate a potential worsening of underlying UC, flexible sigmoidoscopy or colonoscopy may be scheduled if the most recent flexible endoscopy or colonoscopy was ≥ 4 weeks prior to the visit during which worsening was evaluated.

Investigators and participants may opt to intensify care through unscheduled visits. The investigators also have the discretion to have the participant discontinue mirikizumab or the study. However, this study will not include rescue therapy.

7. Discontinuation of Study Intervention and Participant Discontinuation/Withdrawal

Discontinuation of specific sites or of the study as a whole are handled as part of Section 10.1.

7.1. Discontinuation of Study Intervention

Continuing in the study after discontinuation of study intervention

When necessary, a participant may be permanently discontinued from study intervention, per investigator's decision. If so, the participant will discontinue the study intervention (treatment) and will remain in the study and follow procedures for all remaining study visits, as shown in the SoA (Section 1.3).

Further, study discontinuation criteria allow for the participants to be discontinued at any time by investigator or by the decision of participant if they are receiving insufficient benefit from study therapy or require treatment with another therapeutic agent that has been demonstrated to be effective for the treatment of UC.

If the participant is unwilling or unable to return for all applicable study visits, the site should attempt to collect as much follow-up information as possible, especially data collection pertaining to primary and key secondary efficacy endpoints at Week 52 of the treatment period.

7.1.1. Permanent Discontinuation of Study Intervention

Unless indicated otherwise, if a participant meets the criteria for permanent discontinuation, both mirikizumab and tirzepatide should be stopped.

A participant **may** be permanently discontinued from the study intervention for the following reasons:

- **Suicidal ideation and behavior:** The participant
 - answered “yes” to Question 4 or Question 5 on the “Suicidal Ideation” portion of the C-SSRS, **or**
 - answered “yes” to any of the suicide-related behaviors on the “Suicidal Behavior” portion of the C-SSRS.

Note: A psychiatrist or appropriately trained professional may assist in the decision to discontinue the participant.

- **Depression symptoms:** If the participant has a QIDS-SR16 score greater than or equal to 16 on the total score or greater than or equal to 2 on Item 12, the investigator may decide to withhold study intervention or discontinue study intervention. A mental health care professional may assist in the decision.
- **Hypersensitivity reactions:** If the investigator determines that a systemic hypersensitivity reaction has occurred related to study intervention administration, the participant may be permanently discontinued from the study intervention, and the sponsor's designated medical monitor should be notified. If the investigator is uncertain about whether a systemic hypersensitivity reaction has occurred and whether

discontinuation of study intervention is warranted, the investigator may consult the sponsor.

- **Malnutrition:** If the participant develops severe or persistent malnutrition, refractory to treatment despite nutritional guidance and supplementation, consider discontinuing **tirzepatide**, and discuss management of the participant with the sponsor-designated monitor.

A participant **must** be permanently discontinued from the study intervention for the following reasons:

- Participant decision: The participant requests to discontinue study intervention.
The reason(s) for the decision will be documented.
- Investigator decision
 - the investigator decides that the participant should be discontinued from the study medication for safety reasons
- The participant develops anaphylaxis to study intervention.
- The participant is AFAB and of childbearing potential, sexually active, and does not agree to follow the effective contraception defined in Section 10.4.1.
- The participant requires treatment for exacerbation of UC with medication(s)
 - at doses higher than those specified in inclusion criterion [11], for example, prednisone >20 mg/day, or
 - with prohibited medications specified in the exclusion criterion [50], for example, a course of IV corticosteroids, a single dose of infliximab, or IV cyclosporine.
- The participant requires a colectomy, proctocolectomy, or partial colectomy during the study.
- The participant has bariatric surgery.
- The participant has current evidence of an adenomatous polyp with high-grade dysplasia.
- If the participant's BMI falls below 19 kg/m² during the study, the participant must be discontinued from **tirzepatide**.
- Unblinding: If an investigator, site personnel performing assessments, or participant is unblinded from **tirzepatide** treatment, the participant must be discontinued from the **tirzepatide**. In cases where there are ethical reasons to have the participant continue on study intervention, the investigator must obtain specific approval from the sponsor or designee for the participant to continue in the study.
- The participant develops significant GI symptoms despite management (Section 6.6.2.1).

Note: The investigator should contact the sponsor to discuss whether it is medically appropriate for the participant to continue tirzepatide.

- The participant develops any of these diagnoses or conditions during the study:
 - T1DM
 - T2DM requiring insulin
 - new onset diabetic retinopathy requiring therapy
 - intolerable GI symptoms that do not respond to mitigations outlined in Section 6.6.2.1
 - MTC after randomization
 - acute or chronic pancreatitis, confirmed by adjudication, or
 - HBV or HCV: The participant tests positive for HBV DNA (Section 8.2.11) or tests positive for HCV RNA (Section 8.2.12). For additional criteria for permanent discontinuation of **mirikizumab**, see Section 8.2.11.

Note: The HBV DNA result is to be confirmed if initial positive test result is positive but below the level of quantification (Section 8.2.11). Prior to discontinuation of any immunomodulatory and/or immunosuppressive therapy, including mirikizumab, the participant is to be referred to, evaluated, and managed by a specialist physician with expertise in evaluation and management of viral hepatitis. The timing of discontinuation from study intervention relative to the initiation of any antiviral treatment for hepatitis is to be based on the recommendation of the consulting specialist physician, in conjunction with the investigator, and aligned with medical guidelines and standard of care.

- certain AEs:
 - the participant develops HIV infection
 - the participant develops active TB or has untreated LTBI (Section 8.2.10)
- The participant experiences an AE or SAE that, in the opinion of the investigator or sponsor, would preclude them from continuing to receive study drug.
- The participant is diagnosed with malignancy, except for successfully treated basal or squamous cell skin carcinoma.
- The participant develops any condition or contraindication as addressed in the local labelling for mirikizumab or tirzepatide that would preclude the participant from continuing the study intervention.
- If the participant, for any reason, requires treatment with another therapeutic agent that is prohibited by this study (see Section 5.2) and has been demonstrated to be effective for the treatment of UC, obesity, or overweight, discontinuation from study intervention occurs prior to introduction of new agent.
- If the participant is noncompliant with study procedures and/or study intervention administration, the investigator should assess the participant to determine the reason for noncompliance and educate and/or manage the participant as appropriate to improve compliance. If in consultation with sponsor or its designee, the noncompliance is deemed significant, discontinue participant from the study.

- If the participant has any of the following results on 2 consecutive samples taken at least 48 hours, but no more than 1 week, apart:

Test	Result
Absolute neutrophil count	<500 cells/ μ L (<0.5 x 10 ³ / μ L or <0.50 GI/L)
Absolute lymphocyte count	<200 cells/ μ L (<0.2 x 10 ³ / μ L or <0.2 GI/L)
Total WBC	<1000 cells/ μ L (<1.0 x 10 ³ / μ L)
Hemoglobin	<6.5 g/dL (<65 g/L)

7.1.2. Temporary Discontinuation

Temporary discontinuation of both study interventions

In certain situations, after randomization, the investigator may need to temporarily interrupt study interventions. Every effort should be made by the investigator to maintain participants on study intervention and to restart study intervention after any temporary interruption, as soon as it is safe to do so. The investigator should provide guidance to the participant, to consult before any dose interruption, and evaluate participant status prior to resumption.

Cases that may merit temporary withholding of the study intervention should be discussed with the medical monitor. The medical monitor, in consultation with the investigator, will determine when it is appropriate to recommence study intervention.

Both study interventions must be temporarily discontinued if the participant:

- requires major surgery
Note: Administration of the study intervention may be restarted only after adequate wound healing.
- develops depression symptoms (see Section 7.1.1)

Temporary interruption of tirzepatide

This table describes how to manage treatment if there is a temporary interruption of **tirzepatide** study intervention. This may require an additional unscheduled visit to address in a timely manner.

If study intervention interruption is...	Then....
2 consecutive doses or less	participant restarts study intervention at the last dosage taken prior to the interruption and continues escalation, as per escalation schedule (Section 6.1.4.2). If the dose interruption occurred immediately prior to a scheduled dose-escalation visit, the participant should proceed with the next escalation.
3 consecutive doses or more	participant restarts study intervention at 5 mg and repeats dose-escalation scheme.

due to an AE	the event is to be documented and followed according to the procedures in Section 8.3 of this protocol.
due to intolerable persistent GI AE	participant should be treated as suggested in Section 6.6.2.1.

Temporary discontinuation of mirikizumab

If participant meets these criteria...	Then...
<i>Infection-related criteria</i>	
A clinically important extraintestinal or intestinal infection (see exclusion criteria [40-49])	Withhold mirikizumab until resolution of all acute clinical signs and symptoms, and complete of all appropriate anti-infective treatment. If participant is diagnosed with LTBI, see Section 8.2.10 and the discontinuation criterion immediately below. For infections requiring permanent discontinuation, see Section 7.1.1 for further details.
A diagnosis of LTBI during the study	Permanently discontinue from mirikizumab unless the participant is a candidate for LTBI treatment, and is treated for LTBI as follows: • Mirikizumab is temporarily held for at least the first 4 weeks of LTBI treatment. • After receiving at least 4 weeks of appropriate LTBI therapy (as per WHO or US CDC guidelines), if there is no evidence of hepatotoxicity (ALT/AST must remain ≤ 2 times ULN) or other treatment intolerance, mirikizumab may be resumed. • The participant must complete appropriate LTBI therapy to remain eligible to receive study mirikizumab.
HBV DNA results that are reported as positive, or as detecting HBV DNA, but HBV DNA is below the level of quantification	See Section 8.2.11.

<i>Neutrophil-related criteria</i>	
Development of neutrophil and lymphocyte abnormalities during the study: • a confirmed absolute neutrophil count $<1 \times 10^9/L$ ($<1 \times 10^3/\mu L$ or $<1 \text{ GI/L}$; 2 assessments below this threshold) • absolute lymphocyte count $<500 \text{ cells}/\mu L$ ($<0.5 \times 10^3/\mu L$ or $<0.50 \text{ GI/L}$)	After the first retest confirming absolute lymphocyte count $<0.5 \times 10^3/\mu L$ (second assessment below threshold) • discontinue AZA, 6-MP, or MTX, if applicable, and • hold the next dose of mirikizumab. The participant must be retested in approximately 2 weeks. If the absolute lymphocyte count remains $<0.5 \times 10^3/\mu L$, repeat the hematology again in 2 weeks; that is, prior to the next dose of mirikizumab. If the absolute lymphocyte count still remains $<0.5 \times 10^3/\mu L$, the next dose of mirikizumab will not be administered. The participant must be retested again in 2 weeks. If the absolute lymphocyte count still remains $<0.5 \times 10^3/\mu L$, mirikizumab will be permanently discontinued (Section 7.1.1). White blood cell and lymphocyte counts will be followed for these participants until they return to an acceptable level.

7.1.3. Hepatic Criteria for Study Drug Interruption or Discontinuation

See Section 8.2.9.3 for hepatic criteria for study intervention interruption or discontinuation.

See Section 10.5 for further guidance on hepatic monitoring.

7.2. Participant Discontinuation/Withdrawal from the Study

Discontinuation is expected to be uncommon. To minimize the amount of missing data and to enable assessment of study objectives as planned in the study protocol, every attempt will be made to keep participants in the study regardless of study intervention use.

A participant **must** be discontinued from the study intervention and withdrawn from the study if the participant becomes pregnant during the study.

A participant **may** withdraw from the study

- at any time at the participant's own request for any reason or without providing any reason
- at the discretion of the investigator for safety, behavioral, compliance, or administrative reasons, or
- if enrolled in any other clinical study involving an investigational product, or enrolled in any other type of medical research judged not to be scientifically or medically compatible with this study.

At the time of discontinuing from the study, if possible, the participant will complete procedures for an EDV and post-treatment safety follow-up, as shown in the SoA. If the participant has not already discontinued the study intervention, the participant will be permanently discontinued from the study intervention at the time of the decision to discontinue the study.

If the participant withdraws consent for disclosure of future information, the sponsor may retain and continue to use any data collected before such a withdrawal of consent. If a participant

withdraws from the study, the participant may request destruction of any samples taken and not tested, and the investigator must document this in the site study records.

7.3. Lost to Follow-Up

A participant will be considered lost to follow-up if they repeatedly fail to return for scheduled visits and are unable to be contacted by the study site. Site personnel or designee are expected to make diligent attempts to contact participants who fail to return for a scheduled visit or were otherwise unable to be followed up by the site.

8. Study Assessments and Procedures

Study procedures and their timing are summarized in the SoA.

Immediate safety concerns should be discussed with the sponsor immediately upon occurrence or awareness to determine if the participant should continue or discontinue study intervention.

Adherence to the study design requirements, including those specified in the SoA, is essential and required for study conduct.

All screening evaluations must be completed and reviewed to confirm that potential participants meet all eligibility criteria. The investigator will maintain a screening log to record details of all participants screened and to confirm eligibility or record reasons for screening failure, as applicable.

8.1. Efficacy Assessments

This table defines the UC-specific endpoints.

Endpoint	Definition
Clinical remission	mMS of 0 to 2, including the following 3 components - SF subscore = 0 or 1 - RB subscore = 0 - Centrally read ES = 0 or 1; score of 1 modified to exclude friability
Maintenance of clinical remission	Achieving clinical remission at both Week 24 and Week 52
Corticosteroid-free remission	Corticosteroid-free remission at Week 52 defined as: - Clinical remission at Week 52, and - No corticosteroid use for ≥ 12 weeks prior to Week 52
Endoscopic response	ES = 0 or 1, excluding friability
Endoscopic remission	ES = 0
Clinical response	- Decrease in the mMS of ≥ 2 points and $\geq 30\%$ decrease from baseline, and - decrease of ≥ 1 point in the RB subscore from baseline or an RB subscore of 0 or 1
Symptomatic response	$\geq 30\%$ decrease from baseline in the composite clinical endpoint of the sum of SF and RB subscores
Symptomatic remission	- SF subscore = 0 or 1 and - RB subscore = 0
Histologic improvement	Geboes score ≤ 3.1 ; neutrophil infiltration in $< 5\%$ of crypts; no crypt destruction; and no erosions, ulcerations, or granulation tissue
Histologic remission	Geboes score $\leq 2B.0$: no lamina propria neutrophils; no neutrophils in the surface or crypt epithelium; no crypt destruction; and no erosions, ulcerations, or granulation tissue
histologic-endoscopic mucosal improvement plus absence of neutrophils	Achieving both - histologic remission (as defined above), and - endoscopic remission (as defined above)
Bowel Urgency remission	Urgency NRS = 0 or 1

8.1.1. Primary Efficacy Assessments

Primary endpoint

As stated in Section 3, the primary endpoint is the proportion of participants who simultaneously achieve clinical remission and at least 10% weight reduction at Week 52.

Clinical remission is defined in Section 8.1.

Methods for measuring body weight are described in Section 10.9.

8.1.1.1. mMS

The mMS (as provided in the FDA Guidance 2016) is a composite instrument reported by patients and the physician and is comprised of the following 3 subscores:

- Stool Frequency (SF)
- Rectal Bleeding (RB), and
- Endoscopic Subscore (ES).

Stool Frequency (SF)

The SF subscore is a patient-reported measure. This item reports the number of stools in a 24-hour period, relative to the normal number of stools for that participant in the same period, on a 4-point scale (see Section 10.11). A stool is defined as a trip to the toilet when the participant:

- has a bowel movement, or
- passes
 - blood alone, or
 - blood and mucus, or
 - mucus only.

The reference “normal SF” for each participant will be recorded in the eDiary at the screening visit. Normal SF for each participant is based on

- reported SF when the participant was in remission, or
- reported SF before initial onset of signs and symptoms of UC.

Remission refers to a period of time since being diagnosed with UC when the participant is not experiencing any signs or symptoms relating to UC. This period of time may last a few weeks or a few months or may even last several years.

Rectal Bleeding (RB)

The RB subscore is a patient-reported measure. This item reports the most severe amount of blood passed with stool for a given day, on a 4-point scale (see Section 10.11).

Endoscopic Subscore (ES)

The ES is a physician-reported measure that indicates the worst appearance of the mucosa on flexible sigmoidoscopy or colonoscopy, on a 4-point scale. Determination of the ES is further detailed in Section 10.11.

Consistent with current clinical practice and regulatory advice, this study excludes friability from the definition of an ES of 1. Also consistent with best clinical trial practice, endoscopy scores will be determined from blinded central readers.

The local endoscopy score is collected on paper and should be transcribed immediately, no more than 24 hours after reading in the CRF. The entry of the local endoscopy is critical to initiate the blinded central reading process.

The mMS will be calculated using the SF and RB data from the daily eDiary (see the SoA, Section 1.3, and Section 8.1.2.3). The subscores of SF and RB will be calculated by averaging the daily subscores within the 7-day period, excluding the day of bowel preparation and day of endoscopy (for visits that include an endoscopy). The mMS cannot be calculated if the participant is not compliant with the daily eDiary entry of SF and RB data for a minimum of 3 consecutive days or 4 nonconsecutive days of the 7 days before bowel preparation for colonoscopy, or if no bowel preparation, then the day of the colonoscopy. For more details, refer to the SAP.

Scoring

Each mMS subscore is scored on a 4-point scale, ranging from 0 to 3, to give a maximum mMS of 9. See Section 10.11 for more details about the mMS scoring system.

This study will use the mMS as outlined in the FDA Guidance (2016), which will include a sum of the following mMS subscores, with a maximum mMS of 9:

- SF
- RB, and
- ES.

8.1.1.2. Endoscopy: Colonoscopy or Sigmoidoscopy

Sigmoidoscopy or colonoscopy will be used to determine the ES at screening and at other visits, as noted on the SoA. Lilly recommends that colonoscopy be used to document disease activity in all involved segments of the colon and that colonoscopy is preferred. The ES will be determined by

- readings performed by qualified endoscopist who are blinded to induction treatment assignment, and
- central readers who are blinded.

Note: Disagreement between the site reader and central reader will be adjudicated by an additional blinded central reader, as detailed in the Image Review charter.

Participants who are identified as being at increased risk for colon cancer or for adenomatous polyps or have not undergone recent surveillance for dysplasia will undergo a colonoscopy instead of a sigmoidoscopy to allow screening for dysplasia or to assess for the presence of adenomatous polyps, respectively.

A sigmoidoscopy or colonoscopy will be performed on all participants during screening within 21 days prior to randomization and at other visits as specified in the SoA (Section 1.3). The sigmoidoscopy or colonoscopy report and histopathology report must be available in the source documents. Prior to performing the screening sigmoidoscopy or colonoscopy, investigators or designees should ensure that participants have clinically acceptable central laboratory test results, including stool tests that are “negative” for *C. difficile* and other intestinal pathogens (see Section 5).

If a participant undergoes early permanent discontinuation from the study soon after the screening sigmoidoscopy or colonoscopy, there will be no need for EDV sigmoidoscopy or colonoscopy. There is no need for an EDV sigmoidoscopy or colonoscopy if fewer than 12 weeks have elapsed since a prior colonoscopy.

The endoscopist will be a licensed physician, who is qualified by education, training, and experience to perform sigmoidoscopies and colonoscopies. Investigators may delegate sigmoidoscopy or colonoscopy to other authorized site personnel. However, authorized site personnel performing the sigmoidoscopy or colonoscopy must receive training from the sponsor or designee in the determination and calculation of the ES. The endoscopist will determine the site-read ES at each sigmoidoscopy or colonoscopy, and site personnel or their designee will record this in the CRF.

All sigmoidoscopy or colonoscopy procedures will be video recorded in color using a storage medium provided by the sponsor or designee. The video images will be sent for independent central reading. The central imaging vendor will provide a manual that will outline the standard study procedures used to capture and transmit video recordings of endoscopic procedures throughout the study and will outline the qualifications required of the central reader. Recorded sigmoidoscopy or colonoscopy procedures (without participant-identifying information) will be used for research purposes.

The central reader will determine the centrally read ES at each sigmoidoscopy or colonoscopy in a blinded manner, as detailed in the Image Review charter.

The adjudicated ES will be provided to the site personnel prior to randomization to enable the determination of participant eligibility. An endoscopic score, as determined by the Image Review charter, will be used for endpoint determination and clinical responder status.

Lilly may partner with a vendor to use an AI-based continuous scoring system that quantifies the severity and extent of inflammation on endoscopy from endoscopic video recordings in addition to the original central reading of endoscopy.

If a participant becomes pregnant during the study, no additional sigmoidoscopies or colonoscopies will be performed.

Mitigating potential risks of colonoscopy

See Sections [2.3.1](#) and [10.15](#).

8.1.2. Secondary Efficacy Assessments

Several secondary efficacy endpoints are based on

- scores using the mMS described in Section [8.1.1.1](#)
- scores from the Geboes score for the assessment of histological activity, or
- patient-reported outcome instruments.

8.1.2.1. Colonic Biopsies

A histopathology report supporting the diagnosis of UC must be available in the source documents prior to randomization to satisfy study entry criteria.

Biopsies will be obtained at each sigmoidoscopy or colonoscopy to support assessment of the histopathology endpoints in this study and for further research purposes (without participant-identifying information), including the assessment of exploratory biomarkers. These will be sent to the central study laboratory for processing. Histopathologic scoring of these biopsies will be performed by a blinded central reader. The central imaging vendor will provide a manual that will outline the procedures to be used for secure specimen transfer, processing, slide preparation, and digitization of slides for histopathologic scoring. The histopathology results will not be made available to study sites. Investigators should obtain additional biopsies that are sent to the local laboratory for medical management of the participant.

8.1.2.2. Geboes Score

The Geboes Histological Index is based on a comprehensive grading system that evaluates aspects of mucosal injury seen in UC. The Geboes index has these grades:

- Grade 0 - Structural (architectural change)
- Grade 1 - Chronic inflammatory infiltrate
- Grade 2A - Lamina propria eosinophils
- Grade 2B - Lamina propria neutrophils
- Grade 3 - Neutrophils in epithelium
- Grade 4 - Crypt destruction
- Grade 5 - Erosion or ulceration

For each of the histological features assessed, the highest grade in which there is evidence of disease is assigned (Geboes et al. 2000). The Geboes scale is further described in Section 10.12.

8.1.2.3. Patient-Reported Outcomes: Daily Electronic Diary

As stated in the SoA (Section 1.3), the participant's daily electronic diary will include the following items, to rate their UC symptom severity and perception of disease severity:

Daily electronic diary item	Description
Rectal Bleeding (RB)	See Section 8.1.1.1.
Stool Frequency (SF)	See Section 8.1.1.1.
Patient Global Rating of Severity (PGR-S)	The PGR-S is a patient-reported, single-item questionnaire to assess the UC symptom severity in participants over the past 24 hours. Responses are graded on a 6-point scale in which a score of one indicates the participant has no symptoms (that is, "none") and a score of six indicates that the participant's symptom(s) are "very severe."
Urgency NRS	The Urgency NRS is a patient-reported, single-item instrument that measures the severity for the urgency, sudden, or immediate need, to have a bowel movement in the past 24 hours using an 11-point NRS ranging from 0 "no urgency" to 10 "worst possible urgency" (Dubinsky et al. 2022).

For eDiary compliance requirements, see Section 8.1.1.1.

8.1.2.4. Patient-Reported Outcomes: Other

Work Productivity and Activity Impairment Questionnaire: Ulcerative Colitis (WPAI:UC)

The WPAI:UC is a patient-reported instrument that measures the impact on work productivity and regular activities attributable to UC. It contains 6 items that assess

- employment status
- hours missed from work due to the specific health problem (UC)
- hours missed from work for other reasons
- hours actually worked
- degree health affected productivity while working, and
- degree health affected productivity in regular unpaid activities.

Four scores are calculated from the responses to these 6 items:

- absenteeism
- presenteeism
- work productivity loss, and
- activity impairment.

Scores are calculated as impairment percentages (Reilly et al. 1993, 2008), with higher numbers indicating greater impairment and less productivity; that is, worse outcomes.

Inflammatory Bowel Disease Questionnaire (IBDQ)

The IBDQ is a patient-reported, 32-item instrument that measures 4 aspects of the participants' lives (Guyatt et al. 1989; Irvine et al. 1994, 1996, 1999):

- symptoms directly related to the primary bowel disturbance
- systemic symptoms
- emotional function, and
- social function.

Responses are graded on a 7-point Likert scale, in which 7 denotes “not a problem at all” and 1 denotes “a very severe problem.” Scores range from 32 to 224. A higher score indicates a better quality of life (Irvine et al. 1994, 1999).

EQ-5D-5L

The EQ-5D-5L (EuroQol Research Foundation 2019) is a standardized 5-item, self-administered instrument for use as a measure of health outcome. It provides a simple descriptive profile and a single index value for health status that can be used in the clinical and economic evaluation of health care as well as population health surveys. The EQ-5D-5L assesses 5 dimensions of health:

- mobility
- self-care
- usual activities
- pain/discomfort, and
- anxiety/depression.

Each dimension has 5 response levels:

- no problems
- slight problems

- moderate problems
- severe problems, and
- extreme problems.

The VAS records the respondent's self-rated health on a vertical VAS where the endpoints are labeled as "The best health you can imagine" (100) and "The worst health you can imagine" (0).

8.1.2.5. Extraintestinal Manifestations

Review of EIMs will be performed as described in Section 8.2.1.

8.1.2.6. Inflammatory Biomarkers

CRP (high-sensitivity [hsCRP]) is an acute-phase protein expressed by hepatocytes in response to inflammatory cytokines, particularly IL-6, TNF, and IL-1 β (Sands 2015).

Fecal calprotectin is a complex consisting of the calcium-binding proteins S100A8 and S100A9 (Sands 2015). It is expressed by activated neutrophils (and to a lesser extent by macrophages and monocytes), and fecal levels correlate with the number of neutrophils in the gut. It is used as a biomarker of intestinal inflammation in clinical practice.

8.1.3. Exploratory Efficacy Assessment

8.1.3.1. Assessment of Change in Body Fat Mass using Dual-Energy X-Ray Absorptiometry (DXA)

Total body composition assessments for this study

It is recommended that study participants undergo a body composition assessment via DXA scans after meeting all eligibility criteria, including colonoscopy or sigmoidoscopy, measured between Visits 1 and 2 and no later than Visit 3 prior to receiving study intervention and at Week 52 (Visit 15), or at the time of ED (if at least 3 months from the previous DXA scan), with the intent of evaluating changes in body composition associated with weight reduction, including both fat and lean mass compartments and abdominal visceral fat.

Recommendations for DXA body composition scans include the following:

- All scans for a given participant should be performed on the same DXA scanner using the same software.
- All scans for a given participant should be obtained under similar circumstances (e.g., similar time of day, consistent hydration status, and consistent scan status).

DXA scan

DXA total body composition with regional analysis provides a measure of physiological response to obesity interventions superior to simple anthropometrics, such as BMI. It is safe and widely available in clinical practice (Kendler et al. 2013), making it a useful method to detect changes in fat and lean body mass in weight management studies.

The risk associated with DXA scan is low, with a radiation exposure (effective dose) of about 5 μ Sv or 0.005 mSv (millisievert). The typical level of background radiation to which the general population is exposed, not including radiation due to medical procedures, has been estimated to be about 2.5 mSv/year (Lewiecki et al. 2016).

DXA instrument

Investigative sites will perform the body composition DXA scans on either Hologic® or Lunar™ DXA scanners with total body composition capabilities according to the imaging guidelines.

Guidance for conducting DXA scans

To obtain consistent and acceptable quality data, investigators will receive detailed instructions on the DXA scan protocol to be used.

The scans will be read and evaluated by a central reader.

Recommendations

Participants should fast for at least 8 hours without eating or drinking (except water) before each DXA scan. No water should be consumed within 2 hours before each DXA scan.

- Participants will abstain from strenuous exercise for 24 hours before each DXA scan.
- Perform all scans for a given participant using the same DXA scanner and software.
- Perform all scans for a given participant under similar circumstances, for example, using consistent scan status and performing the scan at the same time of day.

DXA scan contraindications

Participants are excluded from the DXA scan if any of the following criteria apply but the participant may still enroll in Study AMCD:

- Have had a recent procedure (within 10 days) where oral contrast or radionuclides were administered (for example, CT scan with contrast or nuclear medicine studies).
- Have any implants, hardware, devices, or other foreign materials in the measurement area that may interfere with the scan, according to the DXA manufacturer's specification.
- Have a body weight, height, and/or width that prohibits the ability to obtain accurate measurements according to the DXA manufacturer's specification.

In these cases, missing DXA data will not be considered a protocol deviation.

8.2. Safety Assessments

Planned time points for all safety assessments are provided in the SoA.

If safety monitoring uncovers GI AE-related issue that needs to be addressed by unblinding from tirzepatide treatment at the individual or treatment group level, additional analyses of the safety data can be performed by members of the external DMC (see Section 10.1.5.1).

8.2.1. Physical Examinations

Physical examinations will be performed when specified in the SoA (Section 1.3). Physical examinations may also be performed as clinically indicated at other visits at the discretion of the investigator.

Complete physical examination

A complete physical examination will include, at a minimum,

- general appearance
- head, ears, eyes, nose, throat
- extremities
- assessments of the skin
- assessment of lymph nodes
- cardiovascular examination
- respiratory examination
- abdominal examination
- neurological systems examination, and
- thyroid examination.

Pelvic, rectal, and breast examinations are not required unless clinically indicated.

The screening physical examination should include the lifetime risk factors for TB infection, TB progression, and symptoms and/or signs of active TB, including measurement of body temperature and assessment of peripheral lymph nodes (Section 8.2.10).

Investigators should pay special attention to clinical signs related to previous serious illnesses.

Height and weight will also be measured and recorded per Section 10.9.

Symptom-directed physical assessments after screening

These examinations are performed based on participant status and standard of care by a physician or other qualified healthcare professional.

Every symptom-directed examination should also include an examination of the heart, lungs, abdomen, neurological system, thyroid, and visual examination of the skin.

Every third month after screening, the physical examination should also include an assessment of TB risk factors and symptoms or signs of TB, including an assessment of peripheral lymph nodes (Section 8.2.10).

Extraintestinal manifestations

EIMs will be reviewed at the time points described in the SoA (Section 1.3).

EIMs include

- | | |
|--------------------------|----------------------------------|
| • ankylosing spondylitis | • osteoporosis |
| • aphthous stomatitis | • pancreatitis |
| • arthralgia | • peripheral arthritis |
| • autoimmune hepatitis | • primary sclerosing cholangitis |
| • cholelithiasis | • pyoderma gangrenosum |
| • deep vein thrombosis | • nephrolithiasis |
| • episcleritis | • sacroiliitis |
| • erythema nodosum | • skin vasculitis, and |

- iritis
- osteopenia
- uveitis

8.2.2. Vital Signs

For each participant, vital signs should be measured according to the SoA (Section 1.3) and Section 10.9. If warranted, additional vital signs may be measured.

Vital signs, including pulse (beats per minute) and blood pressure (mm Hg), will be measured after participant has been sitting for at least 5 minutes and before obtaining an ECG tracing and before collection of blood samples for laboratory testing, at visits where required.

Any clinically significant findings from vital-signs measurement that result in a diagnosis and that occur after the participant receives the first dose of study treatment should be reported to sponsor or its designee as an AE via the CRF.

8.2.3. Electrocardiograms

A 12-lead ECG (local) will be obtained as outlined in the SoA (Section 1.3) using an ECG machine that, preferably, automatically calculates the heart rate and measures PR, QRS, QT, and QTc intervals. The participant should be supine for at least 5 minutes before ECG collection, and remain supine but awake during ECG collection. ECGs should be collected prior to collection of blood samples for laboratory testing.

The ECGs must be interpreted by the investigator or qualified designee at the site as soon after the time of ECG collection as possible, and ideally while the participant is still present, for immediate participant management, if needed. The investigator or designee must document their review of the ECG. If a clinically relevant abnormality is observed on the participant's ECG, then the investigator should assess the participant for symptoms, such as palpitations, near syncope, syncope, or chest pain.

The original ECG must be retained at the investigative site.

8.2.4. Chest Imaging

A high-quality, locally performed chest x-ray (posterior-anterior view and, if needed, a lateral view), interpreted and reported by a radiologist or pulmonologist, will be obtained as specified in the SoA (Section 1.3) for those with a positive TB test (Section 8.2.10).

For participants requiring chest imaging, the chest x-ray films, images, or a radiology report must be available to the investigator for review before the participant is randomly assigned to a treatment in this study.

Conditions for using a previous chest x-ray at screening

Participants do not need to have a chest x-ray at screening if, in the opinion of the investigator, both of these 2 conditions are met:

- the chest x-ray was performed within 6 months before the initial screening, and
- documentation of the prior chest x-ray, read by a qualified radiologist or pulmonologist, is sufficient for TB evaluation according to local standard of care.

Alternatives to chest x-ray

In consultation with the sponsor's medical monitor, results of a chest CT scan or other imaging study similar to a chest x-ray, if performed within the same time window, may be used instead of a chest x-ray for the TB evaluation.

8.2.5. Clinical Safety Laboratory Tests

See Section 10.2 for the list of clinical laboratory tests to be performed and the SoA for the timing and frequency.

Reviewing and recording test results

The investigator must review the laboratory results, document this review, and report any clinically relevant changes occurring during the study as an AE. The laboratory results must be retained with source documents unless a Source Document Agreement or comparable document cites an electronic location that accommodates the expected retention duration. Clinically significant abnormal laboratory findings are those which are not associated with the underlying disease, unless judged by the investigator to be more severe than expected for the participant's condition.

Repeat testing after clinically significant abnormal findings

All laboratory tests with values considered clinically significantly abnormal during participation in the study should be repeated until the values return to normal or baseline or are no longer considered clinically significant by the investigator or medical monitor. For details about managing hepatic laboratory abnormalities, see Section 8.2.9.

If such values do not return to normal/baseline within a period of time judged reasonable by the investigator, the etiology should be identified and the sponsor notified.

All protocol-required laboratory assessments, as defined in Section 10.2, must be conducted in accordance with the SoA, standard collection requirements, and laboratory manual.

If laboratory values from non-protocol specified laboratory assessments performed at an investigator-designated local laboratory require a change in participant management or are considered clinically significant by the investigator, for example, SAE or AE or dose modification, then report the information as an AE.

8.2.6. Pregnancy Testing

Pregnancy testing is to be performed for all IOCBP and will occur as outlined in the SoA (Section 1.3).

Serum and urine pregnancy tests must be performed only for IOCBP, prior to administering study intervention or must be performed with the result available prior to first dose or injection of study intervention(s).

Additional pregnancy tests (beyond those required per the SoA) should be performed at any time during the study

- if a menstrual period is missed
- if there is clinical suspicion of pregnancy, or
- as required by local law or regulation.

If the participant is pregnant, the participant **must** be permanently discontinued from study intervention and be withdrawn from the study (Section 7.2).

Assessment of FSH levels can assist in determining if an individual AFAB meets the definition of “menopausal” as outlined in Section 10.4.1. FSH levels can be optimally obtained at screening or during the study, at the discretion of the investigator to determine menopausal status (Section 1.3).

8.2.7. Suicidal Ideation and Behavior Risk Monitoring

Baseline assessment of suicidal ideation and behavior (SIB) and intervention-emergent SIB will be monitored using the C-SSRS. See Section 8.3.3.7 for more details.

Participants should be monitored appropriately and observed closely for SIB or any other unusual changes in behavior throughout the study.

Participants who experience signs of SIB should undergo a risk assessment. All factors contributing to SIB should be evaluated and consideration should be given to discontinuation of the study intervention (Section 7.1.1).

Family and caregivers of participants should be alerted about the need to monitor participants for the emergence of unusual changes in behavior, as well as the emergence of SIB and to report such symptoms immediately to the study investigator.

8.2.8. Depression Monitoring

Depression will be monitored using the QIDS-SR16 (Section 8.3.3.8) and appropriate actions will be taken (Section 8.3.3.8), including potential discontinuation of study intervention (Section 7.1.1).

Study site personnel should monitor participants receiving study intervention for depression or any other unusual changes in behavior throughout the study.

8.2.9. Hepatic Safety Monitoring, Evaluation, and Criteria for Study Intervention Interruption or Discontinuation

The following tables summarize actions to take based on abnormal hepatic laboratory or clinical changes.

This table summarizes actions to take based on abnormal hepatic laboratory or clinical changes in participants with normal or near-normal baseline (ALT, AST, or ALP <1.5x ULN).

Participants with normal or near normal baseline (ALT, AST, or ALP <1.5x ULN)

If this laboratory value is observed...	Then...		
	initiate or continue close hepatic monitoring	initiate comprehensive evaluation	interrupt or discontinue study intervention
ALT or AST $\geq 3x$ ULN	X		
ALP $\geq 2x$ ULN	X		
TBL $\geq 2x$ ULN ^a	X		
ALT or AST $\geq 5x$ ULN	X	X	
ALP $\geq 2.5x$ ULN	X	X	
ALT or AST $\geq 3x$ ULN with hepatic signs or symptoms ^b	X	X	X
ALT or AST $\geq 5x$ ULN for more than 2 weeks	X	X	X
ALT or AST $\geq 8x$ ULN	X	X	X
ALT or AST $\geq 3x$ ULN and TBL $\geq 2x$ ULN ^a or INR ≥ 1.5	X	X	X
ALP $\geq 3x$ ULN	X	X	X
ALP $\geq 2.5x$ ULN and TBL $\geq 2x$ ULN ^a	X	X	X
ALP $\geq 2.5x$ ULN with hepatic signs or symptoms ^b	X	X	X

Abbreviations: ALP = alkaline phosphatase; ALT = alanine transferase; AST = aspartate aminotransferase;
 INR = international normalized ratio; TBL = total bilirubin; ULN = upper limit of normal.

^a In participants with Gilbert's syndrome, the threshold for TBL will be TBL $\geq 3x$ ULN, if baseline is <1.5x ULN, and TBL $\geq 2x$ baseline, if baseline is $\geq 1.5x$ ULN.

^b Examples of hepatic signs or symptoms: severe fatigue, nausea, vomiting, right upper quadrant abdominal pain, fever, rash, and/or eosinophilia >5%.

Participants with elevated baseline (ALT, AST, or ALP $\geq 1.5x$ ULN)

If this laboratory value is observed...	Then...		
	Initiate or continue close hepatic monitoring	Initiate comprehensive evaluation	Interrupt or discontinue study drug
ALT or AST $\geq 2x$ baseline	X		
ALP $\geq 2x$ baseline	X		
TBL $\geq 2x$ ULN ^a	X		
ALT or AST $\geq 3x$ baseline or ≥ 250 U/L (whichever occurs first)	X	X	
ALP $\geq 2.5x$ baseline	X	X	
ALT or AST $\geq 2x$ baseline or ≥ 250 U/L (whichever occurs first) with hepatic signs or symptoms ^b	X	X	X
ALT or AST $\geq 3x$ baseline or ≥ 250 U/L (whichever occurs first) for more than 2 weeks	X	X	X
ALT or AST $\geq 4x$ baseline or ≥ 400 U/L (whichever occurs first)	X	X	X
ALT or AST $\geq 2x$ baseline or ≥ 250 U/L	X	X	X

If this laboratory value is observed...	Then...		
(whichever occurs first) and TBL $\geq 2x$ ULN ^a or INR ≥ 1.5			
ALP $\geq 3x$ baseline	X	X	X
ALP $\geq 2.5x$ baseline and TBL $\geq 2x$ ULN ^a	X	X	X
ALP $\geq 2.5x$ baseline with hepatic signs or symptoms ^b	X	X	X

Abbreviations: ALP = alkaline phosphatase; ALT = alanine transferase; AST = aspartate aminotransferase; INR = international normalized ratio; TBL = total bilirubin; ULN = upper limit of normal.

^a In participants with Gilbert's syndrome, the threshold for TBL will be TBL $\geq 3x$ ULN, if baseline is $<1.5x$ ULN, and TBL $\geq 2x$ baseline, if baseline is $\geq 1.5x$ ULN.

^b Examples of hepatic signs or symptoms: severe fatigue, nausea, vomiting, right upper quadrant abdominal pain, fever, rash, and/or eosinophilia $>5\%$.

8.2.9.1. Close Hepatic Monitoring

If a participant develops any one of these changes, initiate close hepatic monitoring.

Participants with normal or near normal baseline (ALT, AST, or ALP $<1.5x$ ULN)	Participants with elevated baseline (ALT, AST, or ALP $\geq 1.5x$ ULN)
ALT or AST $\geq 3x$ ULN or	ALT or AST $\geq 2x$ baseline
ALP $\geq 2x$ ULN or	ALP $\geq 2x$ baseline
TBL $\geq 2x$ ULN ^a	TBL $\geq 2x$ ULN ^a

Abbreviations: ALP = alkaline phosphatase; ALT = alanine transferase; AST = aspartate aminotransferase; TBL = total bilirubin; ULN = upper limit of normal.

^a In participants with Gilbert's syndrome, the threshold for TBL will be TBL $\geq 3x$ ULN, if baseline is $<1.5x$ ULN, and TBL $\geq 2x$ baseline, if baseline is $\geq 1.5x$ ULN.

Close hepatic monitoring should include these actions:

- Laboratory tests((Section 10.5), including ALT, AST, ALP, TBL, D. Bil, GGT, CK, and CBC with differential, should be checked within 48 to 72 hours of the detection of elevated liver tests to confirm the abnormality and to determine if it is increasing or decreasing.
- If the abnormality persists, clinical and laboratory monitoring should continue at a frequency of 2 to 3 times weekly until levels start normalizing or return to approximate baseline values. Subsequently, the frequency of monitoring may be lowered to once every 1 to 2 weeks if the participant's clinical condition and lab results stabilize.
- In addition to lab tests, basic evaluation for possible causes of abnormal liver tests should be initiated by the investigator in consultation with the Lilly-designated medical monitor. At a minimum, this evaluation should include physical examination and a thorough medical history, including current symptoms, recent illnesses (for example, heart failure, systemic infection, hypotension, or seizures), recent travel, concomitant medications (including over-the-counter), herbal and dietary supplements, history of alcohol drinking and other substance abuse.

8.2.9.2. Comprehensive Hepatic Evaluation

If a participant develops any one of the following laboratory or clinical changes, initiate a comprehensive hepatic evaluation:

Participants with normal or near-normal baseline (ALT, AST, or ALP <1.5x ULN)	Participants with elevated baseline (ALT, AST, or ALP ≥1.5x ULN)
ALT or AST ≥5x ULN or	ALT or AST ≥3x baseline or ≥250 U/L (whichever occurs first) or
ALP ≥2.5x ULN or	ALP ≥2.5x baseline or
ALT or AST ≥3x ULN with hepatic signs or symptoms ^a or	ALT or AST ≥2x baseline or ≥250 U/L (whichever occurs first) with hepatic signs or symptoms ^a or
ALT or AST ≥3x ULN and TBL ≥2x ULN ^b or INR ≥1.5	ALT or AST ≥2x baseline or ≥250 U/L (whichever occurs first) and TBL ≥2x ULN ^b or INR ≥1.5

Abbreviations: ALP = alkaline phosphatase; ALT = alanine transferase; AST = aspartate aminotransferase; INR = international normalized ratio; TBL = total bilirubin; ULN = upper limit of normal.

^a Examples of hepatic signs or symptoms: severe fatigue, nausea, vomiting, right upper quadrant abdominal pain, fever, rash, and/or eosinophilia >5%.

^b In participants with Gilbert's syndrome, the threshold for TBL will be TBL ≥3x ULN, if baseline is <1.5x ULN, and TBL ≥2x baseline, if baseline is ≥1.5x ULN.

Comprehensive hepatic evaluation should include these actions:

- Clinical and laboratory monitoring should continue at a frequency of 2 to 3 times weekly until levels start normalizing or return to approximate baseline values.
- At a minimum, comprehensive hepatic evaluation should include physical examination and a thorough medical history, as outlined above, as well as tests for PT-INR; tests for viral hepatitis A, B, C, and E; tests for autoimmune hepatitis; and an abdominal imaging study (for example, ultrasound or CT scan). Remaining samples collected for hepatic evaluation may be stored as an exploratory biomarker samples. These stored samples will be used for the limited purpose of additional hepatic evaluations for up to 15 years.
- Based on the participant's history and initial results, further testing should be considered in consultation with the Lilly-designated medical monitor, including tests for hepatitis D virus, cytomegalovirus, Epstein-Barr virus, acetaminophen levels, acetaminophen protein adducts, urine toxicology screen, Wilson's disease, blood alcohol levels, urinary ethyl glucuronide, and blood phosphatidylethanol (PEth).
- For participants with isolated ALP elevation ≥2.5x ULN without GGT elevation, consider testing for ALP isoenzymes and evaluating for extrahepatic sources, such as bone disease (e.g., fractures, bone metastases, osteomalacia, Paget's disease), or pregnancy.
- Based on the circumstances and the investigator's assessment of the participant's clinical condition, the investigator should consider referring the participant for a hepatologist or gastroenterologist consultation and additional tests, including magnetic resonance cholangiopancreatograph, endoscopic retrograde cholangiopancreatograph, cardiac echocardiogram, or a liver biopsy.
- All the medical information and test results related to the hepatic monitoring and comprehensive hepatic evaluation should be collected and recorded in CRFs, including the hepatic safety CRF.

8.2.9.3. Study Drug Interruption or Discontinuation

If a participant develops any one of the following laboratory or clinical changes, interrupt the study drug and continue close monitoring and comprehensive hepatic evaluation as described in Sections 8.2.9.1 and 8.2.9.2.

Participants with normal or near normal baseline (ALT, AST, or ALP <1.5x ULN)	Participants with elevated baseline (ALT, AST, or ALP ≥1.5x ULN)
ALT or AST ≥3x ULN with hepatic signs or symptoms ^a or	ALT or AST ≥2x baseline or ≥250 U/L (whichever occurs first) with hepatic signs or symptoms ^a or
ALT or AST ≥5x ULN for more than 2 weeks or	ALT or AST ≥3x baseline or ≥250 U/L (whichever occurs first) for more than 2 weeks or
ALT or AST ≥8x ULN or	ALT or AST ≥4x baseline or ≥400 U/L (whichever occurs first) or
ALT or AST ≥3x ULN and TBL ≥2x ULN ^b or INR ≥ 1.5 or	ALT or AST ≥2x baseline or ≥250 U/L (whichever occurs first) and TBL ≥2x ULN ^b or INR ≥ 1.5 or
ALP ≥3x ULN or	ALP ≥3x baseline or
ALP ≥2.5x ULN and TBL ≥2x ULN ^b or	ALP ≥2.5x baseline and TBL ≥2x ULN ^b or
ALP ≥2.5x ULN with hepatic signs or symptoms ^a	ALP ≥2.5x baseline with hepatic signs or symptoms ^a

Abbreviations: ALP = alkaline phosphatase; ALT = alanine transferase; AST = aspartate aminotransferase; INR = international normalized ratio; TBL = total bilirubin; ULN = upper limit of normal.

^a Examples of hepatic signs or symptoms: severe fatigue, nausea, vomiting, right upper quadrant abdominal pain, fever, rash, and/or eosinophilia >5%.

^b In participants with Gilbert's syndrome, the threshold for TBL will be TBL ≥3x ULN, if baseline is <1.5x ULN, and TBL ≥2x baseline, if baseline is ≥1.5x ULN.

Interruption or discontinuation of study drug should include these actions:

- While the participant is not receiving the study drug, clinical and laboratory monitoring should continue at a frequency of 1 to 3 times weekly until liver tests normalize or return to approximate baseline values.
- If the hepatic event continues past the anticipated end of the study (that is, data lock) the investigator should consult with the sponsor-designated medical monitor to determine the need for further data collection beyond the end date of the study (that is, data lock date).
- All the medical information and tests results related to the close hepatic monitoring and comprehensive hepatic evaluation should be collected and recorded in a hepatic safety CRF.
- Resumption of the study drug after interruption for a hepatic reason can be considered only in consultation with the sponsor designated medical monitor and only if the liver test results returned to near baseline and if a self-limited non-study-drug etiology is identified.
- Otherwise, both study drugs should be permanently discontinued.

8.2.10. Tuberculosis Testing and Monitoring

Screening

During screening, all participants are to be assessed for risk factors, symptoms, and signs of TB with all of the following:

- thorough history to determine the lifetime risk factors for TB infection, for TB progression, and for symptoms and/or signs of active TB, and
- signs of previous or active TB by means of
 - thorough physical examination for signs of active TB, including measurement of body temperature (Section 8.2.2) and assessment of peripheral lymph nodes (Section 8.2.1), and
 - a high-quality chest x-ray (posterior-anterior view, including a lateral view if needed) or if a participant has a positive TB test, interpreted and reported by a radiologist or pulmonologist (Section 8.2.4).

All participants with no history of LTBI or active TB, and no history of positive Mantoux TST using PPD or positive *M. tuberculosis* IGRA must have one of the following tests:

- PPD TST, or
- IGRA for *M. tuberculosis*.

Retesting

One retest is allowed for participants with an “indeterminate” QuantiFERON-TB Gold assay or “borderline” T-SPOT.TB assay. Participants with 2 indeterminate QuantiFERON-TB Gold assays or 2 borderline T-SPOT.TB assays will be excluded (Section 5.4.1).

Diagnosed LTBI

Participants diagnosed with LTBI are excluded (Section 5.2) unless they are candidates for LTBI treatment, are being treated for LTBI, and the following criteria are met:

- After receiving at least 4 weeks of appropriate LTBI therapy (as per the WHO or US CDC guidelines), there is no evidence of hepatotoxicity (ALT/AST must remain ≤ 2 times ULN) or other treatment intolerance. In this case, the participant may be rescreened (Section 5.4.2) and is not excluded due to LTBI.
- The participant must continue and complete appropriate LTBI therapy to remain eligible to continue receiving study intervention (Section 7.1).

Monitoring during the study

For all participants, monitoring for TB is to be continuous throughout the study. At a minimum, each participant is to have the following documented at least every 3 months:

- thorough history to determine any risk factors for TB infection and for TB progression, and symptoms or signs of active TB, and
- thorough physical examination for signs of active TB, including measurement of body temperature and assessment of peripheral lymph nodes (Sections 8.2.2 and 8.2.1).

8.2.11. Hepatitis B Testing and Monitoring

As specified in the SoA (Section 1.3), initial testing for HBV infection includes HBsAg and anti-HBc.

- If HBsAg is positive, the participant is excluded.
- If HBsAg is negative and anti-HBc is negative, the participant is not excluded.
- If HBsAg is negative and anti-HBc is positive, further testing for HBV DNA is required.
 - If the screening HBV DNA is positive, the participant is excluded.
 - If the screening HBV DNA is negative, the participant is not excluded. Repeat testing for HBV DNA is required at least every 3 months during the study.

Management of participants with treatment-emergent detectable HBV DNA during the study

Follow the guidance in these tables:

<i>Management of participants with previously undetectable HBV DNA</i>	
If ...	Then...
a participant with previously undetectable HBV DNA has treatment-emergent <i>detectable but not quantifiable</i> HBV DNA	• Withhold mirikizumab. • Repeat HBV DNA, HBsAg, ALT, AST, TBL within 2 weeks and again 2 weeks later. • If the repeat HBV DNA is <i>undetectable</i> or <i>detectable but not quantifiable</i> on 2 consecutive tests, and the participant remains asymptomatic^a with no elevation in liver tests^b, resume mirikizumab with close monitoring of HBV DNA and liver tests. • Wait for the results of both HBV DNA tests before making a decision on resuming mirikizumab. • Monitor HBV DNA, ALT, AST, TBL at least once monthly in the first 3 months after the detectable HBV DNA result and every 2-3 months thereafter. • If one of the subsequent HBV DNA results again becomes <i>detectable and not quantifiable</i>, but the participant remains asymptomatic with normal liver tests, continue mirikizumab, and increase monitoring frequency of HBV DNA, ALT, AST, TBL to once monthly. • If one of the repeat HBV DNA results is <i>detectable and quantifiable</i>, follow the instructions below, for participants with

Management of participants with previously undetectable HBV DNA	
	treatment-emergent <i>detectable and quantifiable</i> HBV DNA.
a participant with previously undetectable HBV DNA has treatment-emergent <i>detectable and quantifiable</i> HBV DNA:	
If the participant is asymptomatic^a and liver tests are not elevated^b:	- Withhold mirikizumab. - Repeat HBV DNA, HBsAg, ALT, AST, TBL within 2 weeks. - Inform the sponsor designated medical monitor. - Refer to a hepatologist^c for further management including initiation of a nucleoside/nucleotide analog (NA). - Continue monitoring HBV DNA, HBsAg, ALT, AST, TBL every 2 weeks. If HBV DNA remains undetectable in 3 consecutive tests, and liver tests are not elevated^b, monitoring frequency can decrease to once every 1-2 months. - If NA was initiated, and HBV DNA remains undetectable in 3 consecutive tests, monitoring frequency can be decreased to once every 2-3 months.
If the participant is symptomatic^a, or if ALT, AST, or TBL are elevated^b:	- Withhold mirikizumab. - Repeat HBsAg, ALT, AST, TBL within 48-72 hours to determine if the levels are increasing or decreasing. - Continue monitoring liver tests 1-3 times weekly according to the protocol requirements for treatment-emergent abnormal liver tests. - Inform the sponsor-designated medical monitor. - Refer to a hepatologist^c for further management including initiation of a nucleoside/ nucleotide analog (NA). - Continue monitoring HBV DNA, HBsAg, every 2 weeks. If liver tests are back to baseline, and HBV DNA remains undetectable in 3 consecutive tests, monitoring frequency can decrease to once monthly. - If NA was initiated, liver tests are back to baseline, and HBV DNA remains undetectable in 3 consecutive tests, monitoring frequency can be decreased to once every 2-3 months.

<i>Resumption of mirikizumab in participants with treatment-emergent detectable and quantifiable HBV DNA</i>	
If...	Then...
All of these criteria are met • participants are asymptomatic^a without elevated^b liver tests and • the sponsor-designated medical monitor has been consulted and • participant was seen by a hepatologist^c and NA has been initiated.	Resume mirikizumab 7 days after NA initiation. Notes: • If NA was initiated and mirikizumab was resumed, monitor HBV DNA, HBsAg, ALT, AST, TBL at least monthly. • After HBV DNA is undetectable in 3 consecutive tests, monitoring frequency can be decreased to every 2-3 months.
<i>Duration of HBV DNA monitoring and NA prophylaxis in participant with treatment-emergent detectable and quantifiable HBV DNA</i>	
• HBV DNA monitoring should continue for at least 6 months after discontinuation of mirikizumab. • If NA was initiated and mirikizumab was resumed, NA treatment and HBV DNA monitoring should continue for at least 6 months after discontinuation of mirikizumab. Note: After completion of the study monitoring, refer the participant to a hepatologist^c for continued follow-up and management.	

<i>Permanent discontinuation of mirikizumab in participant with treatment-emergent detectable and quantifiable HBV DNA</i>	
If any of the following occurs...	Then...
• HBV DNA is detectable and quantifiable with concurrent elevation^b of ALT, AST, or TBL. • HBV DNA is detectable and quantifiable with concurrent symptoms^a that are considered by the investigator to be related to HBV reactivation. • HBV DNA viral load is >100 IU/mL on 2 or more tests.	Permanently discontinue mirikizumab.

Abbreviations: ALT = alanine aminotransferase; AST = aspartate aminotransferase; HBsAg = hepatitis B surface antigen; HBV = hepatitis B virus; NA = nucleoside/nucleotide analog; TBL = total bilirubin level; ULN = upper limit of normal.

^a **Symptoms** may include worsening fatigue, nausea, vomiting, abdominal discomfort, jaundice, fever, or rash.

^b **ALT, AST, or TBL elevation** refers to elevations $\geq 2x$ ULN in participants with normal baseline or near normal baseline (that is, $<1.5x$ ULN) or elevation $\geq 2x$ baseline in participants with elevated baseline (that is, $\geq 1.5x$ ULN). . In participants with Gilbert's syndrome, the threshold for TBL will be TBL $\geq 3x$ ULN, if baseline is $<1.5x$ ULN, and TBL $\geq 2x$ baseline, if baseline is $\geq 1.5x$ ULN.

^c **A hepatologist** may include a health care provider with recognized expertise in the assessment and management of viral hepatitis.

8.2.12. Hepatitis C Testing

As specified in the SoA (Section 1.3), initial testing for HCV infection includes testing for anti-HCV.

- If anti-HCV is positive, a test for circulating HCV RNA is required.
- If HCV RNA test is negative, the participant is not excluded.
- If HCV RNA test is positive, the participant is excluded (see Section 5.2).

Participants who have had HCV infection and have been successfully treated, defined as a sustained virologic response (HCV RNA by PCR negative for at least 24 weeks following treatment completion), are not excluded on the basis of HCV as long as HCV RNA test is negative at screening.

If HCV RNA is detected during the study, the study intervention will be discontinued (Section 7.1.1), and the participant should receive appropriate follow-up medical care.

8.2.13. Malnutrition

Malnutrition is a term broadly used to describe deficiencies or excesses in the intake of macronutrients, micronutrients, or energy, as well as the physiological changes that can result.

As stated in Section 5.3.3, medical staff should monitor participants for signs and symptoms of malnutrition. A table of signs and symptoms associated with common nutrient deficiencies is included in Section 10.7. The nutritional panel described in Section 10.2.2 should be collected if

- the PI suspects the participant may have malnutrition, for example, caloric intake is significantly reduced to below 800 to 1200 kcal
- a participant's BMI falls below 22 kg/m², or
- in participants with T2DM, weight loss is >20% or >4 kg/month.

A clinical assessment of dietary intake and nutritional status should also be completed.

Nutritional guidance and, as needed, supplementation, should be considered. For severe or persistent cases, consider discontinuation of tirzepatide (Section 7.1.1), and discuss management of the participant with the medical monitor. The treatment of deficiencies should be managed in line with local clinical guidelines. If additional guidance is required, reach out to the sponsor-designated medical monitor.

8.2.14. Gallbladder Disease and Tirzepatide

Gallbladder problems including gallstones have been reported uncommonly in patients taking tirzepatide and GLP-1 receptor agonists. In the tirzepatide SURMOUNT trials, cholelithiasis was reported in 1.1%, cholecystitis was reported in 0.7%, and cholecystectomy was reported in 0.2% of tirzepatide-treated patients. Risk may be increased if patients have diabetes or obesity and in cases of rapid weight loss. Some gallbladder problems may need surgery. To minimize the risk of gallbladder problems, investigators should aim for gradual weight loss of 1 to 2 pounds per week from tirzepatide in participants, monitor participants for signs of gallbladder disease (such as severe upper abdominal pain), and advise participants to avoid a very high-fat diet, fried or greasy foods.

8.3. Adverse Events, Serious Adverse Events, and Product Complaints

The definitions of the following events can be found in Section 10.3:

- AEs
- SAEs, and
- product complaints.

These events will be reported by the participant, or, when appropriate, by a caregiver or surrogate.

The investigator and any qualified designees are responsible for detecting, documenting, and recording events that meet these definitions and remain responsible for following up events that are serious, considered related to the study intervention or study procedures, or that caused the participant to discontinue the study (see Section 7).

Care will be taken not to introduce bias when detecting events. Open-ended and non-leading verbal questioning of the participant is the preferred method to inquire about event occurrences.

After the initial report, the investigator is required to proactively follow each participant at subsequent visits or contacts. All SAEs and AEs of special interest (as defined in Section 8.3.3)

will be followed until resolution, stabilization, the event is otherwise explained, or the participant is lost to follow-up (as defined in Section [7.3](#)).

For product complaints, the investigator is responsible for ensuring that follow-up includes any supplemental investigations as indicated to elucidate the nature or causality. Further information on follow-up procedures is provided in Section [10.3](#).

8.3.1. Timing and Mechanism for Collecting Events

This table describes the timing, deadlines, and mechanism for collecting events.

Event	Collection Start	Collection Stop	Timing for Reporting to Sponsor or Designee	Mechanism for Reporting	Back-up Method of Reporting
Adverse Event					
AE	Signing of the ICF	Participation in study has ended	As soon as possible upon site awareness	AE CRF	N/A
Serious Adverse Event					
SAE and SAE updates – prior to start of study intervention and deemed reasonably possibly related to study procedures	Signing of the ICF	Start of intervention	Within 24 hours of awareness	SAE CRF	SAE paper form
SAE and SAE updates – after start of study intervention	Start of intervention	Participation in study has ended	Within 24 hours of awareness	SAE CRF	SAE paper form
SAE ^a – after participant’s study participation has ended and the investigator becomes aware	After participant’s study participation has ended	N/A	Promptly	SAE paper form	N/A
Pregnancy					
Pregnancy in participants and partners of participants	After the start of study intervention	10 weeks after the last study intervention for individuals AFAB and AMAB	Within 24 hours (see Section 8.3.2)	Pregnancy CRF	Pregnancy paper form
Product Complaints					
PC associated with an SAE or might have led to an SAE	Start of study intervention	End of study intervention	Within 24 hours of awareness	Product Complaint form	N/A

Event	Collection Start	Collection Stop	Timing for Reporting to Sponsor or Designee	Mechanism for Reporting	Back-up Method of Reporting
PC not associated with an SAE	Start of study intervention	End of study intervention	Within 1 business day of awareness	Product Complaint form	N/A
Updated PC information	—	—	As soon as possible upon site awareness	Originally completed Product Complaint form with all changes signed and dated by the investigator	N/A
PC (if investigator becomes aware)	Participation in study has ended	N/A	Promptly	Product Complaint form	

Abbreviations: AE = adverse event; AFAB = assigned female at birth; AMAB = assigned male at birth; CRF = case report form; ICF = informed consent form; N/A = not applicable; PC = product complaint; SAE = serious adverse event.

^a Serious adverse events should not be reported unless the investigator deems them to be possibly related to study treatment or study participation.

8.3.2. Collection of Pregnancy Information

8.3.2.1. Participants Who Become Pregnant

The investigator will collect pregnancy information on any participant who becomes pregnant while participating in this study. The initial information will be recorded on the appropriate form and submitted to the sponsor within 24 hours of learning of a participant's pregnancy.

The participant will be followed to determine the outcome of the pregnancy. The investigator will collect follow-up information on the participant and the neonate and the information will be forwarded to the sponsor. Follow-up information for pregnancy outcomes for the participant and neonate will be collected until termination of the pregnancy or for at least 12 months post-delivery. Any termination of pregnancy will be reported, regardless of gestational age, fetal status (presence or absence of anomalies) or indication for the procedure.

While pregnancy itself is not considered to be an AE or SAE, any pregnancy complication or elective termination of a pregnancy for medical reasons will be reported as an AE or SAE.

A spontaneous abortion (occurring at <20 weeks gestational age) or still birth (occurring at ≥20 weeks gestational age) is always considered to be an SAE and will be reported as such.

Any post-study pregnancy related SAE considered reasonably related to the study intervention by the investigator will be reported to the sponsor as described in protocol Section 8.3.1. While the investigator is not obligated to actively seek this information in former study participants, the investigator may learn of an SAE through spontaneous reporting.

Any participant who becomes pregnant while participating in the study will discontinue study intervention and be withdrawn from the study. If the participant is discontinued from the study, follow the standard discontinuation process and continue directly to the follow-up phase. The

follow-up on the pregnancy outcome should continue independent of intervention or study discontinuation.

8.3.2.2. Participants With Partners Who Become Pregnant

When to collect pregnancy information

In most circumstances, the investigator will attempt to collect pregnancy information from a participant's partner who becomes pregnant while the participant is in this study.

After learning about a pregnancy in the partner of a study participant, the investigator

- will obtain a consent to release information from the pregnant partner directly, and
- within 24 hours after obtaining this consent, will record pregnancy information on the appropriate form and submit it to the sponsor.

The partner will be followed to determine the outcome of the pregnancy. Information on the status of the mother and neonate will be forwarded to the sponsor. Generally, the follow-up will be no longer than 6 to 8 weeks after the estimated delivery date. Any termination of the pregnancy will be reported regardless of gestational age, fetal status (presence or absence of anomalies) or indication for the procedure.

When not to collect pregnancy information

It is not necessary to collect information about a pregnancy in the partner of a study participant in these circumstances

- the partner of the study participant was not exposed to the study intervention via semen during conception or pregnancy, or
- the participant did not contribute the spermatozoid or ova that resulted in the pregnancy.

8.3.3. Adverse Events of Special Interest

AESIs for this program include:

- hepatic safety
- infections, including opportunistic infections and serious infections
- hypersensitivity reactions
- injection/infusion site reactions
- major adverse cardiovascular events-3
- malignancies
- suicidal ideation and behavior and depression
- hypoglycemia for participants with T2DM at baseline
- serious or severe GI AEs
- pancreatitis, and
- acute kidney injury.

If any of these events are reported, sites will be prompted to collect additional details and data.

8.3.3.1. Hepatic Safety

See Section 8.2.9.

8.3.3.2. Infections, Including Opportunistic Infections and Serious Infections

Completion of the Infection CRF page is required for each infection reported as an AE or SAE. The sponsor will identify infections considered to be opportunistic based on the article by Winthrop et al. (2015; Section 10.10).

8.3.3.3. Hypersensitivity Reactions

Many drugs, including oral agents and biologic agents, carry the risk of systemic hypersensitivity reactions. If such a reaction occurs, additional data should be provided to the sponsor in the designated CRFs.

Sites should have appropriately trained medical staff and appropriate medical equipment available when study participants are receiving study intervention. It is recommended that participants who experience a systemic hypersensitivity reaction be treated per national and international guidelines.

In the case of a suspected systemic hypersensitivity event, additional blood samples should be collected as described in Section 10.2.1. Laboratory results are provided to the sponsor via the central laboratory.

8.3.3.4. Injection/Infusion Site Reactions (ISRs)

Symptoms and signs of a local ISR may include

- erythema
- induration
- pain
- pruritus, and
- edema.

Prespecified ISR assessments will be performed and collected at the time points indicated in the SoA.

In addition, if a spontaneous ISR is reported by a participant or site staff at a nonprespecified time point, the ISR CRF will be used to capture additional information about this reaction, for example,

- injection site pain
- degree and area of erythema
- induration
- pruritus, and
- edema.

8.3.3.5. Major Adverse Cardiovascular Events-3

Potential MACE-3 (cardiovascular death, nonfatal MI, and nonfatal stroke) will be identified by the investigative site or by a medical review conducted by the sponsor or designee. Additional

data about each identified potential event will be provided to the sponsor via specific event endpoint forms. A blinded external CEC (Section 10.1.5.2) will adjudicate the events in a consistent and unbiased manner.

8.3.3.6. Malignancies

Participants will be monitored for symptoms and signs of malignancy after administration of study intervention. All AEs and SAEs will be recorded as described in Section 8.3.

8.3.3.7. Suicidal Ideation or Behavior

Before administering the C-SSRS, study site personnel will question the participant about any change in the preexisting condition(s) and the occurrence and nature of any AEs.

Columbia-Suicide Severity Rating Scale

The C-SSRS captures the occurrence, severity, and frequency of SIB during the assessment period via a semi-structured interview by an appropriately trained healthcare professional.

For this study, the C-SSRS is adapted for the assessment of the ideation and behavior categories only. The Intensity of Ideation and Lethality of Behavior sections are removed.

Timing of collection and AE monitoring

Nonleading (spontaneous) AE collection should occur prior to the collection of the C-SSRS.

Only report a suicide-related event discovered during collection of the C-SSRS on the AE form if it is an AE that leads to discontinuation or an SAE.

Follow standard procedures for reporting SAEs.

8.3.3.8. Depression

Participants will be monitored for depression by using the QIDS-SR16.

If a participant has a QIDS-SR16 score ≥ 16 on the total score or ≥ 2 on Item 12, the investigator may decide to withhold study intervention or discontinue study intervention (Section 7.1.1). A mental health care professional may assist in the decision.

Quick Inventory for Depressive Symptomatology – Self-Report 16 items (QIDS-SR16)

The QIDS-SR16 (Rush et.al. 2000; Reilly et.al. 2015) is a validated, participant-reported instrument designed to assess the diagnostic symptoms for major depressive illness as defined in the Diagnostic and Statistical Manual of Mental Disorders, 5th Edition (APA [WWW]). The QIDS-SR16 also assesses the directionality of the symptom disturbance and considers that some criteria could be met by 1 of 2 symptoms. This scale is validated for use in individuals 18 years of age and older.

The questionnaire assesses the previous week.

Sixteen questions rate the 9 diagnostic symptom domains, with scores ranging from 0 to 3. Higher numbers indicate greater dysfunction.

Diagnostic symptoms	Number of Questions
Mood	1
Anhedonia	1
Appetite and weight disturbance	4
Sleep disturbance	4
Feelings of worthlessness or guilt	1
Psychomotor agitation or retardation	2
Loss of energy	1
Diminished concentration	1
Suicidal thoughts or attempts	1

For domains that have more than 1 question, the highest score of that particular domain is taken.

This table describes the interpretation of results.

Total Score	Interpretation of Depression
0-5	None
6-10	Mild
11-15	Moderate
16-20	Severe
21-27	Very severe

8.3.3.9. Hypoglycemia for Participants with T2DM at Baseline

Participants will be trained about signs and symptoms of hypoglycemia, how to treat hypoglycemia, how to report as AEs, and/or referred to healthcare provider with appropriate expertise per local standard of care. Hypoglycemia is defined as any time a patient feels that he or she is experiencing symptoms and/or signs associated with hypoglycemia and has a blood glucose level of <70 mg/dL (≤ 3.9 mmol/L). Participants with T2DM should check their fasting glucose daily and report any blood glucose level of <70 mg/dL (≤ 3.9 mmol/L) to the investigator.

Incidence of hypoglycemic events with tirzepatide

In line with the ability of tirzepatide to lower blood glucose in a glucose-dependent manner, the overall incidence of clinically significant or severe hypoglycemia attributable to tirzepatide was low. The risk of clinically significant hypoglycemia or severe hypoglycemia was higher when tirzepatide was used in combination with insulin glargine or sulfonylurea, as has been observed with the GLP-1 receptor agonist class (Del Prato et al. 2021; Frías et al. 2021; Ludvik et al. 2021; Rosenstock et al. 2021; Dahl et al. 2022).

Participants who develop persistent or recurrent unexplained hypoglycemia during the treatment period may be asked to reduce the dose or discontinue any concomitant antihyperglycemic medications commonly associated with hypoglycemia according to local standard of care or their primary care physician or healthcare provider managing diabetes mellitus.

Severe hypoglycemia

A severe hypoglycemic event is characterized by altered mental and/or physical status functioning requiring the assistance of another person to actively administer carbohydrate, glucagon, or other resuscitative actions for the treatment of hypoglycemia, irrespective of glucose level.

The determination of an episode of severe hypoglycemia is made by the investigator based on the medical need of the participant to have required assistance and is not based on the report of a participant simply having received assistance.

Examples of severe hypoglycemia in adults are

- altered mental status and the inability to assist in their own care
- semiconscious or unconscious, or
- coma with or without seizures.

Glucose measurements may not be available during such an event, but neurological recovery attributable to the restoration of glucose concentration to normal is considered sufficient evidence that the event was induced by a low glucose concentration.

Reporting of hypoglycemic events

All hypoglycemic events should be reported as an AE.

Severe hypoglycemic event

If a hypoglycemic event meets the criteria of severe hypoglycemia, the investigator should report this as an SAE.

The investigator should also determine if repeated or prolonged episodes of hypoglycemia occurred prior to the severe event.

8.3.3.10. Serious or Severe Gastrointestinal Adverse Events

Tirzepatide has been associated with GI AEs, such as nausea, vomiting, and diarrhea. Information about GI AEs, as well as antiemetic or antidiarrheal use, will be collected in the AE and concomitant medications CRFs, respectively. For detailed information concerning the management of GI AEs, please refer to Section [6.6.2.1](#).

8.3.3.11. Pancreatitis**Diagnosis of acute pancreatitis**

Acute pancreatitis is an AE of interest in all studies with tirzepatide, including this study. The diagnosis of acute pancreatitis requires 2 of the following 3 features (Banks and Freeman 2006, Koizumi et al. 2006):

- abdominal pain, characteristic of acute pancreatitis, that is, epigastric pain radiating to the back, often associated with nausea and vomiting
- serum amylase (total, pancreatic, or both) and/or lipase ≥ 3 times ULN, and
- characteristic findings of acute pancreatitis on CT scan or MRI.

If acute pancreatitis is suspected, the investigator should

- obtain appropriate laboratory tests, including pancreatic amylase (p-amylase) and lipase
 - perform imaging studies, such as abdominal CT scan with or without contrast, or abdominal MRI
- Note:** Abdominal ultrasound may be used as an alternative method only if CT and MRI cannot be performed, and
- evaluate for possible causes of acute pancreatitis, including alcohol use, gallstone/gall bladder disease, hypertriglyceridemia, and concomitant medications.

Discontinuation for acute pancreatitis

If acute pancreatitis is diagnosed, the participant must discontinue use of the study interventions.

Case adjudication and data entry

An independent CEC (Section 10.1.5.2) will adjudicate all suspected cases of acute pancreatitis.

Asymptomatic elevation of serum amylase and/or lipase

Serial measures of pancreatic enzymes have limited clinical value for predicting episodes of acute pancreatitis in asymptomatic patients (Nauck et al. 2016; Steinberg et al. 2017a, 2017b). Therefore, further diagnostic follow-up of cases of asymptomatic elevation of pancreatic enzymes (lipase and/or p-amylase ≥ 3 times ULN) is not mandated but may be performed based on the investigator's clinical judgment and assessment of the participant's overall clinical condition.

8.3.3.12. Acute Kidney Injury

Renal safety will be assessed based on repeated renal functional assessment as well as assessment of AEs suggestive of acute or worsening of chronic renal failure. GI AEs have been reported with tirzepatide, including nausea, diarrhea, and vomiting. This is consistent with other GLP-1R agonists (Aroda and Ratner 2011). These events may lead to dehydration, which could cause a deterioration in renal function, including acute renal failure. Participants should be advised to notify investigators in case of severe nausea, frequent vomiting, or symptoms of dehydration.

8.4. Pharmacokinetics

At the visits and times specified in the SoA, PK blood samples will be collected for all participants.

Instructions for the collection and handling of blood samples will be provided by the sponsor.

The actual date and time (24-hour clock time) of each sampling will be recorded.

In the case of hypersensitivity or ISR, PK samples will be collected, if needed as described in Section 10.2.1.

PK sample retention is described in Section 10.1.12.

8.5. Pharmacodynamics

Refer to Section 10.2 and the SoA (Section 1.3) for fecal calprotectin and serum hsCRP sample collection information. Sample retention is described in Section 10.1.12.

8.6. Genetics

A whole blood sample will be collected for pharmacogenetic analysis where local regulations allow, according to the SoA (Section 1.3) and following procedures detailed in the laboratory manual. See Section 10.13 for information regarding genetic research and Section 10.1.12 for details about sample retention and custody.

8.7. Biomarkers

Blood, stool supernatant, and tissue samples for biomarker research will be collected at the times specified in the SoA (Section 1.3) and as detailed in the laboratory manual.

These samples may be used for research to understand the

- drug target
- disease process
- variable response to treatment with mirikizumab and placebo or mirikizumab concomitantly administered with tirzepatide
- pathways associated with UC
- mechanism of action of mirikizumab and placebo or mirikizumab administered concomitantly with tirzepatide
- research methods in validating diagnostic tools or assays related to UC or to mirikizumab and placebo or mirikizumab concomitantly administered with tirzepatide.

Samples will be coded by the participant number. These samples and any data generated can be linked back to the participant only by the investigator site personnel.

The purpose of retention, the maximum duration of retention, and facility for long-term storage is detailed in Section 10.1.12.

8.8. Immunogenicity Assessments

At the times specified in the SoA (Section 1.3), venous blood samples will be collected and stored for potential future analysis to determine antibody production against mirikizumab and tirzepatide. ADA may be further characterized for their ability to neutralize mirikizumab or for cross-reactive binding to native GIP and GLP-1 and their ability to neutralize the activity of tirzepatide and native GIP and GLP-1.

To interpret the results of immunogenicity, a venous blood sample will be collected for PK analysis at the same time points to determine the serum concentration of mirikizumab and plasma concentration of tirzepatide. All samples for immunogenicity should be taken pre-dose when applicable and possible.

TE ADA are defined in Section 9.3.7.2. In the case of hypersensitivity or ISR, samples will be collected, if needed, as described in Section 10.2.1.

See Section 10.1.12 for details about sample retention.

8.9. Medical Resource Utilization and Health Economics

Sites and participants should provide information regarding healthcare visits, including hospitalizations and surgeries for UC, as instructed on the CRF.

9. Statistical Considerations

The first version of SAP will be finalized prior to first unblinding, and it will include a more technical and detailed description of the statistical analyses described in this section. This section is a summary of the planned statistical analyses of the most important endpoints, including primary and key secondary endpoints.

9.1. Statistical Hypotheses

The null hypothesis corresponding to the primary objective is

Mirikizumab and placebo is not different from mirikizumab and concomitantly administered tirzepatide in simultaneously achieving clinical remission and at least 10% weight reduction at Week 52.

The null hypotheses corresponding to the key secondary objective are as follows:

- Mirikizumab and placebo is not different from mirikizumab and concomitantly administered tirzepatide in achieving clinical remission at Week 52.
- Mirikizumab and placebo is not different from mirikizumab and concomitantly administered tirzepatide in achieving endoscopic response at Week 52.
- Mirikizumab and placebo is not different from mirikizumab and concomitantly administered tirzepatide in achieving histologic-endoscopic mucosal improvement plus absence of neutrophils at Week 52.

Operationally, the hypotheses will be evaluated by 2-sided tests (see Section 9.3.1 for details).

9.1.1. Multiplicity Adjustment

The statistical comparisons for the primary efficacy endpoint and the key secondary endpoints will be carried out based on a graphical approach for multiple comparisons (Bretz et al. 2011). The graphical approach is a closed testing procedure; hence, it strongly controls the familywise Type I error rate (2-sided alpha level of 0.05) across the primary and key secondary objectives (Alosh et al. 2014). Details about the graphical approach will be described in the SAP. If additional estimands are defined in SAP, multiplicity adjustments will be done within each estimand.

9.2. Analyses Sets

This table describes the populations that will be used for statistical analyses:

Analysis	Description
Population	
Modified Intent-to-Treat (mITT) population	All randomly assigned participants who receive at least 1 dose of study intervention. Participants will be analyzed according to their randomly assigned intervention at Visit 2 (Week 0).
Safety population	All randomly assigned participants who are exposed to at least 1 dose of any of the study drugs. Participants will be analyzed according to the intervention they actually received.

9.3. Statistical Analyses

9.3.1. General Considerations

Statistical analysis of this study will be the responsibility of the sponsor or its designee. Any change to the data analysis methods described in the protocol will require an amendment ONLY if it changes a principal feature of the protocol.

Handling of missing, unused, and spurious data is addressed prospectively in the overall statistical methods described in the protocol and in the SAP, where appropriate. Adjustments to the planned analyses are described in the final CSR.

Unless otherwise noted, all tests of treatment effects will be conducted at a 2-sided alpha level of 0.05, and the CI will be calculated at 95% 2-sided.

Unless specified otherwise, efficacy will be assessed using the mITT population and safety will be assessed using the safety population. For the efficacy analyses under primary estimand, participants will be included in the analyses according to the planned study intervention. For safety analyses, participants will be included in the analyses according to the study intervention they received.

Summary of data

Continuous data will be summarized in terms of

- mean
- standard deviation
- minimum
- maximum
- median, and
- number of observations.

Categorical data will be summarized as frequency counts and percentage.

Baseline definition

The baseline mMS is calculated from valid daily diary entries obtained prior to endoscopy during the screening period and the endoscopic appearance of the mucosa at this screening endoscopy (Section [8.1.1.1](#)).

For other efficacy, health outcome and safety assessments, baseline will be defined as the last available non-missing measurement on or prior to the date of the first dose of study intervention (expected at Week 0), unless otherwise specified.

Missing data

Details on statistical methods for the remaining missing data after accounting for ICE strategy will be included in the SAP for the efficacy variables.

Treatment comparison

For binary efficacy outcomes, G-computation via a logistic regression working model with treatment group and a prespecified set of covariates will be performed. The prespecified set of covariates include stratification factors from Section 6.3, and details will be specified in the SAP. The unconditional treatment response rate and treatment difference and their 95% CIs will be reported.

For continuous efficacy outcomes with multiple postbaseline time points, MMRM may be used to estimate the treatment difference. The MMRM will include treatment group, baseline value, randomization stratification factors, visit, and the interactions of treatment-by-visit and baseline-by-visit as fixed effects and a participant-level random intercept term and within participant covariance. Further details will be specified in the SAP. The Kenward-Roger method will be used to estimate the degrees of freedom. Type III sums of squares for the least square means will be used for the statistical treatment comparison. The least square mean difference, standard error, p-value, and 95% CI will be reported.

For continuous efficacy and health outcomes with single post-baseline time point, ANCOVA with robust variance will be performed (Ye et al. 2022; FDA 2023). The ANCOVA with robust variance will include treatment group, baseline value, and a prespecified set of other covariates. The prespecified set of covariates include stratification factors from Section 6.3 and details will be specified in the SAP. The estimates of unconditional mean treatment difference, standard error, p-value, and 95% CI will be reported.

Other statistical methods may be used, as appropriate, and details will be described in the SAP.

9.3.2. Primary Endpoints/Estimand Analysis

As stated in Section 3, the primary endpoint is simultaneously achieving clinical remission and at least 10% weight reduction at Week 52.

A detailed definition of the primary estimand is presented in Section 3. The primary estimand will be estimated using G-computation via a logistic regression working model as described in Section 9.3.1. Additional estimand and sensitivity analyses of this primary endpoint may be specified in the SAP.

9.3.3. Secondary Endpoints/Estimands Analysis

As stated in Section 3, the key secondary endpoints are the proportions of participants who achieve the following at Week 52:

- clinical remission
- endoscopic response, and
- histologic-endoscopic mucosal improvement plus absence of neutrophils.

The secondary estimand is the same as the primary estimand, except the endpoint definition. The analyses methods for treatment comparison will follow the same G-computation method as described in Section 9.3.1.

9.3.4. Other Endpoints/Estimands Analysis

In principle, other endpoints will follow the same estimand strategy as primary estimand. Details will be specified in the SAP.

9.3.5. Safety Analyses

Safety analyses will include AEs, SAEs, AESIs, C-SSRS, vital signs, and laboratory analytes, using the Safety Analysis Set (Section 9.2) descriptively summarized by treatment group. Categorical safety measures will be summarized at a minimum, with frequency and percentages. Continuous safety measures will be summarized as mean change by visit. Exposure to study intervention will be calculated for each participant and summarized by treatment group. AEs will be coded according to the MedDRA and summarized by

- SOC
- preferred term
- severity, and
- relationship to the study intervention.

A TEAE is defined as an event that first occurred or worsened in severity after baseline. Treatment-related TEAEs are defined as events that are indicated by the investigator on the CRF to be related to treatment. AESIs will be identified by a standardized MedDRA query or a Lilly-defined MedDRA preferred term listing. TEAEs (all, by maximum severity), treatment-related TEAEs, SAEs (including deaths), AESIs, AEs that lead to treatment discontinuation, and AEs that lead to study discontinuation will be summarized and analyzed by MedDRA SOC and preferred term. All AEs, including preexisting conditions, will be listed by participant, visit, preferred term, treatment group, severity, and relationship to the treatment.

Analysis of C-SSRS data

Suicide-related thoughts and behaviors will be assessed based on responses to the C-SSRS consistent with the C-SSRS Scoring and Data Analysis Guide (C-SSRS [WWW]).

9.3.6. Pharmacokinetic Analyses

Mirikizumab and tirzepatide concentrations will be summarized by scheduled time points using descriptive statistics. Individual concentration versus time data will be presented if appropriate.

Additional analysis involving mirikizumab and tirzepatide concentration may be conducted, as appropriate.

9.3.7. Other Analyses

9.3.7.1. Subgroup Analyses

Subgroup analyses of the primary endpoint and key secondary endpoints will be made to assess consistency of the study intervention effect across the following subgroups:

- sex (AFAB, AMAB)
- BMI ≥ 27 kg/m² to < 30 kg/m² versus ≥ 30 kg/m²
- previous advanced therapy failure status (yes/no), and
- baseline disease activity (mMS: [5-6] or [7-9]).

Additional subgroup analyses may be specified in the SAP.

9.3.7.2. Evaluation of Immunogenicity

The frequency and percentage of participants with preexisting ADA and with TE ADA against mirikizumab or tirzepatide may be reported.

TE ADA are defined as

- treatment-induced ADA - those with a titer 2-fold (1 dilution) greater than the minimum required dilution if no ADA were detected at baseline, or
- treatment-boosted ADA - those with a 4-fold (2 dilutions) increase in titer compared to baseline if ADA were detected at baseline.

For participants with TE ADA against either study intervention, the corresponding distribution of maximum titers may be reported.

The frequency of mirikizumab TE ADA+ participants with neutralizing antibodies against mirikizumab may be reported.

The frequency of tirzepatide TE ADA+ participants with

- cross-reactive antibodies against native GLP-1 or native GIP
- neutralizing antibodies against tirzepatide activity, and
- neutralizing antibodies against native GLP-1 or native GIP

may be reported.

The relationship between the presence of antibodies and the PK parameters and pharmacodynamic response including safety and efficacy may be assessed.

9.4. Interim Analysis

Two planned interim analyses may be conducted. The first interim analysis may be conducted when approximately 50% of the participants complete Week 24 or discontinue the study early. The second interim analysis may be conducted when approximately 50% of the participants complete Week 52 or discontinue from the study.

The intent of the interim analysis is to inform decisions in other clinical programs. There is no intent to modify or stop the current study for futility or efficacy based on interim efficacy results.

Therefore, no alpha spending is allocated for the interim analysis. Additional details related to statistical methods will be described in the SAP.

If performed, only the DMC that is external to Lilly is authorized to evaluate unblinded interim efficacy and safety analyses. Study sites will receive information about interim results only if they need to know for the safety of their participants. For the DMC description, see Section [10.1.5.1](#).

Unblinding details are specified in the unblinding plan section of the SAP or in a separate unblinding plan document.

9.5. Sample Size Determination

Approximately 350 participants will be randomly assigned to mirikizumab and placebo or mirikizumab and tirzepatide used concomitantly, in a 1:1 randomization ratio.

The sample size calculation is based on the multi-component endpoint of clinical remission and at least 10% weight reduction at Week 52. Below are the assumptions used in the sample size calculation.

The assumed proportions of participants achieving $\geq 10\%$ at Week 52 weight loss are as follows:

- Mirikizumab + Placebo arm: 13%, based on the pooled placebo arms from SURMOUNT-1,-2, and -3 studies, and
- Mirikizumab + Tirzepatide arm: 75%, based on the pooled tirzepatide arm from SURMOUNT-1,-2, and -3 studies.

The assumed proportions of participants achieving clinical remission at Week 52 are as follows:

- Mirikizumab + Placebo arm: 39%, based on the LUCENT-1 and LUCENT-2 studies, and
- Mirikizumab + Tirzepatide arm: 54%, based on a 15% delta considered clinically meaningful.

Assuming independence between weight loss and clinical remission, the estimated proportions of participants achieving the multi-component endpoint are as follows:

- Mirikizumab + Placebo arm: 5% ($39\% \times 13\%$), and
- Mirikizumab + Tirzepatide arm: 40% ($54\% \times 75\%$).

Using 2-sided Chi-squared test with an alpha level of 0.05, a study with an overall sample size of 350 participants will have over 99% power to detect a treatment difference between the 2 study groups for the primary endpoint.

Moreover, given the proportions of participants achieving clinical remission at Week 52 are 39% and 54% for mirikizumab and placebo, and mirikizumab and tirzepatide used concomitantly, the study will have over 80% power to detect a treatment difference in the key secondary endpoint of clinical remission between the 2 study groups using 2-sided Chi-square test with an alpha level of 0.05.

10. Supporting Documentation and Operational Considerations

10.1. Appendix 1: Regulatory, Ethical, and Study Oversight Considerations

10.1.1. Regulatory and Ethical Considerations

This study will be conducted in accordance with the protocol and with the following:

- Consensus ethical principles derived from international guidelines including the Declaration of Helsinki and Council for International Organizations of Medical Sciences (CIOMS) International Ethical Guidelines
- Applicable ICH Good Clinical Practice (GCP) Guidelines
- Applicable laws and regulations.

The protocol, protocol amendments, ICF, Investigator Brochure, and other relevant documents, for example, advertisements, must be submitted to an IRB/IEC by the investigator and reviewed and approved by the IRB/IEC before the study is initiated.

Any amendments to the protocol will require IRB/IEC approval before implementation of changes made to the study design, except for changes necessary to eliminate an immediate hazard to study participants.

Protocols and any substantial amendments to the protocol will require health authority approval prior to initiation except for changes necessary to eliminate an immediate hazard to study participants.

The investigator will be responsible for the following

- providing written summaries of the status of the study to the IRB/IEC annually or more frequently in accordance with the requirements, policies, and procedures established by the IRB/IEC
- notifying the IRB/IEC of SAEs or other significant safety findings as required by IRB/IEC procedures
- providing oversight of study conduct for participants under their responsibility and adherence to requirements of 21 Code of Federal Regulations (CFR), ICH guidelines, the IRB/IEC, European regulation 536/2014 for clinical studies (if applicable), and all other applicable local regulations
- reporting to the sponsor or designee significant issues related to participant safety, participant rights, or data integrity.

Investigator sites are compensated for participation in the study as detailed in the clinical trial agreement.

10.1.2. Financial Disclosure

Investigators and sub-investigators will provide the sponsor with sufficient, accurate financial information as requested to allow the sponsor to submit complete and accurate financial

certification or disclosure statements to the appropriate regulatory authorities. Investigators are responsible for providing information on financial interests during the course of the study and for 1 year after completion of the study.

10.1.3. Informed Consent Process

The investigator or the investigator's representative will explain the nature of the study, including the risks and benefits, to the potential participant and answer all questions regarding the study.

Potential participants must be informed that their participation is voluntary. Participants will be required to sign a statement of informed consent that meets the requirements of 21 CFR 50, local regulations, ICH guidelines, privacy and data protection requirements, where applicable, and the IRB/IEC or study center.

The medical record must include a statement that written informed consent was obtained before the participant was entered in the study and the date the written consent was obtained. The authorized person obtaining the informed consent must also sign the ICF.

Revised consents must be appropriately obtained using the correct approved ICFs for applicable study participants in accordance with sponsor and ERB consenting guidance.

A copy of the ICF(s) must be provided to the participant and is kept on file.

Participants who are rescreened are required to sign a new ICF.

10.1.4. Data Protection

Participants will be assigned a unique identifier by the sponsor to protect the participant's personal data. Any participant information, such as records, datasets or tissue samples that are transferred to the sponsor will contain the identifier only. Participant names or any information which would make the participant identifiable will not be transferred.

The participant must be informed that the participant's personal study-related data will be used by the sponsor in accordance with local data protection law. The level of disclosure must also be explained to the participant who will be required to give consent for their data to be used as described in the informed consent. This is done by the site personnel through the informed consent process.

The participant must be informed through the informed consent by the site personnel that their medical records may be examined by Clinical Quality Assurance auditors or other authorized personnel appointed by the sponsor, by appropriate IRB/IEC members, and by inspectors from regulatory authorities.

The sponsor has processes in place to ensure information security, data integrity, and data protection. These processes address management of data transfer, and prevention and management of unauthorized access, disclosure, dissemination, alteration or loss of information or personal data. These processes include appropriate contingency plan(s) for appropriate and timely response in the event of a data security breach.

The transfer of personal data is subject to appropriate safeguards through contractual agreements and processes. The sponsor's processes are compliant with local privacy laws and relevant legislations including the General Data Protection Regulation (GDPR).

10.1.5. Committees Structure

10.1.5.1. Data Monitoring Committee

One DMC consisting of members external to Lilly will be established for

- periodic monitoring of GI AE-related data from this study, and
- reviewing unblinded efficacy, safety, and PK data from the interim analyses (if conducted).

This committee will consist of a minimum of 3 members, including a physician with expertise in gastroenterology and a statistician.

No member of the DMC may have contact with the study sites. A SAC will prepare and provide unblinded data to the DMC. The SAC members may be Lilly employees or from third-party organizations designated by Lilly. However, they will be external to the study team and will have no contact with sites and no privileges to influence changes to the ongoing studies. The timing and frequency of the periodic clinical trial data review by the DMC will be detailed in the DMC charter.

The DMC will make their recommendation to the Lilly Senior Management Designee, who may order the immediate implementation of the DMC recommendation or may convene an internal review committee, which is independent from the study team, to review the recommendation according to standard Lilly policy. Study sites will receive information about preliminary results ONLY if it is required for the safety of their patients.

Information that may unblind the tirzepatide treatment will not be shared with study sites or the blinded study team until the tirzepatide treatment has been unblinded.

10.1.5.2. Clinical Event Committee

An independent blinded CEC with membership external to the sponsor will be responsible for adjudicating MACE-3 and suspected cases of acute pancreatitis. The committee will include experts with appropriate expertise.

The CEC charter will contain the final membership and detailed event definitions used for adjudication.

10.1.6. Dissemination of Clinical Study Data

Reports

The sponsor will disclose a summary of study information, including tabular study results, on publicly available websites where required by local law or regulation.

The summary of results will be posted within the time frame specified by local law or regulation. If the study remains ongoing in some countries and a statistical analysis of an incomplete dataset would result in analyses lacking scientific rigor (for example, underpowered) or compromise the

integrity of the overall analyses (for example, trial not yet unblinded), the summary of results will be submitted within 1 year after the end of the study globally or as soon as available, whichever is earlier.

Data

The sponsor provides access to all individual participant data collected during the trial, after anonymization, with the exception of PK or genetic data.

Data are available to request 6 months after the indication studied has been approved in the US and EU and after primary publication acceptance, whichever is later. No expiration date of data requests is currently set once data are made available.

Access is provided after a proposal has been approved by an independent review committee identified for this purpose and after receipt of a signed data-sharing agreement.

Data and documents, including the study protocol, statistical analysis plan, clinical study report, and blank or annotated case report forms, will be provided in a secure data-sharing environment for up to 2 years per proposal.

For details on submitting a request, see the instructions provided at www.vivli.org.

10.1.7. Data Quality Assurance

All participant data relating to the study will be recorded on printed or electronic CRFs unless transmitted to the sponsor or designee electronically (for example, laboratory data). The investigator is responsible for verifying that data entries are accurate and correct by physically or electronically signing the CRF.

The investigator must maintain accurate documentation (source data) that supports the information entered in the CRF. This includes laboratory tests, medical records, and clinical notes.

The investigator must review and confirm that data entries are accurate and complete throughout the duration of the study, by physically or electronically signing the CRF, as instructed by the sponsor. All completed CRFs must be signed prior to archival.

The investigator must permit study-related monitoring, audits, IRB/IEC review, and regulatory agency inspections and provide direct or remote access to source documents.

QTLs will be predefined to identify systematic issues that can impact participant safety and/or reliability of study results. These predefined parameters will be monitored during the study and important excursions from the QTLs and remedial actions taken will be summarized in the clinical study report.

Monitoring details describing strategy, for example, risk-based initiatives in operations and quality such as risk management and mitigation strategies and analytical risk-based monitoring, methods, responsibilities and requirements, including handling of noncompliance issues and monitoring techniques are provided in the Monitoring Plan.

The sponsor or designee is responsible for the data management of this study including quality checking of the data.

The sponsor assumes accountability for actions delegated to other individuals, for example, contract research organizations.

The sponsor or designee will perform monitoring to confirm that data transcribed into the CRF by authorized site personnel are accurate, complete, and verifiable from source documents; that the safety and rights of participants are being protected; and that the study is being conducted in accordance with the currently approved protocol and any other study agreements, ICH GCP, and all applicable regulatory requirements.

Records and documents, including signed ICFs, pertaining to the conduct of this study must be retained by the investigator for the time period outlined in the Clinical Trial Agreement unless local regulations or institutional policies require a longer retention period. No records may be destroyed during the retention period without the written approval of the sponsor. No records may be transferred to another location or party without written notification to the sponsor.

In addition, the sponsor or its representatives will periodically check a sample of the participant data recorded against source documents at the study site. The study may be audited by the sponsor or its representatives, or regulatory agencies at any time. Investigators will be given notice before an audit occurs.

Data Capture System

The investigator is responsible for ensuring the accuracy, completeness, legibility, and timeliness of the data reported to the sponsor.

An EDC will be used in this study for the collection of CRF data. The investigator maintains a separate source for the data entered by the investigator or designee into the sponsor-provided EDC system. The investigator is responsible for the identification of any data to be considered source and for the confirmation that data reported are accurate and complete by signing the CRF.

The COA data (participant-focused clinical outcome assessment) will be collected by the authorized study personnel via a paper source document and will be transcribed by the authorized study personnel into the EDC system and will serve as the source documentation.

Additionally, eCOA data (participant-focused electronic clinical outcome assessment) will be directly recorded by the participant into an instrument, for example, hand-held smart phone or tablet. The eCOA data will serve as the source documentation, and the investigator does not maintain a separate written or electronic record of these data.

Data collected via the sponsor-provided data capture system(s) will be stored at third parties. The investigator will have continuous access to the data during the study and until decommissioning of the data capture system(s). Prior to decommissioning, the investigator will receive or access an archival copy of pertinent data for retention.

Data managed by a central vendor, such as laboratory test data, will be stored electronically in the central vendor's database system and reports will be provided to the investigator for review and retention. Data will subsequently be transferred from the central vendor to the sponsor data warehouse.

Data from complaint forms submitted to the sponsor will be encoded and stored in the global product complaint management system.

10.1.8. Source Documents

Source documents provide evidence for the existence of the participant and substantiate the integrity of the data collected. Source documents are filed at the investigator's site.

Data reported on or entered in the CRF and are transcribed from source documents must be consistent with the source documents or the discrepancies must be explained. The investigator may need to request previous medical records or transfer records, depending on the study. Also, current medical records must be available.

Definition of what constitutes source data can be found in the site confirmation of source data.

10.1.9. Study and Site Start and Closure**Study start**

The study start date is the date on which the clinical study will be open for recruitment of participants.

First act of recruitment

The first act of recruitment is the opening of the first site.

Study or site termination

The sponsor or sponsor's designee reserves the right to close the study site or terminate the study at any time for any reason at the sole discretion of the sponsor. Additional reasons that could or should lead to trial discontinuation or termination include: the emergence of an unacceptable risk to participants, a negative shift in the risk-benefit assessment as assessed by the DMC and/or sponsor, or a formal request from a concerned regulatory authority or ethics committee.

Study sites will be closed upon study completion. A study site is considered closed when all required documents and study supplies have been collected and a study-site closure visit has been performed.

The investigator may initiate study-site closure at any time, provided there is reasonable cause and sufficient notice is given in advance of the intended termination.

Reasons for the early closure of a study site by the sponsor or investigator may include but are not limited to

- for study termination due to discontinuation of further study intervention development
- for site termination due to
 - failure of the investigator to comply with the protocol, the requirements of the IRB/IEC or local health authorities, the sponsor's procedures, or GCP guidelines
 - inadequate recruitment, evaluated after a reasonable amount of time of participants by the investigator, or
 - total number of participants included earlier than expected.

If the study is prematurely terminated or suspended, the sponsor shall promptly inform the investigators, the IECs/IRBs, the regulatory authorities, and any contract research organization(s) used in the study of the reason for termination or suspension, as specified by the applicable

regulatory requirements. The investigator shall promptly inform the participant and should assure appropriate participant therapy and/or follow-up.

10.1.10. Publication Policy

In accordance with the sponsor's publication policy, the results of this study will be submitted for publication by a peer-reviewed journal.

10.1.11. Investigator Information

Researchers with appropriate education, training, and experience, as determined by the sponsor, will participate as investigators in this clinical trial.

10.1.12. Sample Retention

Sample retention enables use of new technologies, response to regulatory questions, and investigation of variable response that may not be observed until later in the development of mirikizumab or tirzepatide.

Sample Type	Custodian	Retention Period After Last Participant Visit
Exploratory biomarkers	Sponsor or designee	Up to 15 years
Genetics	Sponsor or designee	Up to 15 years
Pharmacokinetic	Sponsor or designee	1 year
Immunogenicity	Sponsor or designee	Up to 15 years

10.2. Appendix 2: Clinical Laboratory Tests

The tests detailed in the table below will be performed by the central (Lilly-designated) laboratory or by the local laboratory.

In circumstances where the sponsor approves local laboratory testing in lieu of central laboratory testing (in the table below), the local laboratory must be qualified in accordance with applicable local regulations.

Protocol-specific requirements for inclusion or exclusion of participants are detailed in Section 5 of the protocol.

Additional tests may be performed at any time during the study as determined necessary by the investigator or required by local regulations.

Investigators must document their review of the laboratory safety results.

Laboratory/analyte results that could unblind the study will not be reported to investigative sites or other blinded personnel until the study has been unblinded.

Clinical Laboratory Tests	Comments
Hematology	Assayed by Lilly-designated laboratory.
Hemoglobin	
Hematocrit	
Erythrocyte count (red blood cells [RBC])	
Mean cell volume	
Mean cell hemoglobin	
Mean cell hemoglobin concentration	
Leukocytes (white blood cells [WBC])	
Differential	
Absolute neutrophil count (ANC) (segmented and bands) (calculated)	
Absolute count of:	
Neutrophils, segmented	
Neutrophils, bands	Report if detected
Lymphocytes	
Monocytes	
Eosinophils	
Basophils	
Platelet	
Cell morphology (RBC and WBC)	
Clinical chemistry	Assayed by Lilly-designated laboratory.
Sodium	
Potassium	
Chloride	
Bicarbonate	
Total bilirubin (TBL)	
Direct bilirubin	

Alkaline phosphatase (ALP)	
Alanine aminotransferase (ALT)	
Aspartate aminotransferase (AST)	
Gamma-glutamyl transferase (GGT)	
Blood urea nitrogen (BUN) or blood urea	
Creatinine	
Creatine kinase (CK)	
Uric acid	
Total protein	
Albumin	
Calcium	
Glucose	
Additional testing	Assayed by Lilly-designated laboratory.
Calcitonin	
Pancreatic lipase	
Glycated hemoglobin (HbA1c)	
Pancreatic amylase	
High-sensitivity C-reactive protein (hsCRP)	
Cystatin C	
Micronutrient panels	Assayed by Lilly-designated laboratory. Collected at screening for all participants. This planned testing will also include complete blood count, which is already part of the Hematology panel and is thus not specified here. For collecting samples at risk of malnutrition (for example, caloric intake is significantly reduced to below 800 to 1200 kcal), if the participant's BMI falls ≤ 22 kg/m ² , or if a participant with T2DM experiences weight loss of >20% or >4 kg/month, see Section 10.2.2 .
<i>Special tests</i>	
Vitamin A	
Vitamin B1 (Thiamine)	
Vitamin B9 (Folate)	
Vitamin B12	
Vitamin D (25-hydroxy vitamin D)	
Vitamin E	
Vitamin K ₁	Results of Vitamin K ₁ testing may not be available in all countries.
<i>Clinical chemistry</i>	
Magnesium	
Zinc	
Phosphorus	
<i>Iron panel</i>	
Serum iron	
Total iron binding capacity (TIBC)	

Transferrin saturation (%)	
Ferritin	
Lipid panel	Assayed by Lilly-designated laboratory.
Low-density lipoprotein cholesterol (LDL-C)	This value will be calculated. LDL will be directly measured if triglyceride exceeds maximum value for calculation.
High-density lipoprotein cholesterol (HDL-C)	
Very Low-density lipoprotein cholesterol (VLDL-C)	
Total cholesterol	
Triglycerides	
Urinalysis	Assayed by Lilly-designated laboratory. Can be performed at any visit at the discretion of investigator if the participant has symptoms.
Specific gravity	
pH	
Protein	
Glucose	
Ketones	
Bilirubin	
Urobilinogen	
Blood	
Nitrite	
Urine leukocyte esterase	
Microscopic examination of sediment, if indicated	Perform if abnormalities were detected on urinalysis or if participant is symptomatic.
Hormones (individuals AFAB)	
Thyroid-stimulating hormone (TSH)	Assayed by Lilly-designated laboratory
Follicle-stimulating hormone (FSH)	Assayed by Lilly-designated laboratory. For individuals AFAB only.
Serum pregnancy	Assayed by Lilly-designated laboratory. For IOCBP only. See Sections 8.2.6 and 10.4.2.
Urine pregnancy	Evaluated locally. For IOCBP only. See Sections 8.2.6 and 10.4.2. For telehealth visits, home pregnancy tests will be performed.
HIV, Hepatitis Serology, and TB	
Human immunodeficiency virus (HIV) testing	Assayed by Lilly-designated laboratory
HIV antibody	
HIV RNA	
Hepatitis C virus (HCV) testing:	Assayed by Lilly-designated laboratory
HCV antibody	
HCV RNA	Performed only for participants who test positive for hepatitis C antibody.
Hepatitis B virus (HBV) testing:	Assayed by Lilly-designated laboratory
Hepatitis B core total antibody (anti-HBc)	

Hepatitis B surface antigen (HBsAg)	
HBV DNA	Performed only for participants who test positive for anti-HBc at screening.
Tuberculosis (TB) testing	Refer to Section 8.2.10 for more information about TB testing
QuantiFERON®-TB Gold test	Assayed by Lilly-designated laboratory. Preferred central, may be performed locally.
T-SPOT®.TB	May be tested and evaluated locally. Local laboratory must be qualified by local regulations
Tuberculin skin test (TST)	Tested and evaluated locally. Local staff must be qualified to administer and interpret the test
Calculations	Assayed by Lilly-designated laboratory.
Estimated glomerular filtration rate (eGFR)	Calculated using the CKD-EPI Cystatin C Equation 2012 (NKF [WWW])
Pharmacokinetic samples	Assayed by Lilly-designated laboratory. Results will not be provided to the investigative site.
Mirikizumab concentration	
Tirzepatide concentration	
Immunogenicity (ADA) samples	Assayed by Lilly-designated laboratory. Results will not be provided to the investigative sites
Anti-mirikizumab antibodies	
Anti-mirikizumab antibodies neutralization	
Anti-tirzepatide antibodies	
Anti-tirzepatide antibodies neutralization	
Stool samples	Assayed by Lilly-designated laboratory. Additional local testing (for example, for ova and parasites) is allowed at the investigator's discretion
Stool culture	
<i>Clostridioides difficile</i> (<i>C. difficile</i>) testing	Confirmed by a test for <i>C. difficile</i> toxin gene expression
Fecal calprotectin	
Stored samples	Assayed by Lilly-designated laboratory. Results will not be provided to the investigative sites
Genetics sample	
Exploratory biomarker samples	
Serum	
Biopsy tissue	Endoscopic biopsy tissue samples from colon and rectum.
Stool supernatant	

10.2.1. Laboratory Samples to Be Obtained at the Time of a Systemic Hypersensitivity Event

Purpose of collecting samples after a systemic hypersensitivity event

The samples listed in this appendix are not collected for acute study participant management. The sponsor will use the laboratory tests results from these samples to characterize hypersensitivity events across the clinical development program.

When to collect samples after a systemic hypersensitivity event occurs

Collect the samples listed below if a systemic hypersensitivity event is suspected. The timing should be as designated in the table, assuming the participant has been stabilized.

Obtain follow-up pre-dose samples at the next regularly scheduled laboratory sample collection (ideally prior to the next dose after the event) to assess post-event return-to-baseline values.

Timing	Laboratory Test ^a
Collect from 30 minutes to 4 hours after the start of the event. Note: The optimal collection time is from 1 to 2 hours after the start of event.	Total tryptase
	Complements (C3, C3a, and C5a)
	Cytokine panel (IL-6, IL-1 β , IL-10 or any cytokine panel that includes these 3 cytokines)
Collect only if not already collected on the same day as the event. Note: If collecting, collect up to 12 hours after the start of the event.	Mirikizumab and tirzepatide anti-drug antibodies (ADA)
	Mirikizumab and tirzepatide concentration

Abbreviations: ADA = anti-drug antibodies; IL = interleukin.

^a All samples for hypersensitivity testing will be assayed by Lilly-designated laboratory. Results will not be provided to the study site. If samples are not collected or are collected outside the specified time period, this will not be considered a protocol deviation.

What information to record

Record the date and time when the samples are collected.

Allowed additional testing for participant management

The investigator may perform additional tests locally, if clinically indicated, for acute study participant management.

10.2.2. Nutritional Monitoring Panel to Be Obtained in Case of Suspected Malnutrition

Purpose of collecting samples in case of suspected malnutrition

Investigators should monitor participants for signs and symptoms of malnutrition. See Section 8.2.13.

Nutritional guidance and, as needed, supplementation, should be considered. For severe or persistent cases, consider tirzepatide discontinuation, and discuss management of the participant with the sponsor-designated medical monitor (Section 7.1.1).

When to collect samples in case of suspected malnutrition

As stated in the SoA and Sections 5.3.3, 6.6.2.2, and 8.2.13, collect the samples listed below if malnutrition is suspected (for example, caloric intake is significantly reduced to below 800 to 1200 kcal), if a participant's BMI reaches ≤ 22 kg/m², or if a participant with T2DM experiences weight loss of >20% or >4 kg/month.

Category	Laboratory Test
Special tests	Vitamin A
	Vitamin B1 (Thiamine)
	Vitamin B9 (Folate)
	Vitamin B12
	Vitamin D (25-hydroxy vitamin D)
	Vitamin E
	Vitamin K ₁ ^c
Hematology	Complete blood count ^a
Clinical chemistry	Clinical chemistry panel ^b
	Magnesium
	Zinc
	Phosphorus
Iron panel	Serum iron
	Total iron binding capacity (TIBC)
	Transferrin saturation (%)
	Ferritin

^a Refer to Section 10.2 for contents of complete blood count panel (see Hematology panel).

^b Refer to Section 10.2 for contents of Clinical Chemistry panel.

^c Results of Vitamin K₁ testing may not be available in all countries.

Note: Tests will be assayed by a Lilly-designated laboratory and results will be provided to the investigative sites where allowed (or appropriate).

What information to record

Record the date and time when the samples are collected.

Allowed additional testing for participant management

The investigator may perform additional tests locally, if clinically indicated, for acute study participant management.

10.3. Appendix 3: Adverse Events and Serious Adverse Events: Definitions and Procedures for Recording, Evaluating, Follow-up, and Reporting

10.3.1. Definition of AE

An AE is any untoward medical occurrence in a participant administered a pharmaceutical product and which does not necessarily have a causal relationship with the study intervention. An AE can therefore be any unfavourable and unintended sign (including an abnormal laboratory finding), symptom, or disease (new or exacerbated) temporally associated with the use of a medicinal (investigational) product, or investigational combination product, whether or not related to the medicinal (investigational) product or investigational combination product.

Events meeting the AE definition

- Any abnormal laboratory test results (hematology, clinical chemistry, or urinalysis) or other safety assessments, for example, ECG, radiological scans, and vital signs measurements, including those that worsen from baseline, considered clinically significant in the medical and scientific judgment of the investigator, that is, not related to progression of underlying disease.
- Exacerbation of a chronic or intermittent pre-existing condition including either an increase in frequency and/or intensity of the condition.
- New condition detected or diagnosed after study intervention administration even though it may have been present before the start of the study.
- Signs, symptoms, or the clinical sequelae of a suspected drug-drug interaction.
- Medication error, misuse, or abuse of IMP, including signs, symptoms, or clinical sequelae.
- Lack of efficacy or failure of expected pharmacological action per se will not be reported as an AE or SAE. Such instances will be captured in the efficacy assessments. However, the signs, symptoms, and/or clinical sequelae resulting from lack of efficacy will be reported as AE or SAE if they fulfill the definition of an AE or SAE.

Events NOT meeting the AE definition

- Any clinically significant abnormal laboratory findings or other abnormal safety assessments that are associated with the underlying disease, unless judged by the investigator to be more severe than expected for the participant's condition.
- The disease or disorder being studied or expected progression, signs, or symptoms of the disease or disorder being studied, unless more severe than expected for the participant's condition.
- Situations in which an untoward medical occurrence did not occur (social and/or convenience admission to a hospital).
- Anticipated day-to-day fluctuations of pre-existing disease(s) or condition(s) present or detected at the start of the study that do not worsen.

10.3.2. Definition of SAE

An SAE is defined as any untoward medical occurrence that, at any dose, meets one or more of the criteria listed:

- Results in death
- Is life-threatening
 - The term *life-threatening* in the definition of *serious* refers to an event in which the participant was at risk of death at the time of the event. It does not refer to an event, which hypothetically might have caused death, if it were more severe.
- Requires inpatient hospitalization or prolongation of existing hospitalization
 - In general, hospitalization signifies that the participant has been admitted to hospital or emergency ward (usually involving at least an overnight stay) for observation and/or treatment that would not have been appropriate in the physician's office or outpatient setting. Complications that occur during hospitalization are AEs. If a complication prolongs hospitalization or fulfills any other serious criteria, the event is serious. When in doubt as to whether hospitalization occurred or was necessary, the AE should be considered serious.
 - Hospitalization for elective treatment of a pre-existing condition that did not worsen from baseline is not considered an AE.
- Results in persistent disability/incapacity
 - The term disability means a substantial disruption of a person's ability to conduct normal life functions.
 - This definition is not intended to include experiences of relatively minor medical significance, such as uncomplicated headache, nausea, vomiting, diarrhea, influenza, and accidental trauma, for example, sprained ankle, which may interfere with or prevent everyday life functions but do not constitute a substantial disruption.
- Is a congenital anomaly/birth defect
 - Abnormal pregnancy outcomes, for example, spontaneous abortion, fetal death, stillbirth, congenital anomalies, and ectopic pregnancy, are considered SAEs.
- Other situations:
 - Medical or scientific judgment should be exercised by the investigator in deciding whether SAE reporting is appropriate in other situations such as important medical events that may not be immediately life-threatening or result in death or hospitalization but may jeopardize the participant or may require medical or surgical intervention to prevent one of the other outcomes listed in the above definition. These events should usually be considered serious.
 - Examples of such events include invasive or malignant cancers, intensive treatment in an emergency room or at home for allergic bronchospasm, blood

dyscrasias or convulsions that do not result in hospitalization, or development of drug dependency or drug abuse.

10.3.3. Definition of Product Complaints

Product complaint

A product complaint is any written, electronic, or oral communication that alleges deficiencies related to the identity, quality, durability, reliability, safety, effectiveness or performance of a study intervention. When the ability to use the study intervention safely is impacted, the following are also product complaints:

- deficiencies in labeling information, and
- use errors for device or drug-device combination products due to ergonomic design elements of the product.

Product complaints related to study interventions used in clinical trials are collected to ensure the safety of participants, monitor quality, and to facilitate process and product improvements.

If the participant identifies a product complaint or a problem with the study intervention, investigators will instruct participants to contact the site as soon as possible so that the situation can be assessed.

An event may meet the definition of both a product complaint and an AE/SAE. In such cases, it should be reported as both a product complaint and as an AE/SAE.

Device deficiencies are product complaints.

10.3.4. Recording and Follow-Up of AE and/or SAE and Product Complaints

AE, SAE, and product complaint recording

When an AE/SAE/product complaint occurs, it is the responsibility of the investigator to review all documentation (for example, hospital progress notes, laboratory reports, and diagnostics reports) related to the event.

The investigator will then record all relevant AE/SAE/product complaint information in the participant's medical records, in accordance with the investigator's normal clinical practice. AE/SAE information is reported on the appropriate CRF page and product complaint information is reported on the Product Complaint Form.

Note: An event may meet the definition of both a product complaint and an AE/SAE. In such cases, it should be reported as both a product complaint and as an AE/SAE.

It is **not** acceptable for the investigator to send photocopies of the participant's medical records to the sponsor or designee in lieu of completion of the CRF page for AE/SAE and the Product Complaint Form for product complaints.

There may be instances when copies of medical records for certain cases are requested by the sponsor or designee. In this case, all participant identifiers, with the exception of the participant number, will be redacted on the copies of the medical records before submission to the sponsor or designee.

The investigator will attempt to establish a diagnosis of the event based on signs, symptoms, and/or other clinical information. Whenever possible, the diagnosis (not the individual signs/symptoms) will be documented as the AE/SAE.

Assessment of intensity

The investigator will make an assessment of intensity for each AE and SAE reported during the study and assign it to one of the following categories:

- **Mild:** A type of AE that is usually transient and may require only minimal treatment or therapeutic intervention. The event does not generally interfere with usual activities of daily living.
- **Moderate:** A type of AE that is usually alleviated with additional specific therapeutic intervention. The event interferes with usual activities of daily living, causing discomfort but poses no significant or permanent risk of harm to the research participant.
- **Severe:** A type of AE that interrupts usual activities of daily living, or significantly affects clinical status, or may require intensive therapeutic intervention. An AE that is assessed as severe should not be confused with a SAE. Severe is a category utilized for rating the intensity of an event, and both AEs and SAEs can be assessed as severe.

An event is defined as “serious” when it meets at least one of the predefined outcomes as described in the definition of an SAE, NOT when it is rated as severe.

Assessment of causality

The investigator is obligated to assess the relationship between study intervention and each occurrence of each AE/SAE. The investigator will use clinical judgment to determine the relationship.

A “reasonable possibility” of a relationship conveys that there are facts, evidence, and/or arguments to suggest a causal relationship, rather than a relationship cannot be ruled out.

Alternative causes, such as underlying disease(s), concomitant therapy, and other risk factors, as well as the temporal relationship of the event to study intervention administration will be considered and investigated.

The investigator will also consult the IBs and/or prescribing information in their assessment.

The investigator **must** review and provide an assessment of causality for each AE/SAE and document this in the medical notes.

There may be situations in which an SAE has occurred and the investigator has minimal information to include in the initial report to the sponsor or designee. However, it is very important that the investigator always make an assessment of causality for every event before the initial transmission of the SAE data to the sponsor or designee.

The investigator may change their opinion of causality in light of follow-up information and send a SAE follow-up report with the updated causality assessment.

The causality assessment is one of the criteria used when determining regulatory reporting requirements.

Follow-up of AEs and SAEs

- The investigator is obligated to perform or arrange for the conduct of supplemental measurements and/or evaluations as medically indicated or as requested by the sponsor or designee to elucidate the nature and/or causality of the AE or SAE as fully as possible. This may include additional laboratory tests or investigations, histopathological examinations, or consultation with other health care professionals.
- If a participant dies during participation in the study or during a recognized follow-up period, the investigator will provide the sponsor or designee with a copy of any postmortem findings including histopathology.

10.3.5. Reporting of SAEs**SAE reporting via an electronic data collection tool**

The primary mechanism for reporting an SAE will be the electronic data collection tool.

If the electronic system is unavailable, then the site will use the SAE paper form (see next section) in order to report the event within 24 hours.

The site will enter the SAE data into the electronic system as soon as it becomes available.

After the study is completed at a given site, the electronic data collection tool will be taken off-line to prevent the entry of new data or changes to existing data.

If a site receives a report of a new SAE from a study participant or receives updated data on a previously reported SAE after the electronic data collection tool has been taken off-line, then the site can report this information on a SAE paper form (see to next section) or to the sponsor or sponsor's designee by telephone.

Contacts for SAE reporting can be found in the Global Patient Safety Clinical Trial SAE Transmission Cover Sheet and Form.

SAE reporting via paper form

Facsimile transmission of the SAE paper form is the preferred method to transmit this information to the sponsor or sponsor's designee.

Initial notification via telephone does not replace the need for the investigator to complete and sign the SAE CRF pages within the designated reporting time frames.

Contacts for SAE reporting can be found in the Global Patient Safety Clinical Trial SAE Transmission Cover Sheet and Form.

10.3.6. Regulatory Reporting Requirements**SAE regulatory reporting**

Prompt notification by the investigator to the sponsor of a SAE is essential so that legal obligations and ethical responsibilities toward the safety of participants and the safety of a study intervention under clinical investigation are met.

The sponsor has a legal responsibility to notify both the local regulatory authority and other regulatory agencies about the safety of a study intervention under clinical investigation. The

sponsor will evaluate the reported SAEs, including confirmation of relatedness and assessment of expectedness. The sponsor will comply with country-specific regulatory requirements relating to safety reporting to the regulatory authority, IRB/ IEC, and investigators.

An investigator who receives an investigator safety report describing a SAE or other specific safety information (for example, summary or listing of SAEs) from the sponsor will review and then file it along with the IB and will notify the IRB/IEC, if appropriate according to local requirements.

10.4. Appendix 4: Contraceptive and Barrier Guidance

10.4.1. Definitions

Word/Phrase	Definition
Individuals assigned female at birth (AFAB)	Individuals assigned the female sex based on external genitalia and/or genetic or medical information. In addition, if these individuals are of reproductive potential, they are potentially capable of gestating a fetus, and thus are capable of exposing an egg, embryo, or fetus to study drug or drug effects.
Individuals assigned male at birth (AMAB)	Individuals assigned the male sex based on external genitalia and/or genetic or medical information. In addition, if these individuals are of reproductive potential, they are not capable of gestating a fetus, but are capable of exposing a fetus to study drug or drug effects via their semen. Individuals AMAB are considered to be not of reproductive potential if they have had bilateral orchiectomy (orchidectomy) with or without penectomy.
Individuals of childbearing potential (IOCBP) ^a	Adult individuals AFAB are considered IOCBP unless they are INOCBP. Note: Adult individuals AFAB who are receiving hormone therapy as part of gender transition are considered IOCBP unless they meet the conditions outlined below for INOCBP.
Individuals not of childbearing potential (INOCBP) ^b	Individuals AFAB are considered INOCBP if they are not capable of producing ova or embryo, and/or are not capable of potentially gestating a fetus. Such individuals include those who • have a congenital anomaly such as Müllerian agenesis, resulting in confirmed infertility • are infertile due to surgical sterilization, or • are postmenopausal. Acceptable surgical sterilization methods are hysterectomy, bilateral salpingo-oophorectomy, bilateral salpingectomy, or bilateral oophorectomy.
Postmenopausal state ^c	The postmenopausal state is defined as an individual: • at any age at least 6 weeks post-surgical bilateral oophorectomy with or without hysterectomy, or • aged at least 40 years and up to 55 years with an intact uterus, not on hormone therapy^c, who has had cessation of menses for at

	least 12 consecutive months without an alternative medical cause, AND with a follicle-stimulating hormone ≥ 40 mIU/mL, or • 55 years or older not on hormone therapy, who has had at least 12 months of spontaneous amenorrhea, or • aged at least 55 years with a diagnosis of menopause prior to starting hormone replacement therapy.
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- ^a IOCBP is inclusive of the concept of women of childbearing potential (WOCBP or WCBP), a term often used in literature and regulatory guidance documents.
- ^b INOCBP is inclusive of the concept of women not of childbearing potential (WNOCBP).
- ^c The individual **should not** be taking medications during amenorrhea such as oral contraceptives, hormone replacement therapy (HRT), gonadotropin-releasing hormone, anti-estrogens, selective estrogen receptor modulators, or chemotherapy that could induce transient amenorrhea. Individuals on HRT and those whose postmenopausal status cannot be confirmed will be required to comply with the protocol contraception requirements if they wish to continue HRT during the study. Otherwise, they must discontinue HRT to allow confirmation of postmenopausal status before study enrollment.

10.4.2. Contraception Guidance

The decision regarding contraception guidance will be made by the investigator, following local regulations adhering to standard clinical practice and local label.

The investigator will receive training on the available data regarding mirikizumab and tirzepatide use during pregnancy, following the guidelines outlined in the local label.

10.4.2.1. Individuals Assigned Female at Birth

IOCBP who are completely abstinent as their preferred and usual lifestyle, or exclusively engage in sexual relations with other individual(s) who are AFAB as their preferred and usual lifestyle must follow the rules in this table.

Must...	Must not...
agree to either remain abstinent or exclusively engage in sexual relations with other individual(s) who are AFAB, and not plan a pregnancy during the study	• use periodic abstinence methods ○ calendar ○ ovulation ○ symptothermal, or ○ post-ovulation • declare abstinence just for the duration of a trial, or • use the withdrawal method

IOCBP who are NOT completely abstinent as their preferred and usual lifestyle, or who do NOT exclusively engage in sexual relations with other individual(s) who are AFAB as their preferred and usual lifestyle, must follow the rules in this table.

Must...
Agree to use 1 highly effective method of non-oral contraception or add a barrier method of contraception for 4 weeks after initiation of tirzepatide and for 4 weeks after each dose escalation. These methods of contraception must be used during the study and for at least 10 weeks (approximately 2.5 months) after the last dose of the study intervention.

Examples of different methods of contraception^a:

Methods	Examples
Highly effective contraception (less than 1% failure rate)	• fallopian tubal sterilization methods other than bilateral salpingectomy (laparoscopic bipolar electrocoagulation, plastic ring application on the uterine tubes, fallopian tube ligation, hysteroscopic sterilization). Note: Bilateral salpingectomy is indicative of permanent sterilization. Please see the INOCBP definition above. • combined (estrogen and progestogen-containing) hormonal contraception associated with inhibition of ovulation: ○ oral ○ intravaginal (vaginal ring) ○ transdermal (for example, contraceptive patch) (only for individuals <198 pounds or 90 kg) • progestogen-only hormonal contraception associated with inhibition of ovulation ○ oral ○ injectable ○ implantable • total abstinence • sexual relationships exclusively between individuals who are assigned the same sex at birth • vasectomy – for individuals AMAB in clinical trials and for the partner of an individual AFAB (if only sexual partner) • fallopian tube implants (if confirmed by hysterosalpingogram), or • intrauterine devices (IUD) with or without hormone-releasing system

Methods	Examples
Effective contraception	• progestogen-only hormonal contraception, where inhibition of ovulation is not the primary mode of action • penile condom with or without spermicide • vaginal condom with or without spermicide • diaphragm with spermicide • cervical sponge with spermicide, or • cervical cap with spermicide Note: Penile and vaginal condoms should not be used in combination.
Ineffective methods of contraception whether used alone or in any combination	• spermicide alone • periodic abstinence • fertility awareness (calendar method, temperature method, cervical mucus, or symptothermal) • withdrawal • postcoital douche, or • lactational amenorrhea

^a Ablation (endometrial or uterine) is not considered a form of contraception.

10.4.2.2. Individuals Assigned Male at Birth

For individuals AMAB, no contraception is required.

10.5. Appendix 5: Liver Safety: Suggested Actions and Follow-Up Assessments

Hepatic Evaluation Testing

See Section 8.2.9 for guidance on appropriate test selection from the table below.

Note that testing for some analytes may not be available from the central laboratory or local laboratory in certain regions due to local availability of assay or regulatory requirements. If testing is not available, this will not be considered a protocol deviation.

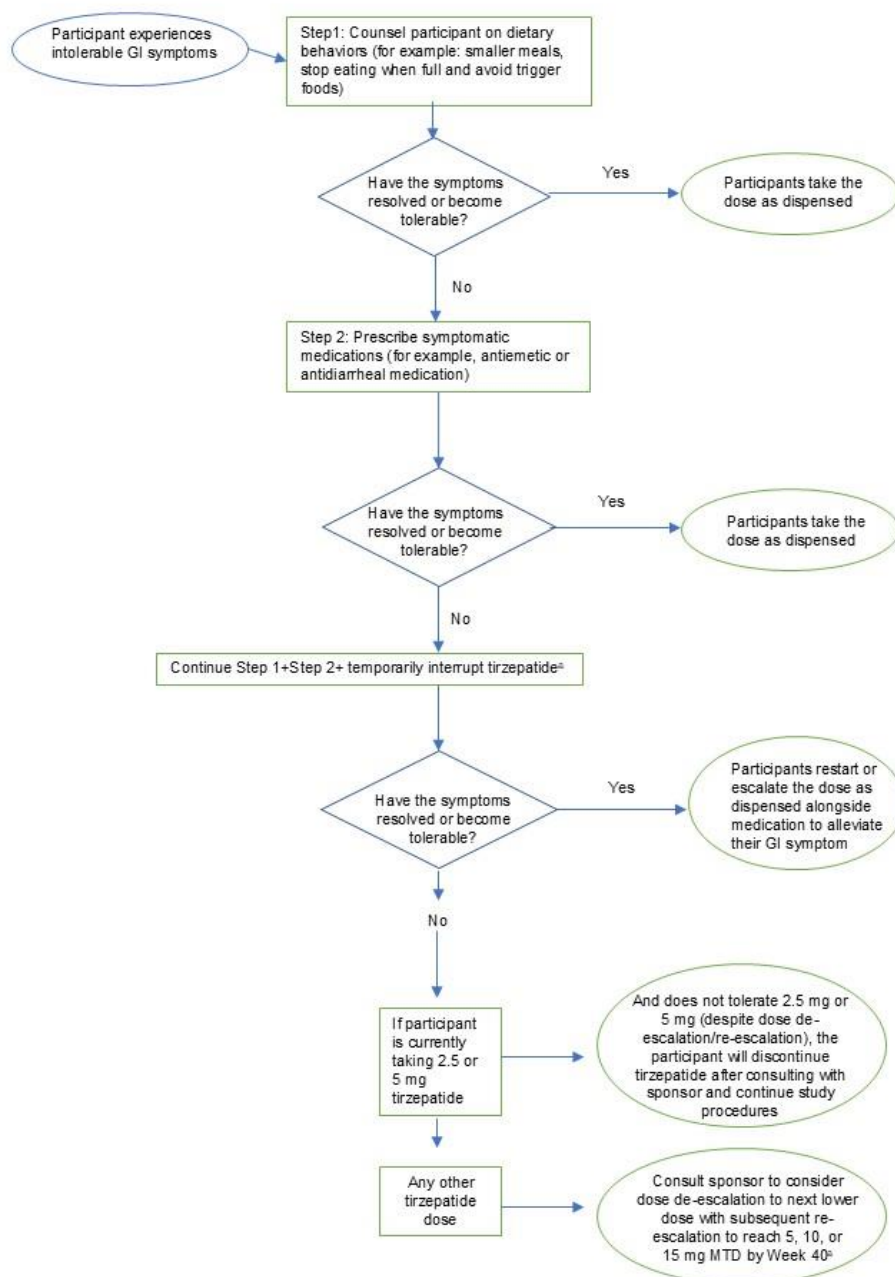
Local testing may be performed *in addition to central testing* when necessary for immediate participant management. If local testing is performed in lieu of central testing, the results should be recorded in the appropriate local laboratory CRF (for example, if immediate participant management circumstances preclude collection of central laboratory samples, such as during an emergency room visit or hospitalization). The local laboratory must be qualified in accordance with applicable local regulations.

TESTS AVAILABLE FROM LILLY-DESIGNATED CENTRAL LABORATORY	
Hepatic Clinical Chemistry Panel	Hepatitis A virus (HAV) testing:
Total bilirubin	HAV total antibody ^a
Direct bilirubin	HAV IgM antibody
Alkaline phosphatase (ALP)	Hepatitis B virus (HBV) testing:
Alanine aminotransferase (ALT)	Hepatitis B surface antigen (HBsAg)
Aspartate aminotransferase (AST)	Hepatitis B surface antibody (anti-HBs)
Gamma-glutamyl transferase (GGT)	Hepatitis B core total antibody (anti-HBc)
Creatine kinase (CK)	Hepatitis B core IgM antibody
Hepatic Coagulation Panel	HBV DNA ^b
Prothrombin time, INR (PT-INR)	Hepatitis C virus (HCV) testing:
Hepatic Hematology Panel	HCV total antibody ^a
Hemoglobin	HCV RNA ^b
Hematocrit	Hepatitis E virus (HEV) testing:
Erythrocytes (RBCs - red blood cells)	HEV IgG antibody
Leukocytes (WBCs - white blood cells)	HEV IgM antibody
Differential:	HEV RNA ^b
Neutrophils	Anti-nuclear antibody (ANA)
Lymphocytes	Anti-smooth muscle antibody (ASMA) or anti-actin antibody
Monocytes	Haptoglobin
Basophils	Immunoglobulin IgA, IgG, IgM (quantitative)
Eosinophils	Urine Chemistry
Platelets	Drug Screen
Cell morphology (RBC and WBC)	

TESTS TO BE PERFORMED BY INVESTIGATOR-DESIGNATED LABORATORY, AS NEEDED	
Acetaminophen	Hepatitis D virus (HDV) testing ^c
Acetaminophen protein adducts	HDV total antibody ^a
Alkaline phosphatase isoenzymes	HDV IgM antibody
Ceruloplasmin	HDV RNA ^b
Copper	Herpes simplex virus (HSV) testing:
Cytomegalovirus (CMV) testing:	HSV (Type 1 and 2) antibody
CMV antibody	HSV (Type 1 and 2) DNA ^b
CMV DNA ^b	Liver kidney microsomal type 1 (LKM-1) antibody
Ethyl alcohol (ethanol, EtOH)	Phosphatidylethanol (PEth)
Ethyl glucuronide (EtG)	Microbiology Culture:
Epstein Barr virus (EBV) testing:	Blood
EBV antibody	Urine
EBV DNAb	

- a If laboratory does not offer total antibody testing, IgG and/ or IgM are acceptable substitutes.
- b Reflex/confirmation dependent on regulatory requirements, testing availability, or both.
- c If HDV testing is not available, HBV testing may be sufficient. If HBV testing is positive, consult with the Lilly-designated medical monitor.

10.6. Appendix 6: Management of Gastrointestinal Symptoms



^a All participants should be on stable dose from Week 40, defined as MTD of 5, 10, or 15 mg. For definition of temporary interruption of tirzepatide, see Section 6.6.2.1.

Note: Assessment of disease worsening is provided in Section 6.9.6.

10.7. Appendix 7: Clinical Signs Associated with Common Nutritional Deficiencies

The following table displays the common clinical signs that are associated with deficiencies of micro- and macro-nutrients (Parrott et al. 2017; Astrup and Bügel 2019; Wilson et al. 2022; Linus Pauling Institute 2023).

Micronutrient	Clinical Signs of Deficiency
A	Vision loss, night blindness, dry, scaly, or itchy skin, frequent respiratory tract infections, infertility, delayed growth and development in children
B1	Peripheral neuropathy, muscle weakness (dry beriberi), tachycardia, heart failure, edema (wet beriberi), ataxia, abnormal eye movements, impaired memory, confusion (cerebral beriberi or Wernicke-Korsakoff syndrome)
B9	Macrocytic anemia, extreme tiredness, shortness of breath, memory problems, irritability, ulcerations in the oral mucosa
B12	Macrocytic anemia, fatigue, shortness of breath, irritability, memory disturbances, depression, disorientation, glossitis, peripheral neuropathy, paresthesia
D	Weak and thinning bones, osteomalacia in adults, rickets in children, muscle weakness and pain
Calcium	Weak and thinning bones, muscle weakness and pain, numbness or tingling in feet and hands, irregular heartbeat
Magnesium	Muscle pain, tremors, cramps, weakness
Phosphorus	Bone pain, fragility, poor development, joint stiffness and pain, weakness, numbness, tingling in limbs, fatigue and loss of appetite, breathing problems, irregular heartbeat
Iron	Microcytic anemia, tiredness, pale skin, shortness of breath, hair loss, brittle nails, cold intolerance
Zinc	Dermatitis, impaired wound healing, impaired taste, decreased immune function
Macronutrient	Clinical Signs of Deficiency
Protein	Muscle wasting, weakness, edema, skin changes, immunosuppression
Fat (essential fatty acid deficiency)	Dry skin, impaired wound healing, susceptibility to infections
Carbohydrate	There is no absolute dietary requirement for carbohydrates, but very low carbohydrate dietary pattern may increase the risk of ketosis, headache, fatigue, halitosis

10.8. Appendix 8: Permitted Concomitant Medications with Dose Stabilization

For permitted antihyperglycemic medications, see Section 6.9.2.

Drug class	Dose modification guidance
Oral 5-ASAs for UC, for example, mesalamine, balsalazide, olsalazine, and sulfasalazine	The prescribed dose must have been stable for at least 2 weeks prior to the screening colonoscopy, with stable dose encouraged during the study unless the medication is discontinued due to a toxicity related to the medication.
Oral corticosteroids for UC (prednisone $\leq$ 20 mg per day or equivalent, budesonide MMX up to 9 mg per day, or beclomethasone dipropionate [gastro-resistant prolonged-release tablet] up to 5 mg per day)	Prescribed dose must have been stable for at least 2 weeks before screening colonoscopy and should remain stable until Week 12. After Week 12, see Section 6.9.5.
Immunomodulators: AZA, 6-MP, or methotrexate	Prescribed at stable dose for at least 8 weeks before screening colonoscopy. Doses should remain stable throughout study unless medication is discontinued due to a toxicity related to the medication
Corticosteroids to treat adrenal insufficiency, or locally administered corticosteroids (for example, inhaled, intranasal, intra-articular, or topical) for non-UC indications	May continue corticosteroids to treat adrenal insufficiency or locally administered corticosteroids during study with stable dose encouraged. Single doses, oral or IV, as premedication in participants with prior IV/other biologic administration reactions are allowed
Antidiarrheals, for example, loperamide or diphenoxylate with atropine	May continue during study with stable doses encouraged
Low-dose or baby aspirin (75 to 162.5 mg)	Daily use for cardiovascular prophylaxis permitted
Non-live (killed, inactivated, or subunit) vaccines	Allowed during the study. The efficacy of non-live vaccinations with administration of mirikizumab alone, mirikizumab concomitantly administered with tirzepatide, or tirzepatide alone, is unknown. If a permitted non-live vaccine is needed, it is recommended that study intervention not be administered on the same day as a vaccination
CBD products, if they are derived exclusively from hemp	Participants who use hemp-based CBD products must be on a stable dose for at least 10 days prior to randomization, and participants must remain on that stable dose during the study

Abbreviations: 5-ASA = 5-aminosalicylates; 6-MP = 6-mercaptopurine; AZA = azathioprine; CBD = cannabidiol; IV = intravenous; MMX = multi-matrix system; UC = ulcerative colitis.

10.9. Appendix 9: Standardized Protocols for the Measurement of Height, Weight, Waist Circumference, BMI, and Vital Signs

The following information has been adapted from standardized physical measurement protocols for the WHO's STEPwise approach to Surveillance (STEPS) Manual (2020).

Measuring height

Step 1. Ask the participant to remove their footwear and any headgear (light headgear worn for religious reasons can remain, but this should be worn by the participant at every clinic visit when their height is measured).

Step 2. Ask the participant to stand on the calibrated height measuring board (stadiometer) or against a wall with their feet together and their knees straight with their heels against the backboard, the stadiometer, or the wall.

Step 3. Ask the participant to look straight ahead without tilting their head up.

Step 4. Ask the participant to breathe in and stand tall. If using a stadiometer or fixed measuring device, move the device's measurement arm gently down onto the top of the participant's head. Record the participant's height.

Measuring weight

- Body weight measurements should be done in a consistent manner using a calibrated electronic scale capable of measuring weight in kilogram (kg).
- All weights for a given participant should be measured using the same scale, whenever possible.
- Participants should be lightly clothed but not wearing shoes while their weight is measured.

Step 1. Ask the participant to remove their footwear, outerwear (coat, jacket, etc.), and any headgear (light headgear worn for religious reasons can remain, but this should be worn by the participant at every clinic visit when weight is measured).

Step 2. Make sure the scale is placed on a firm, flat, even surface (not on carpet, on a sloping surface, or a rough, uneven surface).

Step 3. Ask the participant to step onto the scale with 1 foot on each side of the scale.

Step 4. Ask the participant to stand still with arms by sides and then record weight in kilogram (kg) to the nearest one-tenth kg.

Measuring waist circumference

- Use non-stretchy measuring tape.
- Waist circumference should be measured at midpoint, between lower margin of last palpable rib and top of iliac crest (approximately 1 inch [2.54 cm] above the navel)
- Participants should be lightly clothed.
- Measure to the nearest 0.5 cm.

Step 1. Ask the participant to stand with their feet close together, and arms at their side with their body weight evenly distributed.

Step 2. Ask participant to relax.

Step 3. Measurements should be recorded at the end of a normal expiration.

Calculation of BMI

Height and weight measurements will be used to calculate BMI.

- $BMI = \text{weight (kg)} / (\text{height [m]})^2$

Round to 1 decimal place (NIH [WWJ]). For example, a BMI of 26.6 kg/m² should **not** be rounded to 27.0 kg/m².

Calculation of BMI with amputation or limb loss

In participants with limb amputation or limb loss, use the formula given in the following link: touchcalc.com/calculators/bmi_amputation

Vital sign measurements (blood pressure and heart rate)

- Vital sign measurements, measured by pulse, should be taken before obtaining an ECG tracing and before collection of blood samples for laboratory testing.
- An appropriately sized cuff (cuff bladder encircling at least 80% of the arm) should be used to ensure the accuracy of blood pressure measurements.
- Blood pressure should be taken with an automated blood pressure instrument.
- If blood pressure and pulse measurements are taken separately, pulse should be taken prior to blood pressure.

Step 1. The participant should sit quietly for approximately 5 minutes before vital signs measurements are taken.

Step 2. One measurement will be taken using the nondominant arm.

Step 3. Measurement of sitting pulse and blood pressure needs to be recorded in the CRF.

Note: In the event pulse measurement cannot be taken via an automated blood pressure instrument, the preferred location for measurement of pulse is the radial artery.

10.10. Appendix 10: Examples of Infections that May Be Considered Opportunistic in the Setting of Biologic Therapy

This table lists examples of infections that may be considered opportunistic in the context of biologic therapy. This table is provided to aid the investigator in recognizing infections that may be considered opportunistic. This list is not exhaustive. For data analysis, infections will be categorized by Lilly as opportunistic according to the article by Winthrop et al. (2015).

Examples of Infections That May Be Considered Opportunistic in the Setting of Biologic Therapy

Bacterial
Bartonellosis (disseminated disease only)
Campylobacteriosis (invasive disease only)
Legionellosis
Listeriosis (invasive disease only)
Nocardiosis
Tuberculosis
Non-tuberculous mycobacterial disease
Salmonellosis (invasive disease only)
Shigellosis (invasive disease only)
Vibriosis (invasive disease due to <i>Vibrio vulnificus</i>)
Viral
BK virus disease including polyomavirus-associated nephropathy
Cytomegalovirus disease
Hepatitis B virus reactivation
Hepatitis C virus progression
Herpes simplex (invasive disease only)
Herpes zoster (any form)
Post-transplant lymphoproliferative disorder (Epstein-Barr virus)
Progressive multifocal leukoencephalopathy (PML), John Cunningham (JC) virus
Fungal
Aspergillosis (invasive disease only)
Blastomycosis
Candidiasis (invasive disease or oropharyngeal, esophageal. Not isolated lingual)
Coccidioidomycosis
Cryptococcosis
Histoplasmosis
Paracoccidioides infections
Penicilliosis
Pneumocystosis
Sporotrichosis
Other invasive molds: Mucormycosis (zygomycosis) (<i>Rhizopus</i> , <i>Mucor</i> , and <i>Lichtheimia</i>), <i>Scedosporium/Pseudallescheria boydii</i> , <i>Fusarium</i>
Parasitic
Leishmaniasis (visceral only)
Strongyloidiasis (hyperinfection syndrome or disseminated disease)
Microsporidiosis
Toxoplasmosis

	Trypanosoma cruzi infection (Chagas' disease progression) (disseminated disease only)
	Cryptosporidiosis (chronic disease only)

Source: Based on Winthrop et al. (2015).

10.11. Appendix 11: mMS for the Assessment of Ulcerative Colitis Disease Activity

Mayo Score by Category	
Stool Frequency^a	
0	Normal number of stools for this patient
1	1-2 more stools than normal
2	3-4 more stools than normal
3	5 or more stools more than normal
Rectal Bleeding^b	
0	No blood seen
1	Streaks of blood with stool less than half the time
2	Obvious blood with stool most of the time
3	Blood alone passed
Findings of flexible proctosigmoidoscopy	
0	Normal or inactive disease
1	Mild disease (erythema, decreased vascular pattern ^c)
2	Moderate disease (marked erythema, absent vascular pattern, friability, erosions)
3	Severe disease (spontaneous bleeding, ulceration)

a Each patient served as his or her own control to establish the degree of abnormality of the stool frequency. For details, see Section 8.1.1.1.

b Represents the worst bleeding score for that day.

c Friability is being excluded from the definition for ES=1 per Section 8.1.1.1.

Source: Based on FDA Guidance (2016).

10.12. Appendix 12: Geboes Scoring for the Assessment of Histological Activity

Subgrades	Scoring
Grade 0 Structural (architectural change)	0.0 No abnormality 0.1 Mild abnormality 0.2 Mild or moderate diffuse or multifocal abnormalities 0.3 Severe diffuse or multifocal abnormalities
Grade 1 Chronic inflammatory infiltrate	1.0 No increase 1.1 Mild but unequivocal increase 1.2 Moderate increase 1.3 Marked increase
Grade 2A Lamina propria eosinophils	2A.0 No increase 2A.1 Mild but unequivocal increase 2A.2 Moderate increase 2A.3 Marked increase
Grade 2B Lamina propria neutrophils	2B.0 None 2B.1 Mild but unequivocal increase 2B.2 Moderate increase 2B.3 Marked increase
Grade 3 Neutrophils in epithelium	3.0 None 3.1 <5% crypts involved 3.2 <50% crypts involved 3.3 >50% crypts involved
Grade 4 Crypt destruction	4.0 None 4.1 Probable—local excess of neutrophils in part of crypt 4.2 Probable—marked attenuation 4.3 Unequivocal crypt destruction
Grade 5 Erosion or ulceration	5.0 No erosion, ulceration, or granulation tissue 5.1 Recovering epithelium + adjacent inflammation 5.2 Probable erosion—focally stripped 5.3 Unequivocal erosion 5.4 Ulcer or granulation tissue

Note: For each histological feature assessed, the highest grade in which there is evidence of disease is assigned. For example, if <50% crypts involved is checked (that is, Geboes grade 3 is assigned a 2) and Crypt destruction is noted as “none” (4.0) and Erosion or ulceration is “No erosion, ulceration, or granulation tissue” (5.0), the participant will be assigned a score of 3.2.

Source: Adapted from Geboes et al. (2000).

10.13. Appendix 13: Genetics

Use or analysis of DNA

Genetic variation may impact a participant's response to study intervention and susceptibility to, severity of, and progression of disease. Variable response to study intervention may be due to genetic determinants that impact drug absorption, distribution, metabolism, and excretion; mechanism of action of the drug; disease etiology; molecular subtype of the disease being treated; or a combination of the listed.

Therefore, where local regulations and IRB or IEC allow, a blood sample will be collected for DNA analysis from consenting participants.

DNA samples may be used for research related to mirikizumab and/or tirzepatide or autoimmune/inflammatory diseases. They may also be used to develop tests or assays, including diagnostic tests related to mirikizumab and/or tirzepatide and other autoimmune/inflammatory diseases. Genetic research may consist of the analysis of 1 or more candidate genes or the analysis of genetic markers throughout the genome or analysis of the entire genome (as appropriate).

Analyses may be conducted if it is hypothesized that this may help further understand the clinical data.

The samples may be analyzed as part of a multi-study assessment of genetic factors involved in the response to mirikizumab and/or tirzepatide or study interventions of this class to understand study disease or related conditions.

The sponsor will store the DNA samples in a secure storage space with adequate measures to protect confidentiality.

For sample retention period, see Section [10.1.12](#).

10.14. Appendix 14: Region- or Country-Specific Requirements

10.14.1. All Countries or Regions Outside of the United States

Section 10.4.2 of the protocol, Contraception Guidance

For the OUS countries or regions, the following contraception criterion will apply to IOCBP who are NOT completely abstinent as their preferred and usual lifestyle, or who do NOT exclusively engage in sexual relations with other individual(s) who are AFAB as their preferred and usual lifestyle. The differences from the main protocol are highlighted with underline and ~~strikethrough~~.

Must...
Agree to use 1 highly effective method of contraception <u>Agree to use 1 highly effective method of non-oral contraception or add a barrier method of contraception for 4 weeks after initiation of tirzepatide and for 4 weeks after each dose escalation.</u> These methods of contraception must be used during the study and for at least 10 weeks (approximately 2.5 months) after the last dose of the study intervention.

10.14.2. European Union

The study will be conducted in accordance with the Regulation (EU) No 536/2014.

Regulatory reporting requirements

The sponsor has processes for safety reports for identification, recording, and expedited reporting of SUSARs according to EU regulatory requirements. The sponsor will comply with applicable regulatory requirements relating to safety reporting to the regulatory authority, per European Union Clinical Trial Regulation 536/2014 submission of SUSARs to the EudraVigilance database.

10.14.2.1. Germany

Section 5.2, Exclusion Criteria

Based on the German Drug Law, Section 40a, the following exclusion criterion will apply to the Study AMCD participants in Germany:

- 61. Are submitted to/incarcerated in an institution by virtue of an order of a court of law or a competent governmental body.

Section 10.16. Provisions for Changes in Study Conduct During Exceptional Circumstances

In Germany, the temporary measures described in the “Provisions for Changes in Study Conduct During Exceptional Circumstances” (Section 10.16) are applicable only to “the COVID-19 pandemic.”

10.14.2.2. Hungary

The differences from the main protocol are highlighted with underline and ~~striketrough~~.

Section 8.3.3.7 Suicidal Ideation or Behavior

Columbia-Suicide Severity Rating Scale

The C-SSRS captures the occurrence, severity, and frequency of SIB during the assessment period via a semi-structured interview ~~by an appropriately trained healthcare professional~~. The C-SSRS scale will be administered by a psychiatrist or clinical psychologist with demonstrated expertise in the administration of the scale and assessment of suicidal ideation.

10.14.3. Brazil

This section describes protocol changes applicable for adult participants at study sites in Brazil.

This table describes the changes and provides a rationale for the changes.

Protocol Section Number and Name	Description of the Change	Brief Rationale
10.1.12. Sample Retention	Biological samples will be stored for up to 10 years.	Compliance with Brazilian regulation applicable to sample storage, resolution CNS 441 (Brazil 2011).
10.13. Appendix 13: Genetics	Biological samples obtained to evaluate genetic material (DNA/RNA).	Compliance with Brazilian laws and regulations (CNS 340/2004 and CNS 441/2011).
Patient Access to the Project Benefits	New section specific to Brazil.	Clarifies the sponsor responsibilities to comply with Resolution CNS 466 (Brazil 2012) and RDC 38 (Brazil 2013).
Section 11. References	Addition of Brazil-specific references	The references are specific to the Brazil-specific requirements.

The revised text described below shows the changes applicable to adult participants at study sites in Brazil. Deletions are identified by striketrough format and additions by underlined text.

10.1.12. Sample Retention

In Brazil, the biological samples obtained within this study will be stored for up to 10 years, with possibility of renewal under request, followed by appropriate justification and a report with all activities developed with the biological samples (CNS 441/2011). The sample and any data generated from it can be linked back to the participant only by investigator site personnel. The duration allows the sponsor to respond to regulatory requests related to the study intervention.

10.13. Appendix 13: Genetics

In Brazil, the biological samples from this study used to evaluate genetic material (DNA/RNA) will follow the Brazilian laws and regulations (CNS 340/2004 and CNS 441/2011). The sample and any data generated from it can be linked back to the participant only by investigator site personnel. The duration allows the sponsor to respond to regulatory requests related to the study drug.

Patient Access to the Project Benefits

In Brazil, at the end of their participation in the study, all participants must have assured access to the best proven prophylactic, diagnostic, and therapeutic methods, which may include mirikizumab and tirzepatide, identified through the study (CNS 466/2012 and RDC 38/2013).

11. References

BRASIL, MINISTÉRIO DA SAÚDE. Resolução 251/1997 do Conselho Nacional de Saúde/MS: Normas de Pesquisa com Novos Fármacos, Medicamentos, Vacinas e Testes Diagnósticos Envolvendo Seres Humanos.

BRASIL, MINISTÉRIO DA SAÚDE. Resolução 466/2012 do Conselho Nacional da Saúde / MS: Diretrizes e Normas Regulamentadoras de Pesquisas Envolvendo Seres Humanos. Publicado no Diário Oficial da União em 13 de junho, 2013.

BRASIL, MINISTÉRIO DA SAÚDE. Resolução 340/2004 do Conselho Nacional da Saúde/MS: Diretrizes para Análise Ética e Tramitação dos Projetos de Pesquisa da Área Temática Especial de Genética Humana.

BRASIL, MINISTÉRIO DA SAÚDE. Resolução 441/2011 do Conselho Nacional da Saúde / MS: Diretrizes para análise ética de projetos de pesquisas que envolvam armazenamento de material biológico humano ou uso de material armazenado em pesquisas anteriores.

BRASIL, MINISTÉRIO DA SAÚDE. Resolução-RDC 38/2013 da Agência Nacional de Vigilância Sanitária/MS: Aprova o regulamento para os programas de acesso expandido, usocompassivo e fornecimento de medicamento pós-estudo.

10.15. Appendix 15: Recommendations for the Safe Use of GLP-1 Receptor Agonists During Anesthesia or Deep Sedation

These recommendations are per the Multisociety Clinical Practice Guidance for the Safe Use of Glucagon-like Peptide-1 Receptor Agonists in the Perioperative Period (Kindel et al. 2024; content used with permission) by the

- American Gastroenterological Association
- American Society for Metabolic and Bariatric Surgery
- American Society of Anesthesiologists
- International Society of Perioperative Care of Patients with Obesity, and
- Society of American Gastrointestinal and Endoscopic Surgeons.

Recommendation 1

Use of GLP-1 receptor agonists in the perioperative period should be based on shared decision-making of the patient with procedural, anesthesia, and prescribing care teams balancing the metabolic need for the GLP-1RA with individual patient risk. This can be achieved by developing multidisciplinary protocols/procedures appropriate for individual practices.

a) Care teams should consider the following variables as elevating the risk of delayed gastric emptying and aspiration with the periprocedural use of GLP-1 receptor agonist:

1. *Escalation phase*: The escalation phase, versus the maintenance phase, is associated with a higher risk of delayed gastric emptying with GLP-1 receptor agonist usage (Sorli et al. 2017; AASLD/ACG/AGA/ASGE/NASPGHAN Multisociety Statement 2023; Camilleri et al. 2024; Hashash et al. 2024).
2. *Higher dose*: The higher the dose of GLP-1 receptor agonist, the more likely the risk of gastrointestinal side effects (Sorli et al. 2017; AASLD/ACG/AGA/ASGE/NASPGHAN Multisociety Statement 2023; Camilleri et al. 2024; Hashash et al. 2024).
3. *Weekly dosing*: GI side effects are more common with weekly compared to daily formulation compounds (Ahmann et al. 2018).
4. *Presence of gastrointestinal symptoms*: Symptoms suggestive of delayed gastric emptying and intestinal transit times may include nausea, vomiting, abdominal pain, dyspepsia, and constipation (Silveira et al. 2023).
5. *Medical conditions beyond GLP-1 receptor agonist usage which may also delay gastric emptying*: Patients on GLP-1 receptor agonist should be evaluated for other medical conditions which may exacerbate GI symptoms and delay gastric emptying, such as but not limited to bowel dysmotility, gastroparesis, and Parkinson's disease.

The assessment for these risk factors should occur with enough advance time prior to surgery to allow adjustments in preoperative care if indicated, including diet modification and evaluation of the feasibility of medication bridging if GLP-1 receptor agonist discontinuation is indicated.

b) GLP-1 receptor agonist therapy may be continued preoperatively in patients without elevated risk of delayed gastric emptying and aspiration based on Recommendation 1a. When an elevated risk of delayed gastric emptying and aspiration exist, withholding of GLP-1 receptor agonists should be balanced with the surgical and medical risk of inducing the

potential for a hazardous, metabolic disease state, like hyperglycemia. Further, bridging therapy off a GLP-1 receptor agonist may be resource-intensive, cost or insurance prohibitive, and risk other adverse side effects like hypoglycemia. Finally, withholding GLP-1 receptor agonist perioperatively only for patients with the diseases of overweight and obesity, without an indication as described in Recommendation 1a, could constitute overweight and obesity bias, which should be avoided.

c) If the decision to hold GLP-1 receptor agonists is indicated given an unacceptable safety profile following shared decision-making in the preoperative period, the duration to hold therapy is unknown (Joshi 2024). At this time, it is suggested to follow the original guidance of the American Society of Anesthesiologists, holding the day of surgery for daily formulations, and a week prior to surgery for weekly formulations (Joshi et al. 2023). All patients should still be assessed on the day of procedure for symptoms suggestive of delayed gastric emptying.

Recommendation 2

The safe use of GLP-1 receptor agonists in the perioperative period should include efforts to minimize the aspiration risk of delayed gastric emptying. This can be achieved by preoperative diet modification and/or altering anesthesia plan to consider rapid sequence induction of general anesthesia for tracheal intubation.

a) Preoperative diet modification (preoperative liquid diet for at least 24 hours, as performed in patients undergoing colonoscopy and bariatric surgery) can be utilized in patients when there is concern for delayed gastric emptying based on clinical symptom review as described in Recommendation 1a (Silveira et al. 2023; Hashash et al. 2024; Hiramoto et al. 2024).

b) When clinical concern for retained gastric contents exists on the day of the procedure, point-of-care gastric ultrasound could be used to assess aspiration risk. This technology may be clinically limited based on institutional resources, interuser variability, and credentialing requirements (Joshi et al. 2023; Sen et al. 2024).

c) When clinical concern for retained gastric contents exists or is confirmed on the day of the procedure, providers should engage patients in a shared decision-making model and consider the benefits and risks of rapid sequence induction of general anesthesia for tracheal intubation to minimize aspiration risk versus procedure cancellation (Joshi et al. 2023; Hashash et al. 2024;).

Safe continuation of surgery and GI endoscopy, and prevention of procedure cancellation, for patients on GLP-1 receptor agonists can be prioritized following the recommendations above, as would occur for other patient populations with gastroparesis.

10.16. Appendix 16: Provisions for Changes in Study Conduct During Exceptional Circumstances

Implementation of this appendix

The changes to procedures described in this appendix are temporary measures intended to be used only during specific time periods as directed by the sponsor in partnership with the investigator.

In an exceptional circumstance, the sponsor's procedures ensure that relevant parties are kept informed of changes to the design and conduct of the study.

Exceptional circumstances

Exceptional circumstances are rare events that may cause disruptions to the conduct of the study. Examples include pandemics or natural disasters. These disruptions may limit the ability of the investigators, participants, or both to attend on-site visits or to conduct planned study procedures.

Implementing changes under exceptional circumstances

In an exceptional circumstance, after receiving the sponsor's written approval, sites may implement changes if permitted by local regulations.

After approval by local Ethical Review Boards, regulatory bodies and any other relevant local authorities, implementation of these exceptional circumstance changes will not typically require additional notification to these groups, unless they have specific requirements in which notification is required (for example, upon implementation and suspension of changes). All approvals and notifications must be retained in the study records.

If the sponsor grants written approval for changes in study conduct, the sponsor will also provide additional written guidance, if needed.

Considerations for making a change

The prevailing consideration for making a change is ensuring the safety of study participants. Additional important considerations for making a change are compliance with GCP, enabling participants to continue safely in the study and maintaining the integrity of the study.

Informed consent

Additional consent from the participant will be obtained, if required, for

- participation in remote visits, as defined in Section "Remote visits"
- a change in the method of study intervention administration
- dispensation of additional study intervention during an extended treatment period
- alternate delivery of study intervention and ancillary supplies, and
- provision of their personal or medical information required prior to implementation of these activities.

Changes in study conduct during exceptional circumstances

Changes in study conduct not described in this appendix, or not consistent with applicable local regulations, are not allowed.

The following changes in study conduct will not be considered protocol deviations.

Remote visits***Types of remote visits***

Telemedicine: Telephone or technology-assisted virtual visits, or both, are acceptable to complete appropriate assessments. Assessments to be completed in this manner include, but are not limited to,

- review of
 - AEs and product complaints
 - concomitant medications
 - electronic diary compliance
 - study intervention compliance (for at-home administration of SC mirikizumab and tirzepatide)
- any self-report questionnaires required from the participant at that visit may be collected by site staff verbally along with all other procedures that can be carried out through verbal interaction

Mobile healthcare: Healthcare visits may be performed by a mobile healthcare provider at locations other than the study site when participants cannot travel to the site due to an exceptional circumstance if written approval is provided by the sponsor. Procedures performed at such visits include, but are not limited to, vital signs, weight, physical examinations, evaluations for EIMs, ECGs, collection of blood and urine samples, patient-reported outcomes via the daily eDiary or a web-based collection system, and administration of study intervention.

Other alternative locations: Procedures that could be done at an alternate location in very exceptional circumstances include, but are not limited to, ECGs, chest x-rays, colonoscopies, colonic biopsy sample collections, collections of blood and urine samples, and administration of study intervention, for example, at an infusion center.

Data capture

In source documents and the CRF, the study site should capture the visit method, with a specific explanation for any data missing because of missed in-person site visits.

Safety reporting

Regardless of the type of remote visits implemented, the protocol requirements regarding the reporting of AEs, SAEs, and product complaints remain unchanged.

Return to on-site visits

Every effort should be made to enable participants to return to on-site visits as soon as reasonably possible, while ensuring the safety of both the participants and the site staff.

Local laboratory testing option

Local laboratory testing may be conducted in lieu of central laboratory testing. However, central laboratory testing must be retained for:

- hsCRP
- PK
- immunoglobulin panel
- genetics sample
- tests of exploratory biomarker samples
- fecal calprotectin, and
- tests obtained because of a systemic hypersensitivity event.

The local laboratory must be qualified in accordance with applicable local regulations.

Study intervention and ancillary supplies (including participant diaries)

When a participant is unable to go to the site to receive study supplies during normal on-site visits, the site should work with the sponsor to determine appropriate actions. These actions may include

- asking the participant to go to the site and receive study supplies from site staff without completion of a full study visit
- asking the participant's designee to go to the site and receive study supplies on a participant's behalf
- arranging delivery of study supplies, and
- working with the sponsor to determine how mirikizumab infusion that is typically administered on site will be administered to the participant; for example, during a mobile healthcare visit or at an alternate location such as an infusion center.

These requirements must be met before action is taken:

- Alternate delivery of study intervention should be performed in a manner that does not compromise treatment blinding and ensures product integrity. The existing protocol requirements for product accountability remain unchanged, including verification of participant's receipt of study supplies.
- When delivering supplies to a location other than the study site (for example, participant's home), the investigator, sponsor, or both should ensure oversight of the shipping process to ensure accountability and product quality (that is, storage conditions maintained and intact packaging upon receipt).
- Instructions may be provided to the participant or designee on the final disposition of any unused or completed study supplies.

If study intervention will be administered to the participant during a mobile healthcare visit or at an alternate location, these additional requirements must be met:

- Only authorized study personnel may supply, prepare or administer study intervention.
- Locations must have resuscitation equipment, emergency medications, and appropriately trained personnel available during the infusion and post-infusion monitoring period, as specified in the protocol.

Screening period guidance

To ensure safety of study participants, laboratory values and other eligibility assessments taken at screening visit are valid for a maximum of 35 days.

The following rules will be applied for active, nonrandomized participants whose participation in the study must be interrupted due to exceptional circumstances.

- If screening is interrupted for less than 35 days from screening to randomization visit, the participant will proceed to the next study visit per the usual SoA, provided that randomization visit must be conducted within 35 days from first screening visit. The site should conduct the next visit if the participant's eligibility criteria are confirmed, and the site should document the reason for delay. The site should also reconfirm the impacted participant's consent and document this confirmation in the source documentation.
- If screening is interrupted for more than 35 days from screening visit to randomization visit, the participant must be discontinued because of screening interruption due to an exceptional circumstance. This is documented as a screen failure in the CRF. The participant can reconsent and be rescreened as a new participant. This rescreen is in addition to the 1 or 2 allowed by the main protocol. The screening procedures per the usual SoA should be followed, starting at screening visit to ensure participant eligibility by randomization visit.

Adjustments to visit windows

Whenever possible and safe to do so, as determined by the investigator's discretion, participants should complete the usual SoA. To maximize the possibility that these visits can be conducted as on-site visits, the windows for visits may be adjusted, upon further guidance from the sponsor. This minimizes missing data and preserves the intended conduct of the study.

This table describes the allowed adjustments to visit windows.

For this visit...	Permitted Visit Types	Visit Interval Tolerance ^a
Screening/Baseline and Induction and Dose Escalation Periods (SoA Table 1)		
Visit 1	On-site only	Same as shown in SoA, but flexibility can be considered following consultation with, and with prior approval by, the sponsor
Visit 2	On-site only	Same as shown in the SoA
Visit 3 through Visit 7	Remote	Same as shown in the SoA
Visit 8	Remote	For colonoscopy with biopsy: –10 days relative to the targeted visit date (inclusive) For other procedures: same as shown in the SoA
Maintenance Period (SoA Table 2)		
Visit 9 through Visit 14	Remote	Same as shown in the SoA

For this visit...	Permitted Visit Types	Visit Interval Tolerance ^a
Visit 15	Remote	For colonoscopy with biopsy: –3 to +56 days after the targeted visit date (inclusive) For other procedures: same as shown in the SoA
Early discontinuation visit, unscheduled visit, and Follow-Up Period (SoA Table 2)		
EDV	Remote	For colonoscopy with biopsy: within 56 days after the early discontinuation decision. Note: Colonoscopy and colonic biopsy sample collection are recommended at EDV based on judgment of the investigator and after discussion with the sponsor's designated medical monitor. If not performed at EDV, this will not be considered a protocol deviation. For other procedures: same as shown in the SoA
UV	Remote	Not applicable
Visit 801	Remote	Same as shown in the SoA

Abbreviations: EDV = early discontinuation visit; SoA = schedule of activities; UV = unscheduled visit.

- ^a For participants whose visits have extended windows, additional study intervention may need to be provided to avoid study intervention interruption and maintain overall integrity of the study.

Documentation

Changes to study conduct will be documented

Sites will identify and document the details of how participants, visit types, and conducted activities were affected by exceptional circumstances. Dispensing/shipment records of study intervention and relevant communications, including delegation, should be filed with site study records.

Source documents at alternate locations

Source documents generated at a location other than the study site should be part of the investigator's source documentation and should be transferred to the site in a secure and timely manner.

10.17. Appendix 17: Abbreviations and Definitions

Term	Definition
5-ASA	5-aminosalicylates
6-MP	6-mercaptopurine
abuse	Use of a study intervention for recreational purposes or to maintain an addiction or dependence
ADA	anti-drug antibodies
AE	adverse event
AESI	adverse event of special interest
AFAB	assigned female at birth
ALP	alkaline phosphatase
ALT	alanine aminotransferase
AMAB	assigned male at birth
ANCOVA	analysis of covariance
anti-HBc	anti-hepatitis B core antibody
AST	aspartate aminotransferase
authorized IMP	<i>Applicable to the EU only:</i> a medicinal product authorized in accordance with Regulation (EC) No 726/2004 or in any Member State concerned in accordance with Directive 2001/83/EC, irrespective of changes to the labelling of the medicinal product, which is used as an investigational medicinal product
authorized AxMP	<i>Applicable to the EU only:</i> a medicinal product authorized in accordance with Regulation (EC) No 726/2004, or in any Member State concerned in accordance with Directive 2001/83/EC, irrespective of changes to the labelling of the medicinal product, which is used as an auxiliary medicinal product
AxMP	auxiliary medicinal product. See also NIMP. A medicinal product used for the needs of a clinical trial as described in the protocol, but not as an investigational medicinal product. Examples include rescue medication, challenge agents, agents to assess endpoints in the clinical trial, or background treatment. AxMP does not include investigational medicinal product (IMP) or concomitant medications. Concomitant medications are medications unrelated to the clinical trial and not relevant for the design of the clinical trial
AZA	azathioprine
BCG	Bacillus Calmette-Guerin [vaccination]

Term	Definition
blinding/masking	A single-blind study is one in which the investigator and/or the investigator's staff are aware of the treatment but the participant is not, or vice versa, or when the sponsor is aware of the treatment but the investigator and/the investigator's staff and the participant are not. A double-blind study is one in which neither the participant nor any of the investigator or sponsor staff who are involved in the treatment or clinical evaluation of the participants are aware of the treatment received.
BMI	body mass index
C-SSRS	Columbia-Suicide Severity Rating Scale
<i>C. difficile</i>	<i>Clostridioides difficile</i>
CBD	cannabidiol
CD	Crohn's disease
CEC	Clinical Event Committee
CFR	Code of Federal Regulations
CI	confidence interval
CKD-EPI	Chronic Kidney Disease Epidemiology Collaboration
complaint	A complaint is any written, electronic, or oral communication that alleges deficiencies related to the identity, quality, purity, durability, reliability, safety or effectiveness, or performance of a drug or drug delivery system.
compliance	Adherence to all study-related, good clinical practice (GCP), and applicable regulatory requirements.
CONSORT	Consolidated Standards of Reporting Trials
COVID-19	Coronavirus disease 2019
CRF	case report form; a printed, optical, or electronic document designed to record all of the protocol-required information to be reported to the sponsor for each trial participant.
CT	computed tomography
DMC	data monitoring committee. A data monitoring committee, or data monitoring board (DMB) is a group of independent scientists who are appointed to monitor the safety and scientific integrity of a human research intervention, and to make recommendations to the sponsor regarding the stopping of a study for efficacy, or for harm, or for futility. The composition of the committee is dependent upon the scientific skills and knowledge required for monitoring the particular study.
DPP-4	dipeptidyl peptidase-4

Term	Definition
DXA	dual-energy x-ray absorptiometry
ECG	electrocardiogram
eCOA	electronic Clinical Outcome Assessment
EDC	electronic data capture system
EDV	early discontinuation visit
eGFR	estimated glomerular filtration rate
enroll	The act of assigning a participant to a treatment. Participants who are enrolled in the study are those who have been assigned to a treatment.
enter	Participants entered into a study are those who sign the informed consent form directly.
ES	endoscopic subscore
FSH	follicle-stimulating hormone
GCP	good clinical practice
GDPR	EU General Data Protection Regulation
GI	gastrointestinal
GIP	glucose-dependent insulintropic polypeptide
GLP-1	glucagon-like peptide-1
HBsAg	hepatitis B surface antigen
HBV	hepatitis B virus
HCV	hepatitis C virus
hsCRP	high-sensitivity C-reactive protein
IB	Investigator's Brochure
IBD	inflammatory bowel disease
IBDQ	Inflammatory Bowel Disease Questionnaire
ICE	intercurrent event
ICF	informed consent form
ICH	International Council for Harmonisation

Term	Definition
IEC	independent ethics committee
IGRA	interferon-gamma release assay
IL	interleukin
IMP	Investigational Medicinal Product (see also “investigational product”) A medicinal product which is being tested or used as a reference, including as a placebo, in a clinical trial.
informed consent	A process by which a participant voluntarily confirms their willingness to participate in a particular study, after having been informed of all aspects of the study that are relevant to the participant’s decision to participate. Informed consent is documented by means of a written, signed and dated informed consent form.
investigational product	A pharmaceutical form of an active ingredient or placebo being tested or used as a reference in a clinical trial, including products already on the market when used or assembled (formulated or packaged) in a way different from the authorized form, or marketed products used for an unauthorized indication, or marketed products used to gain further information about the authorized form. See also “IMP.”
INOCBP	individuals not of childbearing potential
IOCBP	individuals of childbearing potential
IRB	independent review board
ISR	injection site reaction or infusion site reaction
IV	intravenous
IWRS	interactive web-response system
JAK	Janus kinase
LTBI	latent tuberculosis infection
MACE	major adverse cardiovascular events
MDD	major depressive disorder
MedDRA	Medical Dictionary for Regulatory Activities

Term	Definition
medication error	Errors in the prescribing, dispensing, or administration of a study intervention, regardless of whether or not the medication is administered to the participant or the error leads to an AE. Medication error generally involve a failure to uphold one or more of the five “rights” of medication use: 1. the right participant 2. the right drug 3. the right dose 4. the right route, 5. at the right time.
misuse	Use of a study intervention for self-treatment that either is inconsistent with the prescribed dosing regimen, indication, or both, or is obtained without a prescription
mITT	modified intent to treat
mMS	modified Mayo score
MMX	multi-matrix system
MRI	magnetic resonance imaging
MMRM	mixed-effects for repeated measures
MTC	medullary thyroid carcinoma
MTD	maximum tolerated dose
NASH	nonalcoholic steatohepatitis
NIMP	Non-investigational Medicinal Product See AxMP. A medicinal product used for the needs of a clinical trial as described in the protocol, but not as an investigational medicinal product. Examples include rescue medication, challenge agents, agents to assess endpoints in the clinical trial, or background treatment.
NRS	numeric rating scale
participant	Equivalent to CDISC term “subject”: an individual who participates in a clinical trial, either as recipient of an investigational medicinal product or as a control
PCR	polymerase chain reaction
PGR-S	Patient’s Global Rating of Severity
PI	principal investigator
PK	pharmacokinetics
PPD	purified protein derivative

Term	Definition
Q4W	every 4 weeks
QIDS-SR16	Quick Inventory for Depressive Symptomatology – Self-Report 16 items
QTc	corrected QT interval
QTL	quality tolerance limit
QW	once weekly
RB	rectal bleeding
SAC	statistical analysis center
SAE	serious adverse event
SAP	statistical analysis plan
SC	subcutaneous
screen	The act of determining if an individual meets minimum requirements to become part of a pool of potential candidates for participation in a clinical study.
SF	Stool Frequency
SIB	suicidal ideation and behavior
SOC	system organ class
SUSAR	suspected unexpected serious adverse reactions Refers to an adverse event that occurs in a clinical trial participant, which is assessed by the sponsor and or study investigator as being unexpected, serious and as having a reasonable possibility of a causal relationship with the study intervention.
T1DM	type 1 diabetes mellitus
T2DM	type 2 diabetes mellitus
TB	tuberculosis
TBL	total bilirubin level
TE	treatment emergent
TEAE	Treatment-emergent adverse event: An untoward medical occurrence that emerges during a defined treatment period, having been absent pretreatment, or worsens relative to the pretreatment state, and does not necessarily have to have a causal relationship with this treatment.
TNF	tumor necrosis factor
TST	tuberculin skin test

Term	Definition
UC	ulcerative colitis
ULN	upper limit of normal
WBC	white blood count
WNOCBP	women not of childbearing potential
WOCBP	women of childbearing potential
WPAI:UC	Work Productivity and Activity Impairment Questionnaire: Ulcerative Colitis

10.18. Appendix 18: Protocol Amendment History

Amendment [a]: (17 Apr 2025)

This amendment is considered to be nonsubstantial.

Overall rationale for the amendment

For consistency with prior mirikizumab studies, the inclusion criteria [12b] is updated with the following maximum baseline liver enzyme levels:

- alanine aminotransferase (ALT) and aspartate aminotransferase (AST) ≤ 2 x ULN
- alkaline phosphatase (ALP) ≤ 1.5 x ULN, and
- total bilirubin level (TBL) ≤ 1.5 x ULN.

To specify that any participant who experiences serious or severe GI AEs that are consistent with worsening of UC and do not improve with the recommendations for management of intolerable GI symptoms related to tirzepatide will be evaluated for UC disease worsening.

To clarify that every symptom-directed examination will also include an examination of the heart, lungs, abdomen, neurological system, thyroid, and visual examination of the skin, consistent with the existing text in the schedule of activities under “Complete Physical Examination.”

To add acute kidney injury as an AESI to the protocol.

To include pregnancy testing every 4 weeks. This can include home urine pregnancy testing.

To update that follow-up information for pregnancy outcomes for participant and neonate will be collected until termination of the pregnancy or for at least 12 months post-delivery.

Section # and Name	Description of Change	Brief Rationale
Section 1.1. Synopsis Section 6.1. Study Intervention(s) Administered	Footnote a (mirikizumab): • added the first sentence to read “Mirikizumab is authorized in EU for the treatment of UC.”	Mirikizumab is approved for treatment of UC in the EU
	Added a sentence “In this study, clinical trial material will be used for mirikizumab”	Clarification on why it is classified as “Authorized and not used according to EU authorization”
	Authorized as defined by EU Clinical Trial Regulation row, Tirzepatide column: revised to read “Authorized and not used according to EU authorization ^b ”	Per addition of footnote ‘b’
Section 1.1. Synopsis Section 6.1. Study Intervention(s) Administered	Footnote ‘b’ added: “Tirzepatide is authorized in EU for the treatment of T2DM and for weight management, including weight loss and weight maintenance, in patients with obesity or overweight with at least one weight-related comorbid condition. In this study, clinical trial material will be used for tirzepatide.”	Clarification on why it is classified as “Authorized and not used according to EU authorization”

Section # and Name	Description of Change	Brief Rationale
Section 1.1. Synopsis Section 5.1. Inclusion criteria Section 8.1.1.2. Endoscopy: colonoscopy or sigmoidoscopy	Updated screening window for endoscopy from “14 days” to “21 days” prior to randomization.	To allow the investigator more time to schedule the procedure.
Section 1.3. Schedule of Activities Section 8.2.1. Physical examination	Updated Symptom-directed physical assessment to include an examination of the heart, lungs, abdomen, neurological system, thyroid, and visual examination of the skin.	To align with the FDA recommendations to include thyroid examination as part of physical examination
Section 1.3. Schedule of Activities	Symptom-directed physical assessment: updated a comment to read “At least every third ...”	For clarity and consistency with Section 8.2.10. Tuberculosis Testing and Monitoring
Section 1.3. Schedule of Activities	Added a comment to Lifestyle counseling Visits “The window period for these visits should be up to 14 days”	To enable investigators to complete lifestyle counseling.
Section 1.3. Schedule of Activities	Added comment for Stool culture as follows “Additional stool testing is allowed throughout the study at the investigator’s discretion.	Clarification
Section 1.3.1. Schedule of Activities for the Screening and Induction/Dose Escalation Period	Updated comment for DXA scan to read as follows: “It is recommended to perform successful baseline scan between Visits 1 and 2 and no later than Visit 3”	Clarification
Section 1.3.1. Schedule of Activities for the Screening and Induction/Dose Escalation Period	Included <i>C. difficile</i> testing at Visit 2 to 8	Clarification
Section 1.3.2. Schedule of Activities for the Maintenance Period, Early Discontinuation Visit, Unscheduled Visits, and Safety Follow-Up	Added urine pregnancy test for telehealth visits on Weeks 28, 36, 44, and 48.	To align with the FDA recommendations for pregnancy testing at virtual visits
Section 1.3.2. Schedule of Activities for the Maintenance Period, Early Discontinuation Visit, Unscheduled Visits, and Safety Follow-Up	Added hematology, clinical chemistry, hsCRP, and fecal calprotectin tests to an unscheduled visit.	To align with the FDA recommendations as evaluation measures for disease progression
Section 1.3.2. Schedule of Activities for the Maintenance Period, Early Discontinuation Visit, Unscheduled Visits, and Safety Follow-Up	Adding fecal calprotectin and residual exploratory stool supernatant at an unscheduled visit.	Clarification
Section 1.3.2. Schedule of Activities for the Maintenance Period, Early Discontinuation Visit, Unscheduled Visits, and Safety Follow-Up	Added sigmoidoscopy or colonoscopy per investigator’s discretion to an unscheduled visit.	To align with the FDA recommendations as evaluation measures for disease progression
Section 4.2. Scientific Rationale for Study Design	Deleted the following statement: “This study design includes investigator and participant masking”	Correction

Section # and Name	Description of Change	Brief Rationale
Section 5.1 Inclusion Criteria Section 5.3.1. Health Promotion and Lifestyle Program	Updated inclusion criterion [6] and Section 5.3.1. providing a qualification for dietitian or nutritionist for dietary counseling.	Clarification
Section 5.1 Inclusion Criteria Section 7.1.2. Temporary Discontinuation	Diagnostic assessments Updated inclusion criterion [12b] for clinical laboratory tests at screening from ‘ALT and AST <5 times ULN’ to ‘ALT and AST ≤2 times ULN’ Updated discontinuation criteria for diagnosis of LTBI during study, to read as follows “...if there is no evidence of hepatotoxicity (ALT/AST must remain ≤2times ULN)..”	To align with the FDA’s request to limit allowable baseline enzyme elevations
Section 5.2. Exclusion Criteria	Exclusion criterion [17] updated to read “Have evidence of toxic megacolon, or stricture or stenosis within the colon that cannot be traversed by a sigmoidoscope or colonoscope”	Correction
Section 5.2. Exclusion Criteria	Added “golimumab biosimilar” to Exclusion criterion [50]	Clarification
Section 5.3.2. Meals and Dietary Restrictions	Updated individualized counseling from ‘Visit 3’ to ‘Visit 2’	Correction
Section 6.1.2. Self-Administration Training and Dosing Diary	Replaced “investigator” with “site personnel” regarding who should perform or document self-administration training to read as follows “The site personnel must document...”	Correction
Section 6.6.2.1. Management of Participants with Gastrointestinal Symptoms	Updated management of GI symptoms in participants.	The FDA recommendations for the management of participants with GI symptoms
Section 6.6.2.2. Management of Participants with Perceived Excessive Weight Reduction	Updated first evaluation statement for malnutrition to read from “...the participant may have malnutrition...” to “...the participant is at risk of malnutrition...”	Clarification
Section 6.6.2.3. Permanent Dose Reduction	Added “or re-escalation attempts” to notes section for recurrent intolerable GI AEs.	Clarification
Section 6.9.6. Disease Progression and Rescue Therapy	Added evaluation measures for disease progression	To align with the FDA recommendations
Section 8.1.2.6. Inflammatory Biomarkers	Deleted investigators and site personnel blinding statements for hsCRP and fecal calprotectin results	Commitment to the FDA
Section 8.1.3.1. Assessment of Change in Body Fat Mass using Dual-Energy X-Ray Absorptiometry	Updated first paragraph for DXA assessment to read as follows: “It is recommended that study participants undergo..., measured between Visits 1 and 2 and no later than Visit 3”	Clarification
Section 8.2.9. Hepatic Safety Monitoring, Evaluation, and Criteria for Study Intervention Interruption or Discontinuation	Updated hepatic monitoring details	Updated hepatic monitoring details to align with the FDA’s request to limit allowable baseline enzyme elevations

Section # and Name	Description of Change	Brief Rationale
Section 8.3.2.1. Participants Who Become Pregnant	Updated follow-up period for pregnancy outcomes from “6 to 8 weeks” to “until termination of the pregnancy or for at least 12 months post-delivery”	To align with the FDA’s, recommendations for follow-up period to determine outcome of pregnancy
Section 8.3.3. Adverse Events of Special Interest	Included new Section 8.3.3.12. to include acute kidney injury as an AESI	As per the FDA’s request, changes made to align with tirzepatide label Warnings and Precautions section
Section 9. Statistical Considerations	Updated the wording of the first sentence from “first participant visit” to “first unblinding”	Clarification
Section 10.2. Appendix 2: Clinical Laboratory Tests	Updated comment in Micronutrient panels row to read from “For collecting samples in case of suspected malnutrition...” to “For collecting samples at risk of malnutrition...”	Clarification
Section 10.2. Appendix 2: Clinical Laboratory Tests	In Urine pregnancy row: added a comment stating that, “For telehealth visits home pregnancy test will be performed”	To align with the FDA recommendations
Section 10.2. Appendix 2: Clinical Laboratory Tests	In Pharmacokinetic samples row: added a comment stating that “Results will not be provided to the investigative sites”	Correction
Section 10.2. Appendix 2: Clinical Laboratory Tests	In Fecal calprotectin row: Deleted comment “Results will not be provided to the investigative sites”	Correction
Section 10.6. Appendix 6	Added footnote providing cross-reference to assessment of disease worsening provided in Section 6.9.6.	Clarification
Throughout protocol	Minor editorial and formatting changes	For consistency and editorial corrections

Amendment [b]

This amendment is considered to be nonsubstantial.

Overall rationale for the amendment

The overall rationale for the amendment is to update the protocol based on response provided to assessment RFIs on the EU Part I dossier.

Section # and Name	Description of Change	Brief Rationale
Section 1.3 Schedule of Activities (SoA)	Added Dosing diary review and return for unscheduled visit	Clarification
Section 2.3.1.2 Study Procedures	Updated table header as per original protocol	Correction of inadvertent deletions
Section 4.2 Scientific Rationale for Study Design	Updated procedure for collection of race and ethnicity information	To clarify self-reporting process for collection of race and ethnicity

Section # and Name	Description of Change	Brief Rationale
Section 6.1 Study Intervention(s) Administered	Updated dosage level details of Tirzepatide and Placebo as per original protocol	Correction of inadvertent deletions
Section 6.1.3 Timing and Location of Doses	Updated study intervention timing details for clarity.	Clarification
Section 6.6.2.1 Management of Participants with Gastrointestinal Symptoms	Updated guidance for use of anti-diarrheal in management of gastrointestinal symptoms	Clarification
Section 7.1.1 Permanent Discontinuation of Study Intervention	Added below criteria for permanent discontinuation “The emergence of an unacceptable risk to participants, a negative shift in the risk-benefit assessment as assessed by the DMC and/or sponsor, or a formal request from a concerned regulatory authority, or ethics committee”	Clarification
Section 8.1 Efficacy Assessments	Updated definition of symptomatic remission	Correction based on updated FDA guidance in 2022
Section 8.1.1.1 mMS	Updated maximum mMS to 9	Correction
Section 8.2.9.3 Study Drug Interruption or Discontinuation	Updated last bullet to read “Otherwise, both study drugs should be permanently discontinued”	Clarification
Section 8.2.14 Gallbladder Disease and Tirzepatide	Added safety assessment for risk of gallbladder disease associated with tirzepatide	Updated risk and measures for its minimization
Section 8.3.1. Timing and Mechanism of Collecting Events	Updated table as per original protocol	Correction of inadvertent deletions
Section 8.3.3 Adverse events of Special interest; Section 8.3.3.5 Major Adverse Cardiovascular Events-3; and Section 10.1.5.2 Clinical Event Committee	Replaced “cerebro-cardiovascular events” with “MACE-3” and “adjudication forms” with “endpoint forms”	Cerebrocardiovascular (CCV) events is a broader terminology and does not specify the exact endpoint that requires adjudication. Hence CCV has been updated with MACE-3 (CV death, MI, and stroke) clarifying the endpoints. Safety monitoring will remain the same.
Section 8.3.3.7 Suicidal Ideation or Behavior	Replaced “trained rater” with “an appropriately trained healthcare professional” for C-SSRS administration.	Clarification
Section 9.3.1 General Considerations	For treatment comparison, reworded details to read “The prespecified set of covariates include stratification factors from Section 6.3, and details will be specified in the SAP”	Clarification
Section 9.4 Interim Analysis	Updated interim analysis language to clarify that there is no intent to modify or stop the trial based on interim efficacy results. Added that no alpha spending will be allocated for interim analysis.	Clarification

Section # and Name	Description of Change	Brief Rationale
Section 9.5 Sample Size Determination	Added details of assumptions used in sample size calculation.	Clarification
Section 10.9 Appendix 9: Standardized Protocols for the Measurement of Height, Weight, Waist Circumference, BMI, and Vital Signs	Updated correct link for calculation of BMI with amputation or limb loss	Correction
Throughout protocol	Minor editorial and formatting changes	For consistency and editorial corrections

Amendment [c] (08 Aug 2025)

This amendment is considered to be nonsubstantial.

Overall rationale for the amendment

The main reason for this amendment is to address regulatory feedback.

Section # and Name	Description of Change	Brief Rationale
4.2. Scientific Rationale for Study Design	Added a new subsection, “Justification for use of placebo”.	Per regulatory request to add justification for the use of placebo.
7.1.1. Permanent Discontinuation of Study Intervention	Deleted the following text and moved it to Section 10.1.9: “The emergence of an unacceptable risk to participants, a negative shift in the risk-benefit assessment as assessed by the DMC and/or sponsor, or a formal request from a concerned regulatory authority, or ethics committee.”	
10.1.9. Study and Site Start and Closure	Added under “Study or site termination”: Additional reasons that could or should lead to trial discontinuation or termination include: the emergence of an unacceptable risk to participants, a negative shift in the risk-benefit assessment as assessed by the DMC and/or sponsor, or a formal request from a concerned regulatory authority or ethics committee.	Per regulatory request, to clarify the criteria for trial termination.

Amendment [d] (10 Sep 2025)

This amendment is considered to be nonsubstantial.

Overall rationale for the amendment

Section # and Name	Description of Change	Brief Rationale
Section 10.14.2.2. Hungary	Updated text to describe administration and evaluation of C-SSRS as below. “The C-SSRS scale will be administered by a psychiatrist or clinical psychologist with demonstrated expertise in the administration of the scale and assessment of suicidal ideation”	Clarification

11. References

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Approval	Shrish Budree Medical Director 04-Nov-2025 14:10:16 GMT+0000
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Approval	David Manner Statistician 04-Nov-2025 14:14:36 GMT+0000
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